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Method of treating an ocular condition by administering multispecific antibodies that activates TIE2 and binds a receptor tyrosine kinase agonist

Abstract

The disclosure provides compositions comprising multi-specific compounds, including a compound that targets a phosphatase and a receptor tyrosine kinase agonist. Also provided are methods for the treatment of conditions associated with angiogenesis, comprising administering a multi-specific compound that targets a phosphatase and a receptor tyrosine kinase agonist.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/117,993, filed Dec. 10, 2020, now U.S. Pat. No. 11,873,334, which is a continuation of U.S. application Ser. No. 16/579,078, filed Sep. 23, 2019, now U.S. Pat. No. 10,894,824, which claims the benefit of U.S. Provisional Application No. 62/832,461, filed Apr. 11, 2019, and U.S. Provisional Application No. 62/735,331, filed Sep. 24, 2018, each of which is incorporated herein by reference in its entirety.

REFERENCE TO A SEQUENCE LISTING SUBMITTED ELECTRONICALLY VIA EFS-WEB

(1) The content of the electronically submitted sequence listing (Name: 4986_0210005_Seqlisting_ST26, Size: 345,321 bytes; and Date of Creation: Dec. 1, 2023) is herein incorporated by reference in its entirety.

BACKGROUND

(2) Individual compounds with the ability to modulate distinct targets can be combined to generate multispecific compounds. Such multispecific compounds can have advantages over the parent compounds administered individually. These advantages can include, for example, a simpler dosing regimen, longer half-life within a subject, or the ability to bind targets in close proximity.

INCORPORATION BY REFERENCE

(3) Each patent, publication, and non-patent literature cited in the application is hereby incorporated by reference in its entirety as if each was incorporated by reference individually.

SUMMARY

(4) In some embodiments, the disclosure provides a compound comprising: (a) a first domain, wherein the first domain modulates a phosphatase, wherein the phosphatase modulates Tie2; and (b) a second domain that specifically binds a receptor tyrosine kinase agonist.

(5) In some embodiments, the disclosure provides a method of treating a condition in a subject in need thereof, the method comprising administering to the subject a therapeutically-effective amount of a compound disclosed herein.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1: Schematic of the basic four chain antibody unit. Light chain sequences are represented by “SEQ A”. Heavy chain sequences are represented by “SEQ B”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(2) FIG. 2: Schematic of a tetravalent, bispecific antibody with sequences appended to the heavy chain C-termini. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B”. Linker sequences are represented by “SEQ C”. Appended antigen binding domain sequences are represented by “SEQ D”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(3) FIG. 3: Schematic of a tetravalent, bispecific antibody with sequences appended to the light chain C-termini. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B”. Linker sequences are represented by “SEQ C”. Appended antigen binding domain sequences are represented by “SEQ D”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(4) FIG. 4: Schematic of a hexavalent, bispecific antibody with sequences appended to the heavy chain and light chain C-termini. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B”. Linker sequences are represented by “SEQ C” and “SEQ E”. Appended antigen binding domain sequences are represented by “SEQ D”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(5) FIG. 5: Schematic of a hexavalent, trispecific antibody with sequences appended to the heavy chain and light chain C-termini. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B”. Linker sequences are represented by “SEQ C” and “SEQ E”. Appended antigen binding domain sequences are represented by “SEQ D” and “SEQ F”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(6) FIG. 6: Schematic of a bivalent, bispecific antibody with two different heavy chain sequences and two different light chain sequences. Light chain sequences are represented by “SEQ A” and “SEQ D”. Heavy chain sequences are represented by “SEQ B” and “SEQ C”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(7) FIG. 7: Schematic of a trivalent, trispecific antibody with two different heavy chain sequences, two different light chain sequences, and a sequence appended to the C-terminus of one light chain. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A” and “SEQ D”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B” and “SEQ C”. A linker sequence is represented by “SEQ E”. An appended antigen binding domain sequence is represented by “SEQ F”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(8) FIG. 8: Schematic of a trivalent, trispecific antibody with two different heavy chain sequences, two different light chain sequences, and a sequence appended to the C-terminus of one heavy chain. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A” and “SEQ D”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B” and “SEQ C”. A linker sequence is represented by “SEQ E”. An appended antigen binding domain sequence is represented by “SEQ F”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(9) FIG. 9: Schematic of a tetravalent antibody with two different heavy chain sequences, two different light chain sequences, and sequences appended to both light chain C-termini. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A” and “SEQ D”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B” and “SEQ C”. Linker sequences are represented by “SEQ E” and “SEQ F”. Appended antigen binding domain sequences are represented by “SEQ G” and “SEQ H”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(10) FIG. 10: Schematic of a tetravalent antibody with two different heavy chain sequences, two different light chain sequences, and sequences appended to both heavy chain C-termini. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A” and “SEQ D”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B” and “SEQ C”. Linker sequences are represented by “SEQ E” and “SEQ F”. Appended antigen binding domain sequences are represented by “SEQ G” and “SEQ H”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(11) FIG. 11: Schematic of a tetravalent antibody with two different heavy chain sequences, two different light chain sequences, and sequences appended to one heavy chain C-terminus and one light chain C-terminus in cis. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A” and “SEQ D”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B” and “SEQ C”. Linker sequences are represented by “SEQ E” and “SEQ F”. Appended antigen binding domain sequences are represented by “SEQ G” and “SEQ H”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(12) FIG. 12: Schematic of a tetravalent antibody with two different heavy chain sequences, two different light chain sequences, and sequences appended to one heavy chain C-terminus and one light chain C-terminus in trans. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A” and “SEQ D”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B” and “SEQ C”. Linker sequences are represented by “SEQ E” and “SEQ F”. Appended antigen binding domain sequences are represented by “SEQ G” and “SEQ H”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(13) FIG. 13: Schematic of a pentavalent antibody with two different heavy chain sequences, two different light chain sequences, and sequences appended to both heavy chain C-termini and one light chain C-terminus. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A” and “SEQ D”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B” and “SEQ C”. Linker sequences are represented by “SEQ E”, “SEQ F”, and “SEQ G”. Appended antigen binding domain sequences are represented by “SEQ H”, “SEQ I”, and “SEQ J”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(14) FIG. 14: Schematic of a pentavalent antibody with two different heavy chain sequences, two different light chain sequences, and sequences appended to one heavy chain C-terminus and both light chain C-termini. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A” and “SEQ D”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B” and “SEQ C”. Linker sequences are represented by “SEQ E”, “SEQ F”, and “SEQ G”. Appended antigen binding domain sequences are represented by “SEQ H”, “SEQ I”, and “SEQ J”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(15) FIG. 15: Schematic of a hexavalent antibody with two different heavy chain sequences, two different light chain sequences, and sequences appended to the heavy chain and light chain C-termini. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A” and “SEQ D”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B” and “SEQ C”. Linker sequences are represented by “SEQ E”, “SEQ F”, “SEQ G”, and “SEQ H”. Appended antigen binding domain sequences are represented by “SEQ I”, “SEQ J”, “SEQ K” and “SEQ L”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(16) FIG. 16: ELISA data demonstrating binding of a tetravalent bispecific antibody HC2-AFL:LC1 to HPTP- β (top panel) and VEGF (bottom panel).

(17) FIG. 17: ELISA data demonstrating binding of a tetravalent bispecific antibody HC2-BRO:LC1 to HPTP- β (top panel) and VEGF (bottom panel).

(18) FIG. 18: ELISA data testing binding of a tetravalent bispecific antibody HC2-RAN:LC1 to HPTP- β (top panel) and VEGF (bottom panel).

(19) FIG. 19: Activation of Tie2 in HUVECs treated with tetravalent bispecific antibodies or hexavalent bispecific antibodies comprising brolucizumab-derived or aflibercept-derived VEGF-binding domains. As demonstrated by immunoprecipitation and western blot, all of the tested bispecific antibodies increased basal Tie2 activation (in the absence of exogenous Ang1 or Ang2), while basal VEGFR2 phosphorylation (in the absence of exogenous VEGF) was not affected.

(20) FIG. 20: Tetravalent bispecific and hexavalent bispecific antibodies comprising brolucizumab-derived or aflibercept-derived VEGF-binding domains enhance Tie2 activation and inhibit VEGFR2 activation in HUVECs treated with Ang1 and VEGF, as demonstrated by immunoprecipitation and western blot.

(21) FIG. 21A, FIG. 21B, and FIG. 21C: Tetravalent bispecific and hexavalent bispecific antibodies comprising abicipar-derived or aflibercept-derived VEGF-binding domains enhance Tie2 activation and inhibit VEGFR2 activation in HUVECs, including HUVECs treated with Ang1 and VEGF. FIG. 21A provides a western blot. FIG. 21B provides quantification of Tie2 activation. FIG. 21C provides quantification of VEGFR2 phosphorylation.

(22) FIG. 22A, FIG. 22B, and FIG. 22C: Hexavalent bispecific antibodies comprising brolocizumab-derived, aflibercept-derived, or abicipar-derived VEGF-binding domains enhance Tie2 activation and inhibit VEGFR2 activation in HUVECs, including HUVECs treated with Ang1 and VEGF. FIG. 22A provides a western blot. FIG. 22B provides quantification of Tie2 activation. FIG. 22C provides quantification of VEGFR2 phosphorylation.

(23) FIG. 23A and FIG. 23B: Tetravalent bispecific and hexavalent bispecific antibodies comprising brolocizumab-derived, aflibercept-derived, or abicipar-derived VEGF-binding domains enhance Tie2 activation and inhibit VEGFR2 activation in HUVECs treated with Ang1 and VEGF, as demonstrated by electrochemiluminescence signal quantification. FIG. 23A provides quantification of Tie2 activation. FIG. 23B provides quantification of VEGFR2 phosphorylation.

(24) FIG. 24A and FIG. 24B: Hexavalent bispecific antibodies comprising brolocizumab-derived, aflibercept-derived, or abicipar-derived VEGF-binding domains enhance Tie2 activation and inhibit VEGFR2 activation in HUVECs treated with Ang1 and VEGF, as demonstrated by electrochemiluminescence signal quantification. FIG. 24A provides quantification of Tie2 activation. FIG. 24B provides quantification of VEGFR2 phosphorylation.

(25) FIG. 25A and FIG. 25B: Tetravalent bispecific and hexavalent bispecific antibodies comprising brolocizumab-derived VEGF-binding domains enhance Tie2 activation and inhibit VEGFR2 activation in HUVECs treated with Ang1 and VEGF, as demonstrated by electrochemiluminescence signal quantification. FIG. 25A provides quantification of Tie2 activation. FIG. 25B provides quantification of VEGFR2 phosphorylation.

(26) FIG. 26A and FIG. 26B: Tetravalent bispecific and hexavalent bispecific antibodies comprising abicipar-derived or aflibercept-derived VEGF-binding domains enhance Tie2 activation and inhibit VEGFR2 activation in HUVECs treated with Ang1 and VEGF, as demonstrated by electrochemiluminescence signal quantification. FIG. 26A provides quantification of Tie2 activation. FIG. 26B provides quantification of VEGFR2 phosphorylation.

(27) FIG. 27A and FIG. 27B: Tetravalent bispecific and hexavalent bispecific antibodies comprising abicipar-derived or aflibercept-derived VEGF-binding domains enhance Tie2 activation and inhibit VEGFR2 activation in HUVECs treated with Ang1 and VEGF, as demonstrated by electrochemiluminescence signal quantification. FIG. 27A provides quantification of Tie2 activation. FIG. 27B provides quantification of VEGFR2 phosphorylation.

DETAILED DESCRIPTION

(28) The present disclosure provides compositions and methods for modulating phosphatases and kinases, for example, receptor tyrosine kinases. Compositions and methods are provided for modulating Tie2, for example, to promote Tie2 phosphorylation, signaling, and/or activation. In some embodiments, the disclosure provides compositions and methods for targeting a phosphatase that modulates Tie2 signaling. In some embodiments, the phosphatase that modulates Tie2 signaling is human protein tyrosine phosphatase-beta (HPTP-0).

(29) Compositions and methods are provided for modulating receptor tyrosine kinases, for example, to reduce receptor tyrosine kinase phosphorylation, signaling, and/or activation. In some embodiments, the disclosure provides compositions and methods for targeting a receptor tyrosine kinase agonist, e.g. vascular endothelial growth factor (VEGF). In some embodiments, the disclosure provides compositions and methods for targeting a receptor tyrosine kinase, e.g. a VEGF receptor.

(30) In some embodiments, the present disclosure provides compositions and methods for targeting human protein tyrosine phosphatase-beta (HPTP- β) or vascular endothelial protein tyrosine

phosphatase (VE-PTP or VEPTP), and vascular endothelial growth factor (VEGF). In some embodiments, the present disclosure provides multispecific compounds, agents, antibodies, fragments, or derivatives thereof that target HPTP- β (VE-PTP) and VEGF.

(31) The agents disclosed herein can be used for the treatment of disorders that are characterized by, for example, vascular instability, angiogenesis, neovascularization, vascular leakage, and/or edema. The agents disclosed herein can be used for the treatment of, for example, vascular disorders, ocular disorders, cancers, renal disorders, and complications of diabetes.

(32) HPTP- β /VE-PTP, Tie2, and Vascular Stability

(33) HPTP- β is a member of the receptor-like family of the protein tyrosine phosphatases (PTPases). HPTP- β is a transmembrane protein found primarily in vascular endothelial cells that displays structural and functional similarity to cell adhesion molecules. Orthologues of HPTP-0 are found in various species including, for example, zebrafish, chicken, dog, mouse, marmoset, and monkey. The orthologues are generally referred to as vascular endothelial protein tyrosine phosphatase (VE-PTP). HPTP- β (VE-PTP) can influence vascular stability through effects on Tie2-mediated signaling.

(34) Tie2 (tyrosine kinase with immunoglobulin and epidermal growth factor homology domains 2) is a membrane receptor tyrosine kinase expressed primarily in vascular endothelial cells. Upstream factors can regulate Tie2 phosphorylation, influencing downstream signaling and vascular stabilization. Non-limiting examples of such factors include angiopoietin 1 (Ang1/Angpt1), angiopoietin 2 (Ang2/Angpt2), and HPTP- β (VE-PTP).

(35) Ang1 is an agonist of Tie2. Binding of Ang1 to Tie2 promotes receptor phosphorylation and downstream signaling to induce vascular stabilization through highly organized angiogenesis, tightening of endothelial cell junctions, enhancement of endothelial viability, reduction of endothelial inflammation, and improvement of endothelial function.

(36) Ang2 acts in a context-dependent antagonist or agonist of Tie2. During angiogenesis, Ang2 acts as a negative regulator of Ang1-Tie2 signaling.

(37) HPTP- β (VE-PTP) is a phosphatase that can modulate Tie2 signaling. HPTP- β (VE-PTP) can dephosphorylate the Tie2 receptor. Under physiological conditions, HPTP- β (VE-PTP) regulates the duration of Tie2 phosphorylation. Inhibition of HPTP- β (VE-PTP), therefore, can result in increased Tie2 phosphorylation, increased Tie2-mediated signaling, and enhanced vascular stability. Inhibitors of HPTP- β (VE-PTP) are Tie2 activators. For example, a compound, inhibitor, antibody, antibody fragment, variant, or derivative thereof that binds HPTP- β (VE-PTP) can promote Tie2 phosphorylation, thereby activating Tie2 downstream signaling, and promoting vascular stability.

(38) By the process described above, HPTP- β (VE-PTP) activity can contribute to, for example, disorders that are characterized by vascular instability, angiogenesis, neovascularization, vascular leakage, and/or edema. For example, HPTP- β (VE-PTP) activity can contribute to vascular disorders, ocular disorders, cancers, renal disorders, complications of diabetes, and other disorders. Inhibition of HPTP- β (VE-PTP) activity can reduce such disorders.

(39) VEGF and Vascular Stability

(40) Vascular endothelial growth factors (VEGFs) are primarily found in endothelial cells, and are implicated in pathological neovascularization in a number of diseases. The VEGFs are members of the cystine-knot growth factor superfamily, the PDGF family, and the VEGF family. The VEGFs can act as pro-angiogenic factors. The VEGF family consists of VEGF-A, VEGF-B, VEGF-C, VEGF-D and placental growth factor (PGF). Nine VEGF-A isoforms exist: VEGF.sub.121, VEGF.sub.145, VEGF.sub.148, VEGF.sub.162, VEGF.sub.16s, VEGF.sub.165b, VEGF.sub.183, VEGF.sub.189, and VEGF.sub.206.

(41) VEGF is a hypoxia-regulated gene, and VEGF levels are increased in hypoxic or ischemic conditions. VEGF is an agonist of VEGF receptors (VEGFRs). VEGFRs are receptor tyrosine kinases; binding of VEGF to a VEGFR can result in phosphorylation of the receptor, and

subsequently of downstream signal transducers. VEGFR-mediated signaling can result in aberrant vasculogenesis, angiogenesis, and permeabilization of blood vessels, contributing to pathologic vascular instability. Thus, inhibition of VEGF can result in decreased VEGFR-mediated signaling and enhanced vascular stability. For example, an inhibitor, antibody, antibody fragment, variant, or derivative thereof that binds VEGF can reduce VEGFR ligation, thereby reducing VEGFR-mediated signaling, and promoting vascular stability. Non-limiting examples of agents that bind VEGF include aflibercept (Eylea®), a recombinant protein comprising the VEGF-binding portions of human VEGF receptors 1 and 2 fused to the Fc portion of human IgG1; brolucizumab, a humanized single-chain antibody fragment (scFv); RTH258, a humanized single-chain antibody fragment (scFv); ranibizumab (Lucentis®), a humanized monoclonal antibody fragment (Fab); bevacizumab (Avastin®), a humanized monoclonal antibody; conbercept, a recombinant fusion protein comprising extracellular domains from VEGF receptors 1 and 2 fused to the Fc portion of human IgG1; Abicipar, a designed ankyrin repeat protein (DARPin); MP0112, a DARPin; MP0250, a DARPin; CT-322, an adnectin; and PRS-050, an anticalin.

(42) By the process described above, VEGF can contribute to, for example, disorders that are characterized by vascular instability, angiogenesis, neovascularization, vascular leakage, and/or edema. For example, VEGF can contribute to vascular disorders, ocular disorders, cancers, renal disorders, complications of diabetes, and other disorders. For example, ischemia in the eye can lead to increased VEGF production, resulting in vascular leakage and pathological neovascularization in the retina. Inhibition of VEGFR-mediated signaling can reduce such disorders.

(43) Receptor Tyrosine Kinases (RTKs) and Receptor Tyrosine Kinase Agonists

(44) Receptor tyrosine kinases (RTKs) are cell surface receptors that participate in the regulation of cell growth, differentiation, and survival. Binding of an agonist to a RTK can cause neighboring RTKs to associate with each other, forming dimers. Dimerization can cause cross-phosphorylation—each RTK in the dimer phosphorylates multiple tyrosine residues on the other RTK. Once cross-phosphorylated, the cytoplasmic tails of the RTKs can initiate signal transduction pathways, for example, by serving as docking platforms for various intracellular proteins. RTK signaling can lead to changes of gene transcription and expression in a cell.

(45) Non-limiting examples of RTKs include AATK, AATYK, AATYK1, AATYK2, ACH, ALK, ARK, AXL, BDB, BDB1, BEK, BFGFR, BREK, Brt, CAK, CCK4, CD115, CD117, CD135, CD136, CD140a, CD140b, CD167, CD202b, CD220, CD221, CD246, CD309, CD331, CD332, CD333, CD334, CDHF12, CDHR16, CDw136, CEK, CEK2, CEK3, c-Eyk, CFD1, C-FMS, C-Kit, cprk, c-ros-1, CSF1R, CSFR, D3S3195, DDR1, DDR2, DFNB97, DKFZp761P1010, Dtk, ECT1, EDDR1, EGFR, EphA10, EphA1-8, EphB1, EphB2, EphB3, EphB4, EphB6, ErbB2, ErbB3, ErbB4, Etk-2, FGFR1, FGFR2, FGFR3, FGFR4, FLG, FLK1, FLK2, FLT, FLT1, FLT2, FLT3, FLT4, FMS, GAS9, H2, H3, H4, H5, HGFR, HSCR1, IGF1R, IGFR, IGFR, INSR, INSR, IRR, JKT5A, JTK11, JTK12, JTK13, JTK14, JTK2, JTK4, JTK5, JWS, KAL2, KDR, KGFR, KIAA0641, KIAA1079, KIAA1883, KIT, KPI2, K-SAM, LMR1, LMR2, LMR3, LMTK1, LMTK2, LMTK3, LTK, MCF3, MEN2A, MEN2B, Mer, MERTK, MET, MGC18216, MST1R, MTC, MTC1, MTRK1, MuSK, NEP, NOK, N-SAM, NTRK2, NTRK3, NTRK4, NTRKR1, NTRKR2, PBT, PCL, PDGFR, PDGFR1, PDGFR2, PDGFRA, PDGFR-alpha, PDGFRB, PDGFR-beta, PPP1R100, PPP1R101, PPP1R77, PTC, PTK3A, PTK7, PTK8, RCCP2, Rek, RET, RET51, RON, ROR1, ROR2, ROS, ROS1, RP38, RSE, RTK6, RYK, Ryk, RYK1, SCFR, Sky, STK, STYK1, SuRTK106, TEK, TIE1, TIE2, Tif, TK14, TK25, TKT, TRK, TrkA, TrkB, TrkC, TYK1, TYKLM3, TYRO10, Tyro12, Tyro3, Tyro7, UFO, VEGFR, VEGFR1, VEGFR2, VEGFR3, VMCM, and VMCM1.

(46) Non-limiting examples of RTK agonists include VEGF, Ang1, Ang2, BDNF, EGF, FGF, HGF, IGF, insulin, MSP, NGF, NT-3, and PDGF.

(47) Antibodies and Antigen-Binding Compounds.

(48) The basic four chain antibody unit comprises two identical heavy chain (H) polypeptide

sequences and two identical light chain (L) polypeptide sequences. Each of the heavy chains can comprise one N-terminal variable (V.sub.H) region and three or four C-terminal constant (C.sub.H1, C.sub.H2, C.sub.H3, and C.sub.H4) regions. Each of the light chains can comprise one N-terminal variable (V.sub.L) region and one C-terminal constant (C.sub.L) region. The light chain variable region is aligned with the heavy chain variable region and the light chain constant region is aligned with heavy chain constant region C.sub.H1. The pairing of a heavy chain variable region and light chain variable region together forms a single antigen-binding site. Each light chain is linked to a heavy chain by one covalent disulfide bond. The two heavy chains are linked to each other by one or more disulfide bonds depending on the heavy chain isotype. Each heavy and light chain also comprises regularly-spaced intrachain disulfide bridges. The C-terminal constant regions of the heavy chains comprise the Fc region of the antibody, which mediate effector functions, for example, through interactions with Fc receptors or complement proteins. FIG. 1 provides a simple representative schematic basic four chain antibody unit; light chain sequences are represented by "SEQ A". Heavy chain sequences are represented by "SEQ B", —S—S— denotes disulfide bonds, N and C denote N- and C-termini, respectively.

(49) The light chain can be designated kappa or lambda based on the amino acid sequence of the constant region. The heavy chain can be designated alpha, delta, epsilon, gamma, or mu based on the amino acid sequence of the constant region. Antibodies are categorized into five immunoglobulin classes, or isotypes, based on the heavy chain. IgA comprises alpha heavy chains, IgD comprises delta heavy chains, IgE comprises epsilon heavy chains, IgG comprises gamma heavy chains, and IgM comprises mu heavy chains. Antibodies of the IgG, IgD, and IgE classes comprise monomers of the four chain unit described above (two heavy and two light chains), while the IgM and IgA classes can comprise multimers of the four chain unit. The alpha and gamma classes are further divided into subclasses on the basis of differences in the sequence and function of the heavy chain constant region. Subclasses of IgA and IgG expressed by humans include IgG1, IgG2, IgG3, IgG4, IgA1 and IgA2.

(50) The constant regions are minimally involved in antigen binding. Rather, the constant regions can mediate various effector functions. Different IgG isotypes or subclasses can be associated with different effector functions or therapeutic characteristics, for example, because of interactions with different Fc receptors and/or complement proteins. Antibodies comprising Fc regions that engage activating Fc receptors can, for example, participate in antibody-dependent cell-mediated cytotoxicity (ADCC), antibody-dependent cellular phagocytosis (ADCP), complement-dependent cytotoxicity (CDC), induction of signaling through immunoreceptor tyrosine-based activation motifs (ITAMs), and induction of cytokine secretion. Antibodies comprising Fc regions that engage inhibitory Fc receptors can, for example, induce signaling through immunoreceptor tyrosine-based inhibitory motifs (ITIMs).

(51) Different antibody subclasses comprise different abilities to elicit immune effector functions. For example, IgG1 and IgG3 can effectively recruit complement to activate CDC, IgG2 elicits minimal ADCC. IgG4 has a lesser ability to trigger immune effector functions. Modifications to the constant regions can also affect antibody characteristics, for example, enhancement or reduction of Fc receptor ligation, enhancement or reduction of ADCC, enhancement or reduction of ADCP, enhancement or reduction of CDC, enhancement or reduction of signaling through ITAMs, enhancement or reduction of cytokine induction, enhancement or reduction of signaling through ITIMs, enhancement or reduction of half-life, or enhancement or reduction of coengagement of antigen with Fc receptors. Modifications can include, for example, amino acid mutations, altering post-translational modifications (e.g., glycosylation), combining domains from different isotypes or subclasses, or a combination thereof.

(52) A compound or antibody of the disclosure can comprise constant regions or Fc regions that are selected or modified to provide suitable antibody characteristics, for example, suitable characteristics for treating a disease or condition as disclosed herein. In some embodiments, IgG1

can be used, for example, to promote immune activation effector functions (e.g., ADCC, ADCP, CDC, ITAM signaling, cytokine induction, or a combination thereof for the treatment of a cancer). In some embodiments, IgG4 can be used, for example, in cases where antagonistic properties of the antibody in the absence of immune effector functions are desirable (e.g., for treatment of ocular disorders).

(53) Non-limiting examples of antibody modifications and their effects are provided in TABLE 1.

(54) TABLE-US-00001 TABLE 1 Effect Isotype Mutation(s)/modification(s) Enhanced ADCC IgG1 F243L/R292P/Y300L/V305I/ P396L Enhanced ADCC IgG1 S239D/I332E Enhanced ADCC IgG1 S239D/I332E/A330L Enhanced ADCC IgG1 S298A/E333A/K334A Enhanced ADCC IgG1 In one heavy chain: L234Y/ L235Q/G236W/S239M/H268D/ D270E/S298A In the opposing heavy chain: D270E/K326D/A330M/K334E Enhanced ADCP IgG1 G236A/S239D/I332E Enhanced CDC IgG1 K326W/E333S Enhanced CDC IgG1 S267E/H268F/S324T Enhanced CDC IgG1, Combination of domains from IgG3 IgG1/IgG3 Enhanced CDC IgG1 E345R/E430G/S440Y Loss of glycosylation, IgG1 N297A or N297Q or N297G reduced effector functions Reduced effector functions IgG1, L235E IgG4 Reduced effector functions IgG1 L234A/L235A Reduced effector functions IgG4 F234A/L235A Reduced effector functions IgG4 F234A/L235A/G237A/P238S Reduced effector functions IgG4 F234A/L235A/ Δ G236/ G237A/P238S Reduced effector functions IgG2, Combination of domains from IgG4 IgG2/IgG4 Reduced effector functions IgG2 H268Q/V309L/A330S/P331S Reduced effector functions IgG2 V234A/G237A/P238S/H268A/ V309L/A330S/P331S Reduced effector functions IgG1 L234A/L235A/G237A/P238S/ H268A/A330S/P331S Increased half-life IgG1 M252Y/S254T/T256E Increased half-life IgG1 M428L/N434S Increased antigen/Fc IgG1 S267E/L328F receptor coengagement Altered antigen/Fc IgG1 N325S/L328F receptor coengagement Reduced Fab arm exchange IgG4 S228P

(55) The variable (V) regions mediate antigen binding and define the specificity of a particular antibody for an antigen. The variable region comprises relatively invariant sequences called framework regions, and hypervariable regions, which differ considerably in sequence among antibodies of different binding specificities. The variable region of each antibody heavy or light chain comprises four framework regions separated by three hypervariable regions. The variable regions of heavy and light chains fold in a manner that brings the hypervariable regions together in close proximity to create an antigen binding site. The four framework regions largely adopt an f3-sheet configuration, while the three hypervariable regions form loops connecting, and in some cases forming part of, the f3-sheet structure.

(56) Within hypervariable regions are amino acid residues that primarily determine the binding specificity of the antibody. Sequences comprising these residues are known as complementarity determining regions (CDRs). One antigen binding site of an antibody comprises six CDRs, three in the hypervariable regions of the light chain, and three in the hypervariable regions of the heavy chain. The CDRs in the light chain are designated L1, L2, and L3, while the CDRs in the heavy chain are designated H1, H2, and H3. CDRs can also be designated LCDR1, LCDR2, LCDR3, HCDR1, HCDR2, and HCDR3, respectively. The contribution of each CDR to antigen binding varies among antibodies. CDRs can vary in length. For example, CDRs are often 5 to 14 residues in length, but CDRs as short as 0 residues or as long as 25 residues or longer exist.

(57) Several methods are used to predict or designate CDR sequences. These methods can use different numbering systems, for example, because sequence insertions and deletions are numbered differently.

(58) The Kabat method was developed by aligning a limited number of antibody sequences and determining the positions of the most variable residues. Based on the alignment, a numbering scheme was introduced for residues in the variable regions. This numbering scheme can be used to determine the positions marking the beginning and the end of each CDR. One iteration of the Kabat numbering system identifies CDRs in the light chain variable region using the following residue positions: LCDR1 around residues 24-34; LCDR2 around residues 50-56; and LCDR3 around

residues 89-97. One iteration of the Kabat numbering system identifies CDRs in the heavy chain variable region using the following residue positions: HCDR1 around residues 31-35; HCDR2 around residues 50-65; and HCDR3 around residues 95-102.

(59) The Chothia method was developed based on analysis of three dimensional antibody structures. The analysis determined that hypervariable loops adopt a restricted set of conformations based on the presence of certain residues at key positions in CDRs and flanking framework regions. This method uses a similar numbering scheme as the Kabat method, but numbers insertions and deletions differently. One iteration of the Chothia numbering system identifies CDRs in the light chain variable region using the following residue positions: LCDR1 around residues 24-34; LCDR2 around residues 50-56; and LCDR3 around residues 89-97. One iteration of the Chothia numbering system identifies CDRs in the heavy chain variable region using the following residue positions: HCDR1 around residues 26-34; HCDR2 around residues 52-56; and HCDR3 around residues 95-102.

(60) The IMGT method (International ImMunoGeneTics database) was developed by integrating existing definitions of framework regions and CDRs, structural data, and data from alignment of antibody variable region sequences. This integration led to the identification of conserved residues in the framework regions that can be used as reference points for identifying CDRs. Examples of conserved residues in variable regions include cysteine at approximately residue 23 (in framework region 1), tryptophan at approximately residue 41 (in framework region 2), a hydrophobic amino acid at approximately residue 89 (in framework region 3), cysteine at approximately residue 104 (in framework region 3), and phenylalanine or tryptophan at approximately residue 118 (in framework region 4). CDRs can be identified in a sequence encoding an antibody variable region of interest by using a computational alignment-based algorithm.

(61) The IMGT method of numbering consistently assigns the same numbers to the conserved amino acids, but the lengths of CDRs and framework regions are permitted to vary. Therefore, IMGT numbering of residues is not necessarily sequential. The length of CDRs identified by the IMGT method can vary. For example, LCDR1 or HCDR1 can be about 5 to about 12 amino acids, LCDR2 or HCDR2 can be about 0 to about 10 amino acids, and LCDR3 or HCDR3 can be about 5 to about 91 amino acids.

(62) The Paratome method was developed based on multiple structural alignments of available antibody-antigen complexes. The structural positions that bind antigen were found to be similar among the examined antibodies, and antibody sequences from the data set were annotated with Antigen Binding Regions (ABRs, similar to CDRs). ABRs in a query sequence can be identified using a computational tool, which first aligns the query sequence against antibodies with solved antibody-antigen structures, then infers the positions of ABRs based on the alignment. Antibodies with solved structures can also have ABRs identified using a structural, rather than sequence-based, alignment method.

(63) A subset of residues within CDRs contacts an antigen. These residues that contact antigen can be referred to as specificity-determining residues (SDRs). However, residues other than SDRs can contribute to binding activity by helping to maintain the conformation of the binding site. The number of SDRs in an antibody can vary based on the size and type of antigen that is recognized, for example, between 0-14 SDRs can be found within a CDR. SDRs can be enriched in some residues, such as tyrosine, serine, tryptophan, and asparagine.

(64) A monoclonal antibody can be obtained from a population of substantially-homogeneous antibodies, i.e., the individual antibodies comprising the population are identical except for possible naturally-occurring mutations that can be present in minor amounts. In contrast to polyclonal antibody preparations, which include different antibodies directed against different epitopes, each monoclonal antibody is directed against a single epitope.

(65) A compound herein can be a monoclonal antibody, for example, a chimeric antibody wherein a portion of the heavy and/or light chain is identical to or homologous to a corresponding sequence

in an antibody derived from a particular species or belonging to a particular antibody class or subclass, while the remainder of the chain(s) is identical or homologous to a corresponding sequence in an antibody derived from another species or belonging to another antibody class or subclass, or an antigen-binding fragment of such an antibody.

(66) An antibody fragment or antigen-binding fragment can comprise a portion of an antibody, for example, the antigen-binding or variable region of the intact antibody. Non-limiting examples of antibody fragments include Fab, Fab', F(ab').sub.2, dimers and trimers of Fab conjugates, Fv, scFv, minibodies, dia-, tria-, and tetrabodies, and linear antibodies. Fab and Fab' are antigen-binding fragments that can comprise the V.sub.H and C.sub.H1 domains of the heavy chain linked to the V.sub.L and CL domains of the light chain via a disulfide bond. A F(ab').sub.2 can comprise two Fab or Fab' that are joined by disulfide bonds. A Fv can comprise the V.sub.H and V.sub.L domains held together by non-covalent interactions. A scFv (single-chain variable fragment) is a fusion protein that can comprise the V.sub.H and V.sub.L domains connected by a peptide linker. Manipulation of the orientation of the V.sub.H and V.sub.L domains and the linker length can be used to create different forms of molecules that can be monomeric, dimeric (diabody), trimeric (triabody), or tetrameric (tetrabody). Minibodies are scFv-C.sub.H3 fusion proteins that assemble into bivalent dimers.

(67) Non-limiting examples of epitopes include amino acids, sugars, lipids, phosphoryl, and sulfonyl groups. An epitope can have specific three-dimensional structural characteristics, and/or specific charge characteristics. Epitopes can be conformational or linear.

(68) For human administration, monoclonal antibodies generated from non-human species can be further optimized by a humanization process to reduce the likelihood of immunogenicity while preserving target specificity. Humanization processes involve the incorporation of human DNA to the genetic sequence of the genes that produce the isolated antibodies. The recombinant DNA is cloned and expressed in cells for large-scale production of the newly humanized antibodies.

(69) An example of a humanized antibody is a modified chimeric antibody. A chimeric antibody can be generated as described above. The chimeric antibody is further mutated outside of the CDRs to substitute non-human sequences in the variable regions with the homologous human sequences. Another example of a humanized antibody is a CDR-grafted antibody, in which non-human CDR sequences are introduced into the human heavy and light chain variable sequences of a human antibody scaffold to replace the corresponding human CDR sequences.

(70) A humanized antibody can be produced in mammalian cells, bioreactors, or transgenic animals, such as mouse, chicken, sheep, goat, pig, or marmoset. The transgenic animal can have a substantial portion of the human antibody-producing genome inserted into the genome of the animal.

(71) In addition to antibodies and antibody fragments, other antigen-binding compounds can also bind target molecules. Non-limiting examples of non-antibody-derived antigen-binding compounds include ankyrin proteins, ankyrin repeat proteins, designed ankyrin repeat proteins (DARPs), affibodies, avimers, adnectins, anticalins, Fynomers, Kunitz domains, knottins, (3-hairpin mimetics, and receptors and derivatives thereof, e.g. VEGF receptors, or the VEGF-binding portions of human VEGF receptors 1 and 2.

(72) Designed ankyrin repeat proteins (DARPs) can be protein scaffolds based on ankyrin repeat proteins. A DARPin can comprise one or more ankyrin repeats that comprise a shared sequence and/or structural motif. The individual ankyrin repeats can comprise a shared sequence and/or structural motif despite comprising mutations, substitutions, additions and/or deletions when compared to one other. A DARPin can comprise, for example, about 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 ankyrin repeats, or more. A DARPin can comprise an N-terminal capping repeat, one or more internal ankyrin repeats, and a C-terminal capping repeat. Each ankyrin repeat can comprise framework residues and protein-interaction residues. The framework residues can contribute to structure or folding topology, for example, the structure of an ankyrin repeat or interaction with a

neighboring ankyrin repeat. Protein-interaction residues can contribute to binding of a target molecule, for example, via direct interaction with the target molecule, or by stabilizing directly-interacting residues in a conformation that allows binding.

(73) Compounds, antibodies, fragments or derivatives thereof, or other compounds that bind target molecules in this disclosure can bind to targets with a $K_{sub.D}$ of, for example, less than about 500 nM, less than about 300 nM, less than about 200 nM, less than about 100 nM, less than about 90 nM, less than about 80 nM, less than about 70 nM, less than about 60 nM, less than about 50 nM, less than about 40 nM, less than about 30 nM, less than about 20 nM, less than about 10 nM, less than about 5 nM, less than about 4 nM, less than about 3 nM, less than about 2 nM, less than about 1 nM, less than about 900 pM, less than about 800 pM, less than about 700 pM, less than about 600 pM, less than about 500 pM, less than about 400 pM, less than about 300 pM, less than about 200 pM, less than about 100 pM, less than about 90 pM, less than about 80 pM, less than about 70 pM, less than about 60 pM, less than about 50 pM, less than about 40 pM, less than about 30 pM, less than about 20 pM, less than about 10 pM, less than about 9 pM, less than about 8 pM, less than about 7 pM, less than about 6 pM, less than about 5 pM, less than about 4 pM, less than about 3 pM, less than about 2 pM, less than about 1 pM, less than about 900 fM, less than about 800 fM, less than about 700 fM, less than about 600 fM, less than about 500 fM, less than about 400 fM, less than about 300 fM, less than about 200 fM, less than about 100 fM, less than about 90 fM, less than about 80 fM, less than about 70 fM, less than about 60 fM, less than about 50 fM, less than about 40 fM, less than about 30 fM, less than about 20 fM, or less than about 10 fM.

(74) In some embodiments, a compound, antibody, fragment or derivatives thereof, or other compound that a bind target molecule in this disclosure can bind to the target with a $K_{sub.D}$ of, for example, about 10 fM to about 500 nM, about 30 fM to about 500 nM, about 30 fM to about 400 nM, about 30 fM to about 300 nM, about 30 fM to about 200 nM, about 30 fM to about 100 nM, about 30 fM to about 90 nM, about 30 fM to about 80 nM, about 30 fM to about 70 nM, about 30 fM to about 60 nM, about 30 fM to about 50 nM, about 30 fM to about 40 nM, about 30 fM to about 30 nM, about 30 fM to about 20 nM, about 30 fM to about 10 nM, about 30 fM to about 9 nM, about 30 fM to about 8 nM, about 30 fM to about 7 nM, about 30 fM to about 6 nM, about 30 fM to about 5 nM, about 30 fM to about 4 nM, about 30 fM to about 3 nM, about 30 fM to about 2 nM, about 30 fM to about 1 nM, about 30 fM to about 900 pM, about 30 fM to about 800 pM, about 30 fM to about 700 pM, about 30 fM to about 600 pM, about 30 fM to about 500 pM, about 30 fM to about 400 pM, about 30 fM to about 300 pM, about 30 fM to about 200 pM, about 30 fM to about 100 pM, about 30 fM to about 90 pM, about 30 fM to about 80 pM, about 30 fM to about 70 pM, about 30 fM to about 60 pM, about 30 fM to about 50 pM, about 30 fM to about 40 pM, about 30 fM to about 30 pM, about 30 fM to about 20 pM, about 30 fM to about 10 pM, about 30 fM to about 1 pM, about 30 fM to about 900 fM, about 30 fM to about 800 fM, about 30 fM to about 700 fM, about 30 fM to about 600 fM, about 30 fM to about 500 fM, about 30 fM to about 400 fM, about 30 fM to about 300 fM, about 30 fM to about 200 fM, about 30 fM to about 100 fM, about 30 fM to about 500 nM, about 30 fM to about 400 nM, about 30 fM to about 300 nM, about 30 fM to about 200 nM, about 30 fM to about 100 nM, about 30 fM to about 90 nM, about 30 fM to about 80 nM, about 30 fM to about 70 nM, about 30 fM to about 60 nM, about 30 fM to about 50 nM, about 30 fM to about 40 nM, about 30 fM to about 30 nM, about 30 fM to about 20 nM, about 30 fM to about 10 nM, about 30 fM to about 9 nM, about 30 fM to about 8 nM, about 30 fM to about 7 nM, about 30 fM to about 6 nM, about 30 fM to about 5 nM, about 30 fM to about 4 nM, about 30 fM to about 3 nM, about 30 fM to about 2 nM, about 30 fM to about 1 nM, about 30 fM to about 900 pM, about 30 fM to about 800 pM, about 30 fM to about 700 pM, about 30 fM to about 600 pM, about 30 fM to about 500 pM, about 30 fM to about 400 pM, about 30 fM to about 300 pM, about 30 fM to about 200 pM, about 30 fM to about 100 pM, about 30 fM to about 90 pM, about 30 fM to about 80 pM, about 30 fM to about 70 pM, about 30 fM to about 60 pM, about 30 fM to about 50 pM, about 30 fM to about 40 pM, about 30 fM to about 30 pM, about 30 fM to

or derivatives thereof, or other compound of the disclosure can bind VEGF with a K.sub.D of about 30 fM to about 1 pM. In some embodiments, a compound, antibody, fragment or derivatives thereof, or other compound of the disclosure can bind VEGF with a K.sub.D of about 30 fM to about 200 fM. In some embodiments, a compound, antibody, fragment or derivatives thereof, or other compound of the disclosure can bind VEGF with a K.sub.D of about 1 pM to about 900 pM. In some embodiments, a compound, antibody, fragment or derivatives thereof, or other compound of the disclosure can bind VEGF with a K.sub.D of about 1 pM to about 600 pM. In some embodiments, a compound, antibody, fragment or derivatives thereof, or other compound of the disclosure can bind VEGF with a K.sub.D of about 1 pM to about 200 pM. In some embodiments, a compound, antibody, fragment or derivatives thereof, or other compound of the disclosure can bind VEGF with a K.sub.D of about 1 pM to about 30 pM. In some embodiments, a compound, antibody, fragment or derivatives thereof, or other compound of the disclosure can bind VEGF with a K.sub.D of about 1 pM to about 40 pM.

(77) In some embodiments, a compound, antibody, fragment or derivatives thereof, or other compound of the disclosure can bind VEGF with a K.sub.D of about 30 fM to about 2 pM. In some embodiments, a compound, antibody, fragment or derivatives thereof, or other compound of the disclosure can bind VEGF with a K.sub.D of about 35 fM to about 200 fM. In some embodiments, a compound, antibody, fragment or derivatives thereof, or other compound of the disclosure can bind VEGF with a K.sub.D of about 100 fM to about 2 pM.

(78) In some embodiments, a compound, antibody, fragment or derivatives thereof, or other compound of the disclosure can bind VEGF with a K.sub.D of about 20 pM to about 1 nM. In some embodiments, a compound, antibody, fragment or derivatives thereof, or other compound of the disclosure can bind VEGF with a K.sub.D of about 20 pM to about 800 pM. In some embodiments, a compound, antibody, fragment or derivatives thereof, or other compound of the disclosure can bind VEGF with a K.sub.D of about 20 pM to about 350 pM.

(79) In some embodiments, a compound, antibody, fragment or derivatives thereof, or other compound of the disclosure can bind VEGF with a K.sub.D of about 30 fM to about 700 pM.

(80) In some embodiments, a compound, antibody, fragment or derivatives thereof, or other compound of the disclosure can bind VEGF with a K.sub.D of about 40 fM to about 520 pM. In some embodiments, a compound, antibody, fragment or derivatives thereof, or other compound of the disclosure can bind VEGF with a K.sub.D of about 40 fM to about 110 pM.

(81) Polyvalent, Multispecific Compounds

(82) Antigen-binding compounds can be combined to generate polyvalent and/or multispecific compounds. Such polyvalent and/or multispecific compounds can have advantages over the parent compounds administered individually. These advantages can include, for example, a simpler dosing regimen, longer half-life within a subject, and the ability to bind target antigens in close proximity.

(83) Multispecific antibodies can be produced by a number of methods. In one method, monospecific antibodies or derivatives thereof can be chemically coupled, for example, via chemical coupling of two IgG antibody units into a conjugate.

(84) In another method, cloning techniques can be used to append additional antigen binding domain(s) to a conventional IgG antibody or derivative thereof. An additional antigen-binding domain can be, for example, a single variable domain (sVD), a single-chain variable fragment (scFv), a single-chain Fab, a peptide, an ankyrin protein, an ankyrin repeat protein, a designed ankyrin repeat protein (DARPin), an affibody, an avimer, an adnectin, an anticalin, a Fynomer, a Kunitz domain, a knottin, a β -hairpin mimetic, a tetrameric polyethylene oxide clustered peptide, a peptide derived from one or more receptors (e.g. VEGF receptors, or the VEGF-binding portions of human VEGF receptors 1 and 2), or a derivative thereof.

(85) The additional antigen-binding domain (e.g., sVD, scFv, single-chain Fab, peptide, ankyrin protein, ankyrin repeat protein, DARPin, affibody, avimer, adnectin, anticalin, Fynomer, Kunitz domain, knottin, β -hairpin mimetic, tetrameric polyethylene oxide clustered peptide, or peptide

derived from one or more receptors) can be appended to the N or C-terminus of the light chain and/or heavy chain of the IgG or Fab, for example, via a peptide linker. In some embodiments, a scFv can be appended to the C-termini of the heavy chains of an IgG to provide a tetravalent bispecific antibody. A scFv can be appended to the C-termini of the light chains of an IgG to provide a tetravalent bispecific antibody. A scFv can be appended to the C-termini of the light chains and the C-termini of the heavy chains of an IgG, to provide a hexavalent bispecific antibody. In some embodiments, a DARPin can be appended to the C-termini of the heavy chains of an IgG to provide a tetravalent bispecific antibody. A DARPin can be appended to the C-termini of the light chains of an IgG to provide a tetravalent bispecific antibody. A DARPin can be appended to the C-termini of the light chains and the C-termini of the heavy chains of an IgG, to provide a hexavalent bispecific antibody. Additional examples of multispecific antibodies produced by cloning techniques include: (i) DVD-IgTM (dual variable domain immunoglobulin, tandem linkage of the second V.sub.H and V.sub.L to the N-termini of HC and LC, respectively), (ii) Tandemab (tandem linkage of 2 V.sub.H-C.sub.H1 in combination of common LC), (iii) DNL (natural association of 2 antibodies or antibody fragments anchored with DDD (dimerization and docking domain) from PKA (protein kinase A) and AD (anchoring domain) from A-kinase anchor protein (AKAP), respectively), (iv) LUZ-Y (leucine zipper tethered at the C-termini of HC and later proteolytically removed), (v) 2-in-1-IgG (same LC and HC capable of dual recognition), and (vi) mAb.sup.2 (engineered loops in C.sub.H3 domain of IgG to obtain second specificity).

(86) Another class of multispecific antibodies can be characterized by structures with variable domains or scFvs as the building blocks. Non-limiting examples of such multispecific antibodies include two V.sub.H domains joined in tandem, diabodies (heterodimers containing 2 polypeptide chains encoding V.sub.LA-V.sub.HB and V.sub.HA-V.sub.LB in the order of V.sub.H-V.sub.L or V.sub.L-V.sub.H with a linker of 5 amino acids), dsDBs (interchain disulfide bond between V.sub.L and V.sub.H of the same antibody), DARTs (dual-affinity re-targeting, interchain disulfide bond between 2 V.sub.L), scDBs (single chain Diabody), tandAbs (Diabody dimer via flexible linkers in between), and 2 scFvs connected in tandem by an adjustable linker.

(87) Another class of multispecific antibodies can contain different antigen binding fragments, while retaining the basic IgG structure. Such antibodies can comprise, for example, two distinct heavy chains and/or two distinct light chains. Various techniques can be used to promote pairing of desirable light and heavy chain combinations, rather than random chain associations. Non-limiting examples of such techniques include use of a common light chain, orthogonal Fab interface (complementary mutations introduced at LC and HC interface in one Fab and no change to the other Fab), CrossMab (wherein one Fab V.sub.H or C.sub.H1 domain(s) can be switched with the partner V.sub.L or C.sub.L domain(s), with the other Fab untouched), and replacing the Fab with a single chain antigen-binding domain. Further examples include engineering strategies that can introduce mutations into the C.sub.H3 domains to promote heterodimerization based on steric or electrostatic complementarity. The “knobs in holes” approach can involve creating a “knob” by replacing threonine at position 366 with a bulky tryptophan residue on one heavy chain, and making a corresponding “hole” by triple mutations (T366S, L368A and Y407V) on the partner heavy chain. Another approach can involve creating alternating human IgG and IgA fragments in C.sub.H3 to provide the so-called SEEDbody (Strand-Exchange Engineered Domain) to guide heavy chain heterodimerization.

(88) Non-limiting schematics of multispecific antibodies are provided in FIGS. 2-15.

(89) FIG. 2 provides a schematic of a tetravalent, bispecific antibody with sequences appended to the heavy chain C-termini. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B”. Linker sequences are represented by “SEQ C”. Appended antigen binding domain sequences are represented by “SEQ D”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(90) FIG. 3 provides a schematic of a tetravalent, bispecific antibody with sequences appended to the light chain C-termini. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B”. Linker sequences are represented by “SEQ C”. Appended antigen binding domain sequences are represented by “SEQ D”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(91) FIG. 4 provides a schematic of a hexavalent, bispecific antibody with sequences appended to the heavy chain and light chain C-termini. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B”. Linker sequences are represented by “SEQ C” and “SEQ E”. Appended antigen binding domain sequences are represented by “SEQ D”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(92) FIG. 5 provides a schematic of a hexavalent, trispecific antibody with sequences appended to the heavy chain and light chain C-termini. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B”. Linker sequences are represented by “SEQ C” and “SEQ E”. Appended antigen binding domain sequences are represented by “SEQ D” and “SEQ F”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(93) FIG. 6 provides a schematic of a bivalent, bispecific antibody with two different heavy chain sequences and two different light chain sequences. Light chain sequences are represented by “SEQ A” and “SEQ D”. Heavy chain sequences are represented by “SEQ B” and “SEQ C”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(94) FIG. 7 provides a schematic of a trivalent, trispecific antibody with two different heavy chain sequences, two different light chain sequences, and a sequence appended to the C-terminus of one light chain. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A” and “SEQ D”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B” and “SEQ C”. A linker sequence is represented by “SEQ E”. An appended antigen binding domain sequence is represented by “SEQ F”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(95) FIG. 8 provides a schematic of a trivalent, trispecific antibody with two different heavy chain sequences, two different light chain sequences, and a sequence appended to the C-terminus of one heavy chain. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A” and “SEQ D”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B” and “SEQ C”. A linker sequence is represented by “SEQ E”. An appended antigen binding domain sequence is represented by “SEQ F”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(96) FIG. 9 provides a schematic of a tetravalent antibody with two different heavy chain sequences, two different light chain sequences, and sequences appended to both light chain C-termini. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A” and “SEQ D”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B” and “SEQ C”. Linker sequences are represented by “SEQ E” and “SEQ F”. Appended antigen binding domain sequences are represented by “SEQ G” and “SEQ H”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(97) FIG. 10 provides a schematic of a tetravalent antibody with two different heavy chain sequences, two different light chain sequences, and sequences appended to both heavy chain C-termini. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A” and “SEQ D”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B” and “SEQ C”. Linker sequences are represented by “SEQ E” and “SEQ F”. Appended antigen binding domain sequences are represented by “SEQ G” and “SEQ H”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(98) FIG. 11 provides a schematic of a tetravalent antibody with two different heavy chain sequences, two different light chain sequences, and sequences appended to one heavy chain C-terminus and one light chain C-terminus in cis. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A” and “SEQ D”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B” and “SEQ C”. Linker sequences are represented by “SEQ E” and “SEQ F”. Appended antigen binding domain sequences are represented by “SEQ G” and “SEQ H”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(99) FIG. 12 provides a schematic of a tetravalent antibody with two different heavy chain sequences, two different light chain sequences, and sequences appended to one heavy chain C-terminus and one light chain C-terminus in trans. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A” and “SEQ D”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B” and “SEQ C”. Linker sequences are represented by “SEQ E” and “SEQ F”. Appended antigen binding domain sequences are represented by “SEQ G” and “SEQ H”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(100) FIG. 13 provides a schematic of a pentavalent antibody with two different heavy chain sequences, two different light chain sequences, and sequences appended to both heavy chain C-termini and one light chain C-terminus. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A” and “SEQ D”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B” and “SEQ C”. Linker sequences are represented by “SEQ E”, “SEQ F”, and “SEQ G”. Appended antigen binding domain sequences are represented by “SEQ H”, “SEQ I”, and “SEQ J”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(101) FIG. 14 provides a schematic of a pentavalent antibody with two different heavy chain sequences, two different light chain sequences, and sequences appended to one heavy chain C-terminus and both light chain C-termini. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A” and “SEQ D”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B” and “SEQ C”. Linker sequences are represented by “SEQ E”, “SEQ F”, and “SEQ G”. Appended antigen binding domain sequences are represented by “SEQ H”, “SEQ I”, and “SEQ J”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(102) FIG. 15 provides a schematic of a hexavalent antibody with two different heavy chain sequences, two different light chain sequences, and sequences appended to the heavy chain and light chain C-termini. Light chain sequences of the IgG isotype antibody unit are represented by “SEQ A” and “SEQ D”. Heavy chain sequences of the IgG isotype antibody unit are represented by “SEQ B” and “SEQ C”. Linker sequences are represented by “SEQ E”, “SEQ F”, “SEQ G”, and “SEQ H”. Appended antigen binding domain sequences are represented by “SEQ I”, “SEQ J”, “SEQ K”, and “SEQ L”. N and C denote N- and C-termini, respectively. —S—S— denotes disulfide bonds.

(103) Antigen-binding compounds specific for, for example, HPTP- β (VE-PTP), can be combined with antigen-binding compounds specific for, for example, a RTK agonist, to provide a polyvalent multispecific compound that can bind both HPTP- β (VE-PTP) and the RTK agonist. For example, an antigen binding compound specific for HPTP- β (VE-PTP) can be combined with an antigen-binding compound specific for VEGF, Ang1, Ang2, BDNF, EGF, FGF, HGF, IGF, insulin, MSP, NGF, NT-3, PDGF, or any combination thereof, to provide a polyvalent multispecific compound that can bind both HPTP- β (VE-PTP) and a RTK agonist.

(104) Antigen-binding compounds specific for, for example, HPTP- β (VE-PTP), can be combined with antigen-binding compounds specific for, for example, VEGF, to provide a polyvalent multispecific compound that can bind both HPTP- β (VE-PTP) and VEGF. Compounds that bind both HPTP- β (VE-PTP) and VEGF can inhibit HPTP- β (VE-PTP), activate Tie2, inhibit VEGF binding to VEGFRs, and inhibit VEGFR signaling.

(105) Sequences derived from aflibercept, a recombinant protein comprising the VEGF-binding portions of human VEGF receptors 1 and 2, can be combined with antigen-binding compounds

specific for HPTP- β (VE-PTP). For example, sequences derived from aflibercept can be fused to the C-termini of the heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 2). Sequences derived from aflibercept can be fused to the C-termini of the light chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 3). Sequences derived from aflibercept can be fused to the C-termini of the light chains and heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a hexavalent, bispecific antibody (FIG. 4).

(106) Sequences derived from brolucizumab, a humanized single-chain antibody fragment (scFv) inhibitor of VEGF that binds to the receptor binding site of VEGF, can be combined with antigen-binding compounds specific for HPTP- β (VE-PTP). For example, sequences from brolucizumab can be fused to the C-termini of the heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 2). Sequences derived from brolucizumab can be fused to the C-termini of the light chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 3). Sequences derived from brolucizumab can be fused to the C-termini of the light chains and heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a hexavalent, bispecific antibody (FIG. 4).

(107) Sequences derived from ranibizumab, a humanized monoclonal antibody fragment (Fab) that binds to and inhibits the activity of VEGF, can be combined with antigen-binding compounds specific for HPTP- β (VE-PTP). For example, the sequences from ranibizumab can be cloned into an scFv, and the ranibizumab-derived scFv can be fused to the C-termini of the heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 2). The ranibizumab-derived scFv can be fused to the C-termini of the light chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 3). The ranibizumab-derived scFv can be fused to the C-termini of the light chains and heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a hexavalent, bispecific antibody (FIG. 4).

(108) Sequences derived from bevacizumab, a humanized monoclonal antibody that binds to and inhibits activity of VEGF, can be combined with antigen-binding compounds specific for HPTP- β (VE-PTP). For example, the sequences from bevacizumab can be cloned into an scFv, and the bevacizumab-derived scFv can be fused to the C-termini of the heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 2). The bevacizumab-derived scFv can be fused to the C-termini of the light chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 3). The bevacizumab-derived scFv can be fused to the C-termini of the light chains and heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a hexavalent, bispecific antibody (FIG. 4).

(109) Sequences derived from conbercept, a recombinant protein comprising the VEGF-binding portions of VEGF receptors 1 and 2, can be combined with antigen-binding compounds specific for HPTP- β (VE-PTP). For example, sequences derived from conbercept can be fused to the C-termini of the heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 2). Sequences derived from conbercept can be fused to the C-termini of the light chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 3). Sequences derived from conbercept can be fused to the C-termini of the light chains and heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a hexavalent, bispecific antibody (FIG. 4).

(110) Sequences derived from abicipar, a VEGF-binding DARPIn, can be combined with antigen-binding compounds specific for HPTP- β (VE-PTP). For example, sequences derived from abicipar can be fused to the C-termini of the heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 2). Sequences derived from abicipar can be fused to the C-termini of the light chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 3). Sequences derived from abicipar can be fused to the C-termini of the light chains and heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a hexavalent, bispecific antibody (FIG. 4).

bispecific antibody (FIG. 4).

(111) Antigen-binding compounds specific for HPTP- β (VE-PTP) can be combined with DARPins or amino acid sequences therefrom, for example, amino acid sequences comprising any one of SEQ ID NOS: 158-217. Amino acid sequences comprising any one of SEQ ID NOS: 158-217 can be fused to the C-termini of the heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 2). Amino acid sequences comprising any one of SEQ ID NOS: 158-217 can be fused to the C-termini of the light chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 3). Amino acid sequences comprising any one of SEQ ID NOS: 158-217 can be fused to the C-termini of the light chains and heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a hexavalent, bispecific antibody (FIG. 4).

(112) Multiple VEGF-specific compounds can be combined with an antigen-binding compound specific for HPTP- β (VE-PTP). For example, one binding domain, (e.g. aflibercept-derived sequences) can be fused to the C-termini of the heavy chains of an anti-HPTP- β (VE-PTP) antibody, and another binding domain (e.g. brolucizumab-derived sequences) be fused to the C-termini of the light chains of the HPTP- β (VE-PTP)-specific antibody, to provide a hexavalent, trispecific antibody (FIG. 5).

(113) If different heavy chains are used in the basic four chain IgG antibody unit (e.g., using the “knobs in holes” approach), antibodies can be generated that are bivalent, trivalent, tetravalent, pentavalent, or hexavalent, and that are monospecific, bispecific, trispecific, tetraspecific, pentaspecific, or hexaspecific.

(114) In one non-limiting example, one arm of the antibody unit can contain CDRs specific for HPTP- β (VE-PTP), while the other arm of the antibody unit can contain CDRs specific for VEGF (FIG. 6).

(115) In one non-limiting example, both antigen binding fragments of the basic antibody unit can be HPTP- β (VE-PTP)-specific, and a binding domain specific for VEGF (e.g. aflibercept-derived sequence, brolucizumab-derived sequence, ranibizumab-derived sequence, or a bevacizumab-derived sequence) can be fused to one light chain, to provide a trivalent bispecific antibody (FIG. 7).

(116) In one non-limiting example, both antigen binding fragments of the basic antibody unit can be HPTP- β (VE-PTP)-specific, and a binding domain specific for VEGF (e.g. aflibercept-derived sequence, brolucizumab-derived sequence, ranibizumab-derived sequence, or a bevacizumab-derived sequence) can be fused to one heavy chain, to provide a trivalent bispecific antibody (FIG. 8).

(117) In one non-limiting example, both antigen binding fragments of the basic antibody unit can be HPTP- β (VE-PTP)-specific, and a binding domain specific for VEGF (e.g. aflibercept-derived sequence, brolucizumab-derived sequence, ranibizumab-derived sequence, or a bevacizumab-derived sequence) can be fused to one light chain, and a second, different binding domain specific for VEGF can be fused to the other light chain, to provide a tetravalent trispecific antibody (FIG. 9).

(118) In one non-limiting example, both antigen binding fragments of the basic antibody unit can be HPTP- β (VE-PTP)-specific, a binding domain specific for VEGF (e.g. aflibercept-derived sequence, brolucizumab-derived sequence, ranibizumab-derived sequence, or a bevacizumab-derived sequence) can be fused to one heavy chain, and a second, different binding domain specific for VEGF can be fused to the other heavy chain, to provide a tetravalent trispecific antibody (FIG. 10).

(119) In one non-limiting example, both antigen binding fragments of the basic antibody unit can be HPTP- β (VE-PTP)-specific, and a binding domain specific for VEGF (e.g. aflibercept-derived sequence, brolucizumab-derived sequence, ranibizumab-derived sequence, or a bevacizumab-derived sequence) can be fused to one heavy chain, and a second, different binding domain specific

for VEGF can be fused to one light chain in cis, to provide a tetravalent trispecific antibody (FIG. 11).

(120) In one non-limiting example, both antigen binding fragments of the basic antibody unit can be HPTP- β (VE-PTP)-specific, a binding domain specific for VEGF (e.g. aflibercept-derived sequence, brolucizumab-derived sequence, ranibizumab-derived sequence, or a bevacizumab-derived sequence) can be fused to one heavy chain, and a second, different binding domain specific for VEGF can be fused to one light chain in trans, to provide a tetravalent trispecific antibody (FIG. 12).

(121) In one non-limiting example, both antigen binding fragments of the basic antibody unit can be HPTP- β (VE-PTP)-specific, a binding domain specific for VEGF (e.g. aflibercept-derived sequence, brolucizumab-derived sequence, ranibizumab-derived sequence, or a bevacizumab-derived sequence) can be fused to one heavy chain, a second, different binding domain specific for VEGF can be fused to a second heavy chain, and a third, different binding domain specific for VEGF can be fused to one light chain, to provide a pentavalent tetraspecific antibody (FIG. 13).

(122) In one non-limiting example, both antigen binding fragments of the basic antibody unit can be HPTP- β (VE-PTP)-specific, a binding domain specific for VEGF (e.g. aflibercept-derived sequence, brolucizumab-derived sequence, ranibizumab-derived sequence, or a bevacizumab-derived sequence) can be fused to one heavy chain, a second, different binding domain specific for VEGF can be fused to a one light chain, and a third, different binding domain specific for VEGF can be fused to the other light chain, to provide a pentavalent tetraspecific antibody (FIG. 14).

(123) In one non-limiting example, both antigen binding fragments of the basic antibody unit can be HPTP- β (VE-PTP)-specific, a binding domain specific for VEGF (e.g. aflibercept-derived sequence, brolucizumab-derived sequence, ranibizumab-derived sequence, or a bevacizumab-derived sequence) can be fused to one heavy chain, a second, different binding domain specific for VEGF can be fused to the other heavy chain, a third, different binding domain specific for VEGF can be fused to one light chain, and a fourth binding domain specific for VEGF can be fused to the other light chain, to provide a hexavalent pentaspecific antibody (FIG. 15).

(124) In one non-limiting example, both antigen binding fragments of the basic antibody unit can be HPTP- β (VE-PTP)-specific, one binding domain, (e.g. an aflibercept-derived sequence) can be fused to the C-terminus of one heavy chain, and second binding domain (e.g. a brolucizumab-derived sequence) be fused to the C-terminus of the other heavy chain, to provide a tetravalent, trispecific antibody (FIG. 10). In one non-limiting example, both antigen binding fragments of the basic antibody unit can be HPTP- β (VE-PTP)-specific, an aflibercept-derived sequence can be fused to the C-terminus of one light chain, and a brolucizumab-derived sequence be fused to the C-terminus of another light chain, to provide a tetravalent, trispecific antibody (FIG. 9). In one non-limiting example, both antigen binding fragments of the basic antibody unit can be HPTP- β (VE-PTP)-specific, an aflibercept-derived sequence can be fused to the C-terminus of one heavy chain, and a brolucizumab-derived sequence be fused to the C-terminus of one light chain, to provide a tetravalent, trispecific antibody (FIG. 11, FIG. 12). In one non-limiting example, both antigen binding fragments of the basic antibody unit can be HPTP- β (VE-PTP)-specific, an aflibercept-derived sequence can be fused to the C-terminus of one heavy chain, a brolucizumab-derived sequence can be fused to the C-terminus of another heavy chain, and ranibizumab-derived sequences can be fused to the C-termini of both light chains, to provide a hexavalent, tetraspecific antibody (FIG. 15). In one non-limiting example, both antigen binding fragments of the basic antibody unit can be HPTP- β (VE-PTP)-specific, an aflibercept-derived sequence can be fused to the C-terminus of one heavy chain, a brolucizumab-derived sequence can be fused to the C-terminus of another heavy chain, a ranibizumab-derived sequence can be fused to the C-terminus of one light chain, and a bevacizumab-derived sequence can be fused to the C-terminus of another light chain, to provide a hexavalent, pentaspecific antibody (FIG. 15).

(125) Antigen-binding compounds specific for, for example, HPTP- β (VE-PTP), can be combined

with other amino acid sequences to provide polyvalent multispecific compounds that, for example, enhance Tie2 activation, enhance Tie2 phosphorylation, enhance Tie2 signaling, reduce VEGFR activation, reduce VEGFR phosphorylation, reduce VEGFR signaling, or a combination thereof.

(126) Antigen-binding compounds specific for HPTP- β (VE-PTP) can be combined with collagen IV-derived biomimetic peptides, for example, amino acid sequences comprising SEQ ID NO: 152 or SEQ ID NO: 153. Amino acid sequences comprising SEQ ID NO: 152 or SEQ ID NO: 153 can be fused to the C-termini of the heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 2). Amino acid sequences comprising SEQ ID NO: 152 or SEQ ID NO: 153 can be fused to the C-termini of the light chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 3). Amino acid sequences comprising SEQ ID NO: 152 or SEQ ID NO: 153 the C-termini of the light chains and heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a hexavalent, bispecific antibody (FIG. 4).

(127) Antigen-binding compounds specific for HPTP- β (VE-PTP) can be combined with Ang1 mimetics, for example, vasculotide. Sequences derived from vasculotide can be fused to the C-termini of the heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 2). Sequences derived from vasculotide can be fused to the C-termini of the light chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 3). Sequences derived from vasculotide can be fused to the C-termini of the light chains and heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a hexavalent, bispecific antibody (FIG. 4).

(128) Antigen-binding compounds specific for, for example, HPTP- β (VE-PTP), can be combined with antigen-binding compounds specific for, for example, a RTK, to provide a polyvalent multispecific compound that can bind both HPTP- β (VE-PTP) and the RTK. For example, an antigen binding compound specific for HPTP- β (VE-PTP) can be combined with an antigen-binding compound specific for VEGFR (e.g. VEGFR2), to provide a polyvalent multispecific compound that can bind both HPTP- β (VE-PTP) and VEGFR.

(129) Compounds that bind both HPTP- β (VE-PTP) and VEGFR can inhibit HPTP- β (VE-PTP), activate Tie2, inhibit VEGF binding to VEGFR, and inhibit VEGFR signaling.

(130) Sequences derived from ramucirumab, a humanized monoclonal antibody that binds an extracellular domain of VEGFR2 and inhibits VEGFR2 signaling, can be combined with antigen-binding compounds specific for HPTP- β (VE-PTP). For example, the sequences from ramucirumab can be cloned into an scFv, and the ramucirumab-derived scFv can be fused to can be fused to the C-termini of the heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 2). The ramucirumab-derived scFv can be fused to the C-termini of the light chains of a HPTP- β (VE-PTP)-specific antibody, to provide a tetravalent, bispecific antibody (FIG. 3). The ramucirumab-derived scFv can be fused to the C-termini of the light chains and heavy chains of a HPTP- β (VE-PTP)-specific antibody, to provide a hexavalent, bispecific antibody (FIG. 4).

(131) Multiple compounds that enhance Tie2 activation, enhance Tie2 phosphorylation, enhance Tie2 signaling, reduce VEGFR activation, reduce VEGFR phosphorylation, reduce VEGFR signaling, or a combination thereof, can be combined with an antigen-binding compound specific for HPTP- β (VE-PTP). For example, one binding domain, (e.g. brotuzumab-derived sequences) can be fused to the C-termini of the heavy chains of an anti-HPTP- β (VE-PTP) antibody, and another binding domain (e.g. vasculotide-derived sequences) be fused to the C-termini of the light chains of the HPTP- β (VE-PTP)-specific antibody, to provide a hexavalent, trispecific antibody (FIG. 5).

(132) Compounds, antibodies, or derivatives thereof disclosed herein can have 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 domains or more. Each domain can modulate, bind, antagonize, inhibit or activate any targets disclosed herein, for example, a phosphatase, a phosphatase that modulates Tie2 signaling, a protein tyrosine

phosphatase, a receptor-like protein tyrosine phosphatase, a Tie2 modulator, HPTP- β (VE-PTP), an extracellular domain of HPTP- β (VE-PTP), the first FN3 repeat of an extracellular domain of HPTP- β (VE-PTP), a kinase, a tyrosine kinase, a receptor tyrosine kinase, a receptor tyrosine kinase activator, a receptor tyrosine kinase agonist, a growth factor, a growth factor receptor activator, a growth factor receptor agonist, a cysteine-knot growth factor superfamily member, a pro-angiogenic factor, a PDGF family member, a VEGF receptor, a VEGF receptor agonist, a VEGF family member, a VEGF, VEGF-A, VEGF-B, VEGF-C, VEGF-D, PGF, VEGF.sub.121, VEGF.sub.145, VEGF.sub.148, VEGF.sub.162, VEGF.sub.165, VEGF.sub.165b, VEGF.sub.183, VEGF.sub.189 or VEGF.sub.206.

(133) Inhibitors, Activators, Modulators, and Binding Agents

(134) A HPTP- β (VE-PTP) inhibitor, modulator, or binding agent of the disclosure can include a compound, a recombinant protein, an antibody, antigen-binding fragment, variant, or derivative thereof, a Tie2-peptomimetic, a tetrameric polyethylene oxide clustered peptide, a collagen IV-biomimetic peptide, a DARPin or derivative thereof, an affibody, an avimer, an adnectin, an anticalin, a Fynomer, a Kunitz domain, a knottin, a β -hairpin mimetic, or a peptide derived from one or more receptors, either alone or in combination with another amino acid sequence or multiple other amino acid sequences. The inhibitor, modulator, or binding agent can undergo modifications, for example, enzymatic cleavage or posttranslational modifications.

(135) In some embodiments, a HPTP- β (VE-PTP) inhibitor, modulator, or binding agent of the disclosure can inhibit HPTP- β (VE-PTP) by interfering with the interaction of HPTP- β (VE-PTP) and Tie2. In some embodiments, a HPTP- β (VE-PTP) inhibitor, modulator, or binding agent of the disclosure can inhibit HPTP- β (VE-PTP) by stabilizing HPTP- β (VE-PTP) in an inactive conformation. In some embodiments, a HPTP- β (VE-PTP) inhibitor, modulator, or binding agent of the disclosure can inhibit HPTP- β (VE-PTP) by promoting internalization of HPTP- β (VE-PTP) (e.g., promoting endocytosis and degradation of HPTP- β (VE-PTP)). In some embodiments, a HPTP- β (VE-PTP) inhibitor, modulator, or binding agent of the disclosure can inhibit HPTP- β (VE-PTP) by blocking binding of a ligand that activates HPTP- β (VE-PTP). In some embodiments, a HPTP- β (VE-PTP) inhibitor, modulator, or binding agent of the disclosure can inhibit HPTP- β (VE-PTP) by modulating oligomerization of HPTP- β (VE-PTP).

(136) In some embodiments, a HPTP- β (VE-PTP) inhibitor, modulator, or binding agent of the disclosure can bind to a dominant-negative isoform of HPTP- β (VE-PTP). A dominant-negative isoform can correspond to a form of HPTP- β (VE-PTP) deficient in phosphatase activity that can compete with endogenous HPTP- β (VE-PTP).

(137) A HPTP- β (VE-PTP) inhibitor, modulator, or binding agent of the disclosure can comprise a plurality of HPTP- β (VE-PTP) binding sites. In some embodiments, a HPTP- β (VE-PTP) inhibitor, modulator, or binding agent can bind to two HPTP- β (VE-PTP) molecules simultaneously, thereby bringing the two HPTP- β (VE-PTP) molecules into close proximity. In some embodiments, a HPTP- β (VE-PTP) inhibitor, modulator, or binding agent can bind to three HPTP- β (VE-PTP) molecules simultaneously, thereby bringing the three HPTP- β (VE-PTP) molecules into close proximity. In some embodiments, a HPTP- β (VE-PTP) inhibitor, modulator, or binding agent can bind to four HPTP- β (VE-PTP) molecules simultaneously, thereby bringing the four HPTP- β (VE-PTP) molecules into close proximity.

(138) A HPTP- β (VE-PTP) inhibitor, modulator, or binding agent of the disclosure can be covalently or non-covalently conjugated to another moiety or vehicle. A moiety or vehicle can, for example, provide binding specificity for an additional target, inhibit degradation, increase half-life, increase absorption, reduce toxicity, reduce immunogenicity, and/or increase biological activity of the inhibitor, modulator, or binding agent. Non-limiting examples of the moiety to which the inhibitor, modulator, or binding agent can be conjugated include a Fc domain of an immunoglobulin, a peptide, a lipid, a carbohydrate, a dendrimer, an oligosaccharide, a cholesterol group such as a steroid, and a polymer such as a polyethylene glycol (PEG), a polylysine, or a

dextran.

(139) A compound of the present disclosure can be used for targeting HPTP- β (VE-PTP) to restore Tie2 activity. A HPTP- β (VE-PTP) inhibitor, modulator, or binding agent can thus be a Tie2 activator. In some embodiments, a compound of the present disclosure can initiate or inhibit a signaling cascade downstream of HPTP- β (VE-PTP) or Tie2, for example, Akt/PI3-K signaling, Rac1 signaling, MAPK/Ras signaling, or NF- κ B signaling.

(140) Inhibition of HPTP- β (VE-PTP) can lead to vascular stabilization, which can be beneficial for the treatment of, for example, disorders that are characterized by vascular instability, angiogenesis, neovascularization, vascular leakage, and/or edema. For example, inhibition of HPTP- β (VE-PTP) can be beneficial for the treatment of vascular disorders, ocular disorders, cancers, renal disorders, complications of diabetes, and other disorders. In some embodiments, a HPTP- β (VE-PTP) inhibitor, modulator, or binding agent of the disclosure can be used to treat, for example, diabetic retinopathy, non-proliferative diabetic retinopathy (NPDR), glaucoma, intraocular pressure, ocular edema, ocular hemorrhage, ocular hypertension, ocular inflammation, ocular neovascularization, ocular vascular leak, retinal perfusion, or retinopathy.

(141) A Tie2 activator, modulator, or binding agent of the disclosure can include, for example, a compound, a recombinant protein, a peptide, an antibody, an antigen-binding fragment, variant, or derivative thereof, an angiopoietin 1 recombinant protein, an Ang1 mimetic, a Tie2 agonist, a HPTP- β (VE-PTP) phosphatase inhibitor, a Tie2-peptomimetic, a tetrameric polyethylene oxide clustered peptide, a collagen IV-biomimetic peptide, a DARPIn or derivative thereof, an affinibody, an avimer, an adnectin, an anticalin, a Fynomer, a Kunitz domain, a knottin, a β -hairpin mimetic, or a peptide derived from one or more receptors, either alone or in combination with another amino acid sequence or multiple other amino acid sequences. The activator, modulator, or binding agent can undergo modifications, for example, enzymatic cleavage or posttranslational modifications.

(142) A Tie2 activator, modulator, or binding agent of the disclosure can be covalently or non-covalently conjugated to another moiety or vehicle. A moiety or vehicle can, for example, provide binding specificity for an additional target, inhibit degradation, increase half-life, increase absorption, reduce toxicity, reduce immunogenicity, and/or increase biological activity of the activator, modulator, or binding agent. Non-limiting examples of the moiety to which the activator, modulator, or binding agent can be conjugated include a Fc domain of an immunoglobulin, a peptide, a lipid, a carbohydrate, a dendrimer, an oligosaccharide, a cholesterol group such as a steroid, and a polymer such as a polyethylene glycol (PEG), a polylysine, or a dextran.

(143) In some embodiments, a compound of the present disclosure can initiate or inhibit a signaling cascade downstream of Tie2, for example, Akt/PI3-K signaling, Rac1 signaling, MAPK/Ras signaling, or NF- κ B signaling.

(144) The activation of Tie2 can lead to vascular stabilization, which can be beneficial for the treatment of, for example, disorders that are characterized by vascular instability, angiogenesis, neovascularization, vascular leakage, and/or edema. For example, activation of Tie2 can be beneficial for the treatment of ocular disorders, cancers, renal disorders, complications of diabetes, and other disorders. In some embodiments, a Tie2 activator, modulator, or binding agent of the disclosure can be used to treat diabetic retinopathy, non-proliferative diabetic retinopathy (NPDR), glaucoma, intraocular pressure, ocular edema, ocular hemorrhage, ocular hypertension, ocular inflammation, ocular neovascularization, ocular vascular leak, retinal perfusion, or retinopathy.

(145) A VEGF inhibitor, modulator, or binding agent of the disclosure can include a compound, a recombinant protein, a peptide, an antibody, antigen-binding fragment, variant, or derivative thereof, a DARPIn or derivative thereof, an affinibody, an avimer, an adnectin, an anticalin, a Fynomer, a Kunitz domain, a knottin, a β -hairpin mimetic, a tetrameric polyethylene oxide clustered peptide, a collagen IV-biomimetic peptide, or a peptide derived from one or more receptors (e.g. VEGF receptors, or the VEGF-binding portions of human VEGF receptors 1 and 2), either alone or in combination with another amino acid sequence or multiple other amino acid

sequences. The inhibitor, modulator, or binding agent can undergo modifications, for example, enzymatic cleavage or posttranslational modifications.

(146) A VEGF inhibitor, modulator, or binding agent of the disclosure can comprise a plurality of VEGF binding sites. In some embodiments, a VEGF inhibitor, modulator, or binding agent can bind to two VEGF molecules simultaneously, thereby bringing the two VEGF molecules into close proximity. In some embodiments, a VEGF inhibitor, modulator, or binding agent can bind to three VEGF molecules simultaneously, thereby bringing the three VEGF molecules into close proximity. In some embodiments, a VEGF inhibitor, modulator, or binding agent can bind to four VEGF molecules simultaneously, thereby bringing the four VEGF molecules into close proximity.

(147) A VEGF inhibitor, modulator, or binding agent of the disclosure can be covalently or non-covalently conjugated to another moiety or vehicle. A moiety or vehicle can, for example, provide binding specificity for an additional target, inhibit degradation, increase half-life, increase absorption, reduce toxicity, reduce immunogenicity, and/or increase biological activity of the inhibitor, modulator, or binding agent. Non-limiting examples of the moiety to which the inhibitor, modulator, or binding agent can be conjugated include a Fc domain of an immunoglobulin, a peptide, a lipid, a carbohydrate, a dendrimer, an oligosaccharide, a cholesterol group such as a steroid, and a polymer such as a polyethylene glycol (PEG), a polylysine, or a dextran.

(148) A VEGFR inhibitor, modulator, or binding agent of the disclosure can include a compound, a recombinant protein, an antibody, an antigen-binding fragment, variant, or derivative thereof, a tetrameric polyethylene oxide clustered peptide, a collagen IV-biomimetic peptide, a DARPin or derivative thereof, an affibody, an avimer, an adnectin, an anticalin, a Fynomer, a Kunitz domain, a knottin, a β -hairpin mimetic, or a peptide derived from one or more receptors, either alone or in combination with another amino acid sequence or multiple other amino acid sequences. The inhibitor, modulator, or binding agent can undergo modifications, for example, enzymatic cleavage or posttranslational modifications.

(149) A VEGFR inhibitor, modulator, or binding agent of the disclosure can comprise a plurality of VEGFR binding sites. In some embodiments, a VEGFR inhibitor, modulator, or binding agent can bind to two VEGFR molecules simultaneously, thereby bringing the two VEGFR molecules into close proximity. In some embodiments, a VEGFR inhibitor, modulator, or binding agent can bind to three VEGFR molecules simultaneously, thereby bringing the three VEGFR molecules into close proximity. In some embodiments, a VEGFR inhibitor, modulator, or binding agent can bind to four VEGFR molecules simultaneously, thereby bringing the four VEGFR molecules into close proximity.

(150) A VEGFR inhibitor, modulator, or binding agent of the disclosure can be covalently or non-covalently conjugated to another moiety or vehicle. A moiety or vehicle can, for example, provide binding specificity for an additional target, inhibit degradation, increase half-life, increase absorption, reduce toxicity, reduce immunogenicity, and/or increase biological activity of the inhibitor, modulator, or binding agent. Non-limiting examples of the moiety to which the inhibitor, modulator, or binding agent can be conjugated include a Fc domain of an immunoglobulin, a peptide, a lipid, a carbohydrate, a dendrimer, an oligosaccharide, a cholesterol group such as a steroid, and a polymer such as a polyethylene glycol (PEG), a polylysine, or a dextran.

(151) In some embodiments, a VEGFR inhibitor, modulator, or binding agent of the disclosure can inhibit VEGFR by stabilizing VEGFR in an inactive conformation. In some embodiments, a VEGFR inhibitor, modulator, or binding agent of the disclosure can inhibit VEGFR by promoting internalization of VEGFR (e.g., promoting endocytosis and degradation of VEGFR). In some embodiments, a VEGFR inhibitor, modulator, or binding agent of the disclosure can inhibit VEGFR by blocking binding of a ligand that activates VEGFR (e.g., blocking VEGF ligation of VEGFR). In some embodiments, a VEGFR inhibitor, modulator, or binding agent of the disclosure can inhibit VEGFR by modulating oligomerization of VEGFR (e.g., preventing or reducing the likelihood of dimerization or oligomerization of VEGFR).

(152) A compound of the present disclosure can be used for interfering with the interaction of VEGF and VEGFR, thereby reducing VEGFR phosphorylation and downstream signaling. In some embodiments, inhibition of VEGF can reduce aberrant vasculogenesis, angiogenesis, or blood vessel permeabilization, thereby reducing pathologic vascular instability. In some embodiments, inhibition of VEGF can be beneficial for the treatment of disorders that are characterized by vascular instability, angiogenesis, neovascularization, vascular leakage, and/or edema. For example, inhibition of VEGF can be beneficial for the treatment of vascular disorders, ocular disorders, cancers, renal disorders, complications of diabetes, and other disorders. In some embodiments, a VEGF inhibitor, modulator, or binding agent of the disclosure can be used to treat diabetic retinopathy, non-proliferative diabetic retinopathy (NPDR), glaucoma, intraocular pressure, ocular edema, ocular hemorrhage, ocular hypertension, ocular inflammation, ocular neovascularization, ocular vascular leak, retinal perfusion, or retinopathy. In some embodiments, inhibition of VEGF can reduce cancer.

(153) Methods

(154) The promotion of Tie2-signaling and inhibition of VEGFR signaling can lead to vascular stabilization, which can be beneficial for the treatment of a condition with a vascular component. A compound disclosed herein can be used to treat, for example, a disease characterized by changes in the vasculature, a disease characterized by decreased Tie2 activation, or a disease characterized by involvement of VEGF in pathogenesis, whether progressive or non-progressive, acute or chronic.

(155) In some embodiments, a compound disclosed herein can be used to treat an ocular disorder. A compound disclosed herein can be used to treat, for example, age-related macular degeneration (dry form), age-related macular degeneration (wet form), atopic keratitis, Bests disease, blepharitis, blurry vision, choroidal neovascularization, chronic retinal detachment, chronic uveitis/vitritis, choroiditis, conjunctivitis, contact lens overwear, corneal graft neovascularization, corneal graft rejection, corneal neovascularization, cystoid macular edema, diabetic macular edema, double vision, diabetic retinopathy, diseases associated with rubeosis (neovascularization of the angle), diseases caused by abnormal proliferation of fibrovascular or fibrous tissue including all forms of proliferative vitreoretinopathy, Eales disease, elevated intraocular pressure, epidemic keratoconjunctivitis, floaters, glaucoma, hard yellow exudates within 500 μm of the center of the fovea with adjacent retinal thickening, hyperviscosity syndromes, infections causing choroiditis, infections causing retinitis, iris neovascularization, ischemic retinopathy, loss of contrast, macular telangiectasia, marginal keratolysis, multifocal choroiditis, myopia, neovascular glaucoma, non-proliferative diabetic retinopathy (NPDR), ocular edema, ocular hemorrhage, ocular histoplasmosis, ocular hypertension, ocular inflammation, ocular ischemia, ocular neovascularization, ocular trauma, ocular vascular leak, optic pits, papilloedema, pars planitis, phlyctenulosis, polypoidal choroidal vasculopathy, post-laser complications, proliferative diabetic retinopathy, pterygium keratitis sicca, radial keratotomy, retinal angiomatous proliferation, retinal degeneration, retinal edema (including macular edema), retinal neovascularization, retinal perfusion, retinal thickening within 1 disc diameter of the center of the fovea, retinal thickening within 500 μm of the center of the fovea, retinal vein occlusion (central or branch), retinitis, retinitis pigmentosa, retinopathy, retinopathy of prematurity, scleritis, Stargarts disease, superior limbic keratitis, surgery induced edema, surgery induced neovascularization, Terrien's marginal degeneration, trachoma, trauma, uveitis, vasculitis (e.g. central retinal vein occlusion), or other ophthalmic diseases wherein the eye disease or disorder is associated with ocular neovascularization, vascular leakage, or retinal edema, or a combination thereof.

(156) In some embodiments, a compound disclosed herein can be used to treat a complication of diabetes (e.g., a comorbidity of diabetes). A compound disclosed herein can be used to treat, for example, Acute glomerulonephritis, Acute myocardial infarction, Amputation, Amyotrophy, Aneurysm, Angina pectoris, Aortic aneurysm, Aortic dissection, Atherosclerosis, Atherosclerotic cardiovascular disease, Atrial fibrillation, Autonomic neuropathy, Blindness, Cardiovascular

complications of diabetes, Cerebrovascular complications of diabetes, Charcot's arthropathy, Chronic glomerulonephritis, Chronic renal failure, Claudication, Clinically significant macular edema, Coronary artery disease, Cranial nerve palsy, Cystoid macular degeneration, Cystoid macular edema, Diabetic cardiomyopathy, Diabetic cheiroarthropathy, Diabetic coma, Diabetic encephalopathy, Diabetic foot wound, Diabetic hyperglycemia, Diabetic hyperlipidemia, Diabetic hyperosmolar syndrome, Diabetic hypoglycemia, Diabetic ketoacidosis, Diabetic myonecrosis, Diabetic nephropathy, Diabetic neuropathy, Diabetic ophthalmologic disease, Diabetic peripheral vascular disease, Diabetic retinopathy, Diffuse idiopathic skeletal hyperostosis, Dupuytren's contracture, Embolism, End-stage renal disease, Erectile dysfunction, Forestier disease, Gangrene, Gas gangrene, Gastroparesis/diarrhea, Heart failure, Hyperglycemic crisis, Hypertension, Ischemic heart disease, Ketoacidosis, Lipohypertrophy, Metabolic complications of diabetes, Mononeuropathy, Myocardial infarction, Nephritis, Nephropathy, Nephrosis, Nephrotic syndrome, Neurogenic bladder, Neuropathic arthropathy, Neuropathy, Orthostatic hypotension, Osteoarthritis, Osteoporosis, Periodontal disease, Peripheral vascular disease, Polyneuropathy, Proliferative retinopathy, Renal failure, Renal insufficiency, Restrictive lung disease, Retinal detachment, Retinal edema, Retinopathy, Stroke, Thrombosis, Transient ischemic attack, Ulceration, Ventricular fibrillation, Vitreous hemorrhage, or a combination thereof.

(157) In some embodiments, a compound disclosed herein can be used to treat a renal disorder. A compound disclosed herein can be used to treat, for example, Acute kidney injury, Acute proliferative glomerulonephritis, Adenine phosphoribosyltransferase deficiency, Alport syndrome, Analgesic nephropathy, Autosomal dominant polycystic kidney disease, Autosomal recessive polycystic kidney disease, Balkan endemic nephropathy, Benign nephrosclerosis, Bright's disease, Cardiorenal syndrome, CFHR5 nephropathy, Chronic kidney disease, Chronic kidney disease-mineral and bone disorder, Congenital nephrotic syndrome, Conorenal syndrome, Contrast-induced nephropathy, Cystic kidney disease, Dents disease, Diabetic nephropathy, Diffuse proliferative nephritis, Distal renal tubular acidosis, Diuresis, EAST syndrome, End Stage Renal Disease, Epithelial-mesenchymal transition, Epstein syndrome, Fanconi syndrome, Fechtner syndrome, Focal proliferative nephritis, Focal segmental glomerulosclerosis, Fraley syndrome, Galloway Mowat syndrome, Gitelman syndrome, Glomerulocystic kidney disease, Glomerulopathy, Goodpasture syndrome, High anion gap metabolic acidosis, HIV-associated nephropathy, Horseshoe kidney, Hydronephrosis, Hypertensive kidney disease, IgA nephropathy, Interstitial nephritis, Juvenile nephronophthisis, Kidney cancer, Kidney disease, Kidney stone disease, Lightwood-Albright syndrome, Lupus nephritis, Malarial nephropathy, Medullary cystic kidney disease, Medullary sponge kidney, Membranous glomerulonephritis, Mesoamerican nephropathy, Milk-alkali syndrome, Minimal mesangial glomerulonephritis, Multicystic dysplastic kidney, Nephritis, Nephrocalcinosis, Nephrogenic diabetes insipidus, Nephromegaly, Nephroptosis, Nephrosis, Nephrotic syndrome, Nutcracker syndrome, Papillorenal syndrome, Phosphate nephropathy, Polycystic kidney disease, Primary hyperoxaluria, Proximal renal tubular acidosis, Pyelonephritis, Pyonephrosis, Rapidly progressive glomerulonephritis, Renal agenesis, Renal angina, Renal artery stenosis, Renal cyst, Renal ischemia, Renal osteodystrophy, Renal papillary necrosis, Renal tubular acidosis, Renal vein thrombosis, Secondary hypertension, Serpentine fibula-polycystic kidney syndrome, Shunt nephritis, Sick cell nephropathy, Thin basement membrane disease, Transplant glomerulopathy, Tubulointerstitial nephritis and uveitis, Tubulopathy, Uremia, Uremic frost, Wunderlich syndrome, or a combination thereof.

(158) In some embodiments, a compound disclosed herein can be used to treat a cancer. A compound disclosed herein can be used to treat, for example, acute leukemia, astrocytomas, biliary cancer (cholangiocarcinoma), bone cancer, breast cancer, brain stem glioma, bronchioloalveolar cell lung cancer, cancer of the adrenal gland, cancer of the anal region, cancer of the bladder, cancer of the endocrine system, cancer of the esophagus, cancer of the head or neck, cancer of the kidney, cancer of the parathyroid gland, cancer of the penis, cancer of the pleural/peritoneal

membranes, cancer of the salivary gland, cancer of the small intestine, cancer of the thyroid gland, cancer of the ureter, cancer of the urethra, carcinoma of the cervix, carcinoma of the endometrium, carcinoma of the fallopian tubes, carcinoma of the renal pelvis, carcinoma of the vagina, carcinoma of the vulva, cervical cancer, chronic leukemia, colon cancer, colorectal cancer, cutaneous melanoma, ependymoma, epidermoid tumors, Ewings sarcoma, gastric cancer, glioblastoma, glioblastoma multiforme, glioma, hematologic malignancies, hepatocellular (liver) carcinoma, hepatoma, Hodgkin's Disease, intraocular melanoma, Kaposi sarcoma, lung cancer, lymphomas, medulloblastoma, melanoma, meningioma, mesothelioma, multiple myeloma, muscle cancer, neoplasms of the central nervous system (CNS), neuronal cancer, non-small cell lung cancer, osteosarcoma, ovarian cancer, pancreatic cancer, pediatric malignancies, pituitary adenoma, prostate cancer, rectal cancer, renal cell carcinoma, sarcoma of soft tissue, schwannoma, skin cancer, spinal axis tumors, squamous cell carcinomas, stomach cancer, synovial sarcoma, testicular cancer, uterine cancer, or tumors and their metastases, including refractory versions of any of the above cancers, or a combination thereof.

(159) In some embodiments, a compound disclosed herein can be used to treat a disease characterized by changes in the vasculature, a disease characterized by decreased Tie2 activation, or a disease characterized by involvement of VEGF in pathogenesis. A compound disclosed herein can be used to treat, for example, acne rosacea, acute lung injury, acute respiratory distress syndrome (ARDS), adhesion formation from abdominal surgery, adipositas, albuminuria, allergic edema, allergy, angina, angiofibroma, arteriosclerosis, artery occlusion, ascites, atheroma, atherosclerosis, asthma, avascular necrosis, bacterial ulcers, *Bartonella bacilliformis* infection, Behcet's disease, Buerger's disease (thromboangiitis obliterans), cardiac fibrosis, cardiac hypertrophy, cardiomyopathy, carotid obstructive disease, cerebral infarction, chemical burns, COPD, Crohn's disease, cytokine-induced vascular leak, destabilized blood flow, diabetes (including non-insulin dependent diabetes mellitus), dysfunctional uterine bleeding, endometriosis, Epstein-Barr virus infection, erectile dysfunction, excessive hair growth, follicular cysts, foot ulcer (e.g., diabetic foot ulcer), fungal ulcers, giant cell arteritis, glomerulosclerosis, Graves' disease, Hashimoto's autoimmune thyroiditis, hemangioma, hemangioendothelioma, hemophilic joints, hemorrhage, hepatitis C, hereditary hemorrhagic telangiectasia (HHT), Herpes simplex infections, Herpes zoster infections, hypertension, idiopathic thrombocytopenic purpura, impaired wound healing, inflammatory and infectious processes (e.g. hepatitis, pneumonia, glomerulonephritis), interstitial fibrosis, ischemia, kidney disease, leishmaniasis, leukomalacia, lipid degeneration, liver regeneration, Lupus nephritis, Lyme disease, lymphoproliferative disorders, malaria (*Plasmodium* infection), Mooren ulcer, multiple sclerosis, mycobacterial infections, myocardial infarction, nasal polyps, nephropathy, neuronal inflammation, neuropathy, obesity, osteomyelitis, osteophyte, ovarian hyperstimulation, Paget's disease, pannus growth, peripheral artery disease, peritoneal sclerosis, pemphigoid, polyarteritis, protozoan infections, pseudoxanthoma elasticum, psoriasis, pulmonary hypertension, pyogenic granulomas, renal fibrosis, respiratory distress, rheumatoid arthritis, rickettsial infection, scar keloids, sepsis, sickle cell anemia, Stevens-Johnson disease, stroke, synovitis, systemic lupus erythematosus, syphilis, thyroid enlargement, thyroiditis, toxic shock syndrome, toxoplasmosis, trauma, ulcerative colitis, vascular leak, vascular leak syndrome, vascular malformations (e.g. Osler-Weber syndrome), vein occlusion, viral hemorrhagic fevers (e.g., dengue fever), vitamin A deficiency, warts, or Wegener's sarcoidosis, or a combination thereof.

(160) Sequences

(161) As used herein, the abbreviations for the L-enantiomeric and D-enantiomeric amino acids are as follows: alanine (A, Ala); arginine (R, Arg); asparagine (N, Asn); aspartic acid (D, Asp); cysteine (C, Cys); glutamic acid (E, Glu); glutamine (Q, Gln); glycine (G, Gly); histidine (H, His); isoleucine (I, Ile); leucine (L, Leu); lysine (K, Lys); methionine (M, Met); phenylalanine (F, Phe); proline (P, Pro); serine (S, Ser); threonine (T, Thr); tryptophan (W, Trp); tyrosine (Y, Tyr); valine

(V, Val). In some embodiments, the amino acid is a L-enantiomer. In some embodiments, the amino acid is a D-enantiomer.

(162) TABLE 2 shows the amino acid sequences of humanized V.sub.H antibody regions that bind HPTP- β (VE-PTP). SEQ ID NO: 1 is V.sub.H1, SEQ ID NO: 2 is V.sub.H2, SEQ ID NO: 3 is V.sub.H3, and SEQ ID NO: 4 is V.sub.H4.

(163) TABLE-US-00002 TABLE 2 SEQ ID NO: Name Amino acid sequence 1 V.sub.H1
EVQLVESGGGLVQPGGSLKLSCAASGFTFNA
NAMNWVRQASGKGLEWVGRIRTKSNNYATYY
AGSVKDRFTISRDDSKNTAYLQMNSLKTEDT
AAYYCVRDYYGSSAWITYWGQGLTVTVSS 2 V.sub.H2
EVQLVESGGGLVQPGGSLRLSCAASGFTFNA
NAMNWVRQAPGKGLEWVGRIRTKSNNYATYY
AGSVKDRFTISRDDSKNSLYLQMNSLKTEDT AVYYCVRDYYGSSAWITYWGQGLTVTVSS
3 V.sub.H3 EVQLVESGGGLVQPGGSLRLSCTASGFTFNA
NAMNWVRQAPGKGLEWVGRIRTKSNNYATYY
AGSVKDRFTISRDDSKNIAYLQMNSLKTEDT AVYYCVRDYYGSSAWITYWGQGLTVTVSS
4 V.sub.H4 LVQLVESGGGLVKPGGSLRLSCAASGFTFNA
NAMNWIRQAPGKGLEWVSRIRTKSNNYATYY
AGSVKDRFTISRDNALNSLYLQMNSLRAEDT
AVHYCVRDYYGSSAWITYWGQGLTVTVSS

(164) TABLE 3 shows the amino acid sequences of humanized V.sub.L antibody regions that bind HPTP- β (VE-PTP). SEQ ID NO: 5 is V.sub.L1, SEQ ID NO: 6 is V.sub.L2, SEQ ID NO: 7 is V.sub.L3, and SEQ ID NO: 8 is V.sub.L4.

(165) TABLE-US-00003 TABLE 3 SEQ ID NO: Name Amino acid sequence 5 V.sub.L1
DVVMTQSPSFLSASVGDRVTITCKASQHVGTAVAWYQ
QRPGKAPKLLIYWASTRHTGVPSRFSGSGSGTEFTLT
ISSLPEDFATYFCQQYSSYPFTFGGGTKLEIK 6 V.sub.L2
DIVMTQSPDSLAVSLGERATINCKASQHVGTAVAWYQ
QKPGQPPKLLIYWASTRHTGVDPDRFSGSGSGTDFTLT
ISSLQAEDVAVYYCQQYSSYPFTFGQGTKLEIK 7 V.sub.L3
DIQMTQSPFSLSASVGDRVTITCKASQHVGTAVAWYQ
QKPGKAPKLLIYWASTRHTGVPSRFSGSGSGTDFTLT
ISSLPEDFATYFCQQYSSYPFTFGGGTKLEIK 8 V.sub.L4
DIVMTQSPDSLAVSLGERATINCKASQHVGTAVAWYQ
QKPEQPPKLLIYWASTRHTGVDPDRFSGSGSGTDFTLT
ISSLQAEDVAVYYCQQYSSYPFTFGGGTKVEIK

(166) Each of the V.sub.H domains can be synthesized in-frame with a constant domain sequence, for example, a human IgG1, IgG2, IgG3, IgG4, IgE, IgA1, IgA2, IgM, or IgD sequence. A DNA sequence encoding the entire heavy chain sequence can be codon optimized and verified.

(167) Illustrative amino acid sequences of constant domain sequences are provided in TABLE 4. In some embodiments, a V.sub.H domain disclosed herein is synthesized in-frame with a human IgG1 constant domain sequence. A human IgG1 constant domain sequence can comprise SEQ ID NO: 220. In some embodiments, a V.sub.H domain disclosed herein is synthesized in-frame with a human IgG2 constant domain sequence. A human IgG2 constant domain sequence can comprise SEQ ID NO: 221. In some embodiments, a V.sub.H domain disclosed herein is synthesized in-frame with a human IgG3 constant domain sequence. A human IgG3 constant domain sequence can comprise SEQ ID NO: 222. In some embodiments, a V.sub.H domain disclosed herein is synthesized in-frame with a human IgG4 constant domain sequence. The human IgG4 isotype constant domain sequence can be mutated to a proline rather than a serine at position 228 to reduce Fab-arm exchange (stabilizing S228P mutation). An amino acid sequence of the IgG4 constant

domain with S228P mutation can comprise SEQ ID NO: 9. In some embodiments, a V.sub.H domain disclosed herein is synthesized in-frame with a human IgE constant domain sequence. A human IgE constant domain sequence can comprise SEQ ID NO: 223. In some embodiments, a V.sub.H domain disclosed herein is synthesized in-frame with a human IgA1 constant domain sequence. A human IgA1 constant domain sequence can comprise SEQ ID NO: 224. In some embodiments, a V.sub.H domain disclosed herein is synthesized in-frame with a human IgA2 constant domain sequence. A human IgA2 constant domain sequence can comprise SEQ ID NO: 225. In some embodiments, a V.sub.H domain disclosed herein is synthesized in-frame with a human IgM constant domain sequence. A human IgM constant domain sequence can comprise SEQ ID NO: 226. In some embodiments, a V.sub.H domain disclosed herein is synthesized in-frame with a human IgD constant domain sequence. A human IgD constant domain sequence can comprise SEQ ID NO: 227.

(168) TABLE-US-00004 TABLE 4 SEQ ID NO: Name Amino acid sequence 220 IgG1

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNS constant

GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVN
HKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPK
PKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKAL
PAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKS
RWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK 221 IgG2

ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNS constant

GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSNFGTQTYTCNVD
HKPSNTKVDKTVKCCVECPPCPAPPVAGPSVFLFPPKPKDT
LMISRTPEVTCVVVDVSHEDPEVQFNWYVDGVEVHNAKTKP
REEQFNSTFRVVSIVLTVVHQDWLNGKEYKCKVSNKGLPAPIE
KTISKTKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSD
ISVEWESNGQPENNYKTTTPMLDSDGSFFLYSKLTVDKSRWQ
QGNVFSCSVMHEALHNHYTQKSLSLSPGK 222 IgG3

ASTKGPSVFPLAPCSRSTSGGTAALGCLVKDYFPEPVTVSWNS constant

GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYTCNVN
HKPSNTKVDKRVELKTPLGDTTHTCPRCPEPKSCDTPPPCPRC
PEPKSCDTPPPCPRCPEPKSCDTPPPCPRCPAPELLGGPSVFLFP
PKPKDTLMISRTPEVTCVVVDVSHEDPEVQFKWYVDGVEVH
NAKTKPREEQYNSTFRVVSIVLTVLHQDWLNGKEYKCKVSNK
ALPAPIEKTISKTKGQPREPQVYTLPPSREEMTKNQVSLTCLV
KGFYPSDIAVEWESSGQPENNYNTTPMLDSDGSFFLYSKLTV
DKSRWQQGNIFSCSVMHEALHNRFTQKSLSLSPGK 9 IgG4

ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNS constant

GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYTCNVD
HKPSNTKVDKRVESKYGPPCPAPAEFLGGPSVFLFPPKPKD
TLMISRTPEVTCVVVDVSQEDPEVQFNWYVDGVEVHNAKTK
PREEQFNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSI
EKTISKAKGQPREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPS
DIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSRLTVDKSRW
QEGNVFSCSVMHEALHNHYTQKSLSLGLGK 223 IgE

ASTQSPSVFPLTRCCKNIPSNATSVTLGCLATGYFPEPVMVTW constant

DTGSLNGTTMTLPTTLTSLGHYATISLLTVSGAWAKQMFTC
RVAHTPSSTDWVDNKTFSVCSRDFTPPTVKILQSSCDGGGHFP
PTIQLLCLVSGYTPGTINITWLEDGQVMDVDLSTASTTQEGEL

ASTQSELTSLQKHWLSDFTQVTYQGHTFEDSTKKCADSN
 PRGVSAYLRSRSPFDLFIRKSPTITCLVVDLAPSKGTVNLTWSR
 ASGKPVNHSTRKEEKQRNGTLTVTSTLPGTRDWIEGETYQC
 RVTHPHLPRALMRSTTKTSGPRAAPEVYAFATPEWPGSRDKR
 TLACLIQNFMPEDISVQWLHNEVQLPDARHSTTQPRKTKGSG
 FFVFSRLEVTRAWEQKDEFICRAVHEAASPSQTVQRAVSVN PGK 224 IgA1
 ASPTSPKVFPLSLCSTQPDGNVVIACLVQGFFPQEPLSVTWSES constant
 GQGVTARNFPPSQDASGDLYTTSSQLTLPATQCLAGKSVTCH
 VKHYTNPSQDVTVPVPCVPSTPPTPSPSTPPTPSPSCCHPRLSLH
 RPALEDLLLGSSEANLTCTLTGLRDASGVTFTWTPSSGKSAVQ
 GPPERDLCGCYSVSSVLPGCAEPWNHKGKTFCTAAYPESKTP
 LTATLSKSGNTFRPEVHLLPPPSEELALNELVTLTCLARGFSPK
 DVLVRWLQGSQELPREKYLTWASRQEPSQGTTFFAVTSILRV
 AAEDWKKGDTFSCMVGHEALPLAFTQKTIDRLAGKPTHVNV SVVMAEVDGTCY 225
 IgGA2 ASPTSPKVFPLSLDSTPQDGNVVVACLVQGFFPQEPLSVTWSE constant
 SGQNVTARNFPPSQDASGDLYTTSSQLTLPATQCPDGKSVTC
 HVKHYTNSSQDVTVPCRVPVPPPPCCHPRLSLHRPALEDLLLGS
 EANLTCTLTGLRDASGATFTWTPSSGKSAVQGPPPERDLCGCY
 SVSSVLPGCAQPWNHGETFTCTAAHPELKTPLTANITKSGNTF
 RPEVHLLPPPSEELALNELVTLTCLARGFSPKDVLVRWLQGSQ
 ELPREKYLTWASRQEPSQGTTTYAVTSILRVAAEDWKKGETF
 SCMVGHEALPLAFTQKTIDRMAGKPTHINVSVVMAEADGTC Y 226 IgM
 GSASAPTLFPLVSCENSPSDTSSVAVGCLAQDFLPDSITFSWKY constant
 KNNSDISSTRGFPSVLRGGKYAATSQVLLPSKDVMQGTDEHV
 VCKVQHPNGNKEKNVPLPVIAELPPKVSFVPPRDGFFGNPR
 KSKLICQATGFSPRQIQVSWLREGKQVGSVTTDQVQAEAKE
 SGPTTYKVTSTLTIKESDWLGQSMFTCRVDHRGLTFQQNASS
 MCVPDQDTAIRVFAIPPSFASIFLTKSTKLTCLVTDLTITYDSVTI
 SWTRQNGEAVKTHTNISESHPNATFSAVGEASICEDDWNNGE
 RFTCTVTHTDLPSPKQTI SRPKGVALHRPDVYLLPPAREQLN
 LRESATITCLVTGFSPADV FVQWMQRGQPLSPEKYVTSAPMP
 EPQAPGRYFAHSILTVSEEEWNTGETYTCVVAHEALPNRVTE
 RTVDKSTGKPTLYNVSLVMSDTAGTCY 227 IgD
 APTKAPDVFPIISGCRHPKDNSPVVLACLITGYHPTSVTVTWY constant
 MGTQSQPQRTEFPEIQRRDSYYMTSSQLSTPLQQWRQGEYKCV
 VQHTASKSKKEIFRWPESPKAQASSVPTAQPQAEGSLAKATT
 APATTRNTGRGGEEKKKEKEKEEQEERETKTPECPSHTQPLG
 VYLLTPAVQDLWLRDKATFTCFVVGSDLKDAHLTWEVAGK
 VPTGGVEEGLLERHSNGSQSQHSRLTLPRSLWNAGTSVTCTL
 NHPSLPPQRLMALREPAAPVKLSLNLASSDPPEAASWLL
 CEVSGFSPPNILLMWLEDQREVNTSGFAPARPPPQPRSTTFWA
 WSVLRVPAPPSPQPATYTCVVSHEDSRTLLNASRSLEVSYVTD HGPMK

(169) Each of the V.sub.L domains can be synthesized in-frame with a human light chain constant domain sequence, e.g. a kappa (IgK) or lambda (IgL) chain. The entire light chain sequence can then be codon optimized, and the DNA sequence can be verified. TABLE 5 provides example light chain constant domain sequences.

(170) In some embodiments, a V.sub.L domain disclosed herein is synthesized in-frame with a human IgK constant domain sequence. A human IgK constant domain sequence can comprise SEQ ID NO: 10. In some embodiments, a V.sub.L domain disclosed herein is synthesized in-frame with a human IgL constant domain sequence. A human IgL constant domain sequence can comprise

SEQ ID NO: 228.

(171) TABLE-US-00005 TABLE 5 SEQ ID NO: Name Amino acid sequence 10 IgK
TVAAPSVFIFPPSDEQLKSGTASVVCLLNNFY constant
PREAKVQWKVDNALQSGNSQESVTEQDSKDST
YLSSTLTLSKADYEKHKVYACEVTHQGLSSP VTKSFNRGEC 228 IgL
GQPKANPTVTLFPPSSEELQANKATLVCLISD constant
FYPGAVTVAWKADGSPVKAGVETTKPSKQSN
KYAASSYLSLTPEQWKSHRSYSCQVTHEGSTV EKTVAAPTECS

(172) Signal peptides can result in higher protein expression and/or secretion by a cell. The following signal peptides can be appended to all the constructs disclosed herein.

(173) TABLE-US-00006 Heavy chain signal peptide (SEQ ID NO: 11):

MGWTLVFLFLLSVTAGVHS Light chain signal peptide (SEQ ID NO: 12):
MVSSAQFLGLLLLCFQGTRC

(174) Signal peptidases can cleave a signal peptide off a protein, for example, during a secretion process, generating a mature protein that does not comprise the signal peptide sequence. In some embodiments, a signal peptide is cleaved off a compound or antibody of the disclosure. In some embodiments, a mature compound or antibody of the disclosure does not comprise a signal peptide.

(175) TABLE 6 and TABLE 7 list the full amino acid sequences of humanized heavy and light chains that bind HPTP- β (VE-PTP), respectively. HC1, HC2, HC3 and HC4 are the human IgG4 isotype constant domain with stabilizing S228P mutation joined to V.sub.H1, V.sub.H2, V.sub.H3 and V.sub.H4, respectively, and with signal peptide appended. Amino acids 1-19 of SEQ ID NOs: 13, 14, 15, and 16 are the heavy chain signal peptide (SEQ ID NO: 11). Also shown are HC1, HC2, HC3, and HC4 without the signal peptide appended (SEQ ID NOs: 246, 247, 248, and 249). In some embodiments, a mature compound or antibody of the disclosure does not comprise the signal peptide(s).

(176) LC1, LC2, LC3 and LC4 are the human IgK isotype constant domain joined to V.sub.L1, V.sub.L2, V.sub.L3, and V.sub.L4, respectively, and with signal peptide appended. Amino acids 1-20 of SEQ ID NOs: 17, 18, 19, and 20 are the light chain signal peptide (SEQ ID NO: 12). Also shown are LC1, LC2, LC3, and LC4 without the signal peptide appended (SEQ ID NOs: 250, 251, 252, and 253). In some embodiments, a mature compound or antibody of the disclosure does not comprise the signal peptide(s).

(177) TABLE-US-00007 TABLE 6 SEQ ID NO: Name Amino acid sequence 13 signal
MGWTLVFLFLLSVTAGVHSEVQLVESGGGLVQPGGSLKLSCA peptide -

ASGFTFNANAMNWVRQASGKGLEWVGRIRTKSNNYATYYAG HC1
SVKDRFTISRDDSKNTAYLQMNSLKTEDTAAYYCVRDYYGSS
AWITYWGQGTTLTVSSASTKGPSVFPLAPCSRSTSESTAALGCL
VKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTV
PSSSLGKTYTCNVDPKPSNTKVDKRVESKYGPPCPPCPAPEFL
GGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSQEDPEVQFNWY
VDGVEVHNAKTKPREEQFNSTYRVVSVLTVLHQDWLNGKEY
KCKVSNKGLPSSIEKTISKAKGQPREPQVYTLPPSQEEMTKNQV
SLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLY
SRLTVDKSRWQEGNVFSCSVMHEALHNHYTQKSLSLGLK 246 HC1
EVQLVESGGGLVQPGGSLKLSCAASGFTFNANAMNWVRQAS
GKGLEWVGRIRTKSNNYATYYAGSVKDRFTISRDDSKNTAYL
QMNSLKTEDTAAYYCVRDYYGSSAWITYWGQGTTLTVSSAST
KGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALT
SGVHTFPAVLQSSGLYSLSSVTVPSDDLTKTYTCNVDPKPSN
TKVDKRVESKYGPPCPPCPAPEFLGGPSVFLFPPKPKDTLMISR
TPEVTCVVVDVSQEDPEVQFNWYVDGVEVHNAKTKPREEQFN

STYRVLSVLTQKCKVSNKGLPSSIEKTISKAK
GQPREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWES
NGQPENNYKTTPPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCS
VMHEALHNHYTQKSLSLGLK 14 signal
MGWTLVFLFLLSVTAGVHSEVQLVESGGGLVQPGGSLRLSCA peptide -
ASGFTFNANAMNWVRQAPGKGLEWVGRIRTKSNNYATYYAG HC2
SVKDRFTISRDDSKNSLYLQMNSLKTEDTAVYYCVRDYYGSS
AWITYWGQGTLLTVSSASTKGPSVFPLAPCSRSTSESTAALGCL
VKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTV
PSSSLGTKTYTCNVDPHKPSNTKVDKRVESKYGPPCPPCPAPEFL
GGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSDQEDPEVQFNWY
VDGVEVHNAKTKPREEQFNSTYRVVSVLTVHLHQLDNLNGKEY
KCKVSNKGLPSSIEKTISKAKGQPREPQVYTLPPSQEEMTKNQV
SLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLY
SRLTVDKSRWQEGNVFSCSVMHEALHNHYTQKSLSLGLK 247 HC2
EVQLVESGGGLVQPGGSLRLSCAASGFTFNANAMNWVRQAPG
KGLEWVGRIRTKSNNYATYYAGSVKDRFTISRDDSKNSLYLQ
MNSLKTEDTAVYYCVRDYYGSSAWITYWGQGTLLTVSSASTK
GPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALT
SGVHTFPAVLQSSGLYSLSSVTVPSDDLGTKTYTCNVDPHKPSNT
KVDKRVESKYGPPCPPCPAPEFLGGPSVFLFPPKPKDTLMISRT
PEVTCVVVDVSDQEDPEVQFNWYVDGVEVHNAKTKPREEQFNST
YRVVSVLTVHLHQLDNLNGKEYKCKVSNKGLPSSIEKTISKAKGQ
PREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTPPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVM
HEALHNHYTQKSLSLGLK 15 signal
MGWTLVFLFLLSVTAGVHSEVQLVESGGGLVQGRSLRLSCT peptide -
ASGFTFNANAMNWVRQAPGKGLEWVGRIRTKSNNYATYYAG HC3
SVKDRFTISRDDSKNIAYLQMNSLKTEDTAVYYCVRDYYGSSA
WITYWGQGTLLTVSSASTKGPSVFPLAPCSRSTSESTAALGCLV
KDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTV
PSSSLGTKTYTCNVDPHKPSNTKVDKRVESKYGPPCPPCPAPEFL
GGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSDQEDPEVQFNWY
VDGVEVHNAKTKPREEQFNSTYRVVSVLTVHLHQLDNLNGKEY
KCKVSNKGLPSSIEKTISKAKGQPREPQVYTLPPSQEEMTKNQV
SLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLY
SRLTVDKSRWQEGNVFSCSVMHEALHNHYTQKSLSLGLK 248 HC3
EVQLVESGGGLVQGRSLRLSCTASGFTFNANAMNWVRQAPG
KGLEWVGRIRTKSNNYATYYAGSVKDRFTISRDDSKNIAYLQ
MNSLKTEDTAVYYCVRDYYGSSAWITYWGQGTLLTVSSASTK
GPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALT
SGVHTFPAVLQSSGLYSLSSVTVPSDDLGTKTYTCNVDPHKPSNT
KVDKRVESKYGPPCPPCPAPEFLGGPSVFLFPPKPKDTLMISRT
PEVTCVVVDVSDQEDPEVQFNWYVDGVEVHNAKTKPREEQFNST
YRVVSVLTVHLHQLDNLNGKEYKCKVSNKGLPSSIEKTISKAKGQ
PREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTPPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVM
HEALHNHYTQKSLSLGLK 16 signal
MGWTLVFLFLLSVTAGVHSLVQLVESGGGLVKPGGSLRLSCA peptide -
ASGFTFNANAMNWIRQAPGKGLEWVSRIRTKSNNYATYYAGS HC4

VKDRFTISRAEDNTSLYQMNRAEDTAVHYCVRDYYGSSA
WITYWGQGTTLVTVSSASTKGPSVFPLAPCSRSTSESTAALGCLV
KDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVP
SSSLGTKTYTCNVDHKPSNTKVDKRVESKYGPPCPPCPAPEFL
GGPSVFLEPPKPKDTLMISRTPEVTCVVVDVSQEDPEVQFNWY
VDGVEVHNAKTKPREEQFNSTYRVVSVLTVLHQDWLNGKEY
KCKVSNKGLPSSIEKTISKAKGQPREPQVYTLPPSQEEMTKNQV
SLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLY
SRLTVDKSRWQEGNVFSCSVMHEALHNHYTQKSLSLGLK 249 HC4
LVQLVESGGGLVLPKGGSLRLSCAASGFTENANAMNWIRQAPG
KGLEWVSRIRTKSNNYATYYAGSVKDRFTISRDNAKNSLYLQ
MNSLRAEDTAVHYCVRDYYGSSAWITYWGQGTTLVTVSSASTK
GPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVVTVPSSSLGTKTYTCNVDHKPSNT
KVDKRVESKYGPPCPPCPAPEFLGGPSVFLEPPKPKDTLMISRTP
EVTCTVVVDVSQEDPEVQFNWYVDGVEVHNAKTKPREEQFNST
YRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTISKAKGQ
PREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTPPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVM
HEALHNHYTQKSLSLGLK

(178) TABLE-US-00008 TABLE 7 SEQ ID NO: Name Amino acid sequence 17 signal

MVSSAQFLGLLLLCFQGTRCDVVMTQSPSFLSASVGDRVTIT peptide-
CKASQHVGTAVAWYQQRP GKAPKLLIYWASTRHTGVPSRF LC1
SGSGSGTEFTLTISLQPEDFATYFCQQYSSYPFTFGGGTKLEI
KRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQ
WKVDNALQSGNSQESVTEQDSKDYSLSTLTLSKADYEK
HKVYACEVTHQGLSSPVTKSFNRGEC 250 LC1

DVVMTQSPSFLSASVGDRVTITCKASQHVGTAVAWYQQRP
GKAPKLLIYWASTRHTGVPSRFSGSGSGTEFTLTISLQPEDF
ATYFCQQYSSYPFTFGGGTKLEIKRTVAAPSVFIFPPSDEQLK
SGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQ
DSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTK SFNRGEC 18 signal

MVSSAQFLGLLLLCFQGTRCDIVMTQSPDSLAVSLGERATIN peptide-
CKASQHVGTAVAWYQQKPGQPPKLLIYWASTRHTGVPDF LC2
SGSGSGTDFTLTISLQAEDVAVYYCQQYSSYPFTFGQGTKL
EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKV
QWKVDNALQSGNSQESVTEQDSKDYSLSTLTLSKADYE
KHKVYACEVTHQGLSSPVTKSFNRGEC 251 LC2

DIVMTQSPDSLAVSLGERATINCKASQHVGTAVAWYQQKP
GQPPKLLIYWASTRHTGVPDFSGSGSGTDFTLTISLQAED
VAVYYCQQYSSYPFTFGQGTKLEIKRTVAAPSVFIFPPSDEQ
LKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVT
EQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPV TKSFNRGEC 19 signal

MVSSAQFLGLLLLCFQGTRCDIQMTQSPFSLSASVGDRVTIT peptide-
CKASQHVGTAVAWYQQKPGKAPKLLIYWASTRHTGVPSRF LC3
SGSGSGTDFTLTISLQPEDFATYFCQQYSSYPFTFGGGTKLE
IKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQ
WKVDNALQSGNSQESVTEQDSKDYSLSTLTLSKADYEK
HKVYACEVTHQGLSSPVTKSFNRGEC 252 LC3
DIQMTQSPFSLSASVGDRVTITCKASQHVGTAVAWYQQKPG

KAPKLLIYWASTRHTGVPSRFSGSGTDFTLTISLQPEDFA
 TYFCQQYSSYPFTFGGGTKLEIKRTVAAPSVFIFPPSDEQLKS
 GTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQD
 SKDSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSF NRGE 20 signal
 MVSSAQFLGLLLLCFQGTRCDIVMTQSPDSLAVSLGERATIN peptide-
 CKASQHVGTAVAWYQQKPEQPPKLLIYWASTRHTGVPDRF LC4
 SGSGSGTDFTLTISLQAEDVAVYYCQQYSSYPFTFGGGTKV
 EIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKV
 QWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYE
 KHKVYACEVTHQGLSSPVTKSFNRGE 253 LC4

DIVMTQSPDSLAVSLGERATINCKASQHVGTAVAWYQQK
 EQPPKLLIYWASTRHTGVPDRFSGSGSGTDFTLTISLQAED
 VAVYYCQQYSSYPFTFGGGTKVEIKRTVAAPSVFIFPPSDEQ
 LKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVT
 EQDSKDSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPV TKSFNREGC

(179) Any of the V.sub.H regions disclosed herein can be combined with any of the V.sub.L regions disclosed herein, with or without additional sequences appended, to produce compounds that bind anti-HPTP- β (VE-PTP). For example, signal peptides and constant region sequences can be appended to V.sub.H and V.sub.L regions as shown in TABLE 6 and TABLE 7 respectively, and the resulting heavy and light chains can be paired in any combination to form antibodies. TABLE 8 shows 16 possible pairings of HC1-4 and LC1-4 that bind HPTP- β (VE-PTP). Transfection and expression of each of the anti-HPTP- β (VE-PTP) antibodies in TABLE 8 can be pursued.

(180) TABLE-US-00009 TABLE 8 HC1:LC1 HC1:LC2 HC1:LC3 HC1:LC4 HC2:LC1 HC2:LC2
 HC2:LC3 HC2:LC4 HC3:LC1 HC3:LC2 HC3:LC3 HC3:LC4 HC4:LC1 HC4:LC2 HC4:LC3
 HC4:LC4

(181) Antibodies or antigen-binding compounds specific for HPTP- β (VE-PTP) can be combined with antibodies or compounds that activate Tie2, inhibit VEGF, or inhibit VEGFR, to form multispecific compounds.

(182) TABLE 9 provides sequences of collagen IV-derived biomimetic peptides. SEQ ID NO: 152 is AXT-107, an integrin-targeting collagen IV-derived biomimetic peptide that can inhibit VEGFR phosphorylation/activation/signaling and promote Tie2 phosphorylation/activation/signaling. SEQ ID NO: 153 provides a consensus sequence, wherein X is any standard amino acid or non-genetically encoded amino acid. In some embodiments, X at position 7 is M, A, or G; X at position 9 is F, A, Y, or G; X at position 0 is M, A, G, D-Alanine (dA), or norleucine (Nle); X at position 11 is F, A, Y, G, or 4-chlorophenylalanine (4-CiPhe); X at position 12 and position 18 are independently selected from 2-Aminobutyric acid (Abu), G, S, A, V, T, I, L, or Allylglycine (AllylGly).

(183) TABLE-US-00010 TABLE 9 SEQ ID NO: Amino acid sequence 152
 LRRFSTAPFAFIDINDVIN 153 LRRFSTXPXXXXNINNVXNF

(184) TABLE 10 provides sequences related to vasculotide, a synthetic Ang1 mimetic peptide that can act as a Tie2 agonist and activate Tie2 signaling. SEQ ID NO: 229 provides the sequence of a synthetic 7-mer that binds the Tie2 receptor. SEQ ID NO: 230 provides an 8-mer with a cysteine residue added at the N-terminus, allowing covalent tethering to a polyethylene glycol backbone to generate a tetrameric polyethylene oxide clustered version of the peptide.

(185) TABLE-US-00011 TABLE 10 SEQ ID NO: Amino acid sequence 229 HHHRHSF
 230 CHHHRHSF

(186) Antibodies or antigen-binding compounds specific for HPTP- β (VE-PTP) can be combined with antibodies or compounds specific to VEGF or VEGFR to form multispecific compounds that bind HPTP- β (VE-PTP) and VEGF or VEGFR. TABLE 11, TABLE 12, TABLE 13, TABLE 14, TABLE 15, and TABLE 17 provide example sequences of compounds specific for VEGF. TABLE

16 provides sequences of an antibody that binds VEGFR.

(187) TABLE 11 provides sequences related to aflibercept, a recombinant protein comprising VEGF-binding portions of human VEGF receptors 1 and 2 fused to the Fc portion of human IgG1. SEQ ID NO: 21 is the full amino acid sequence of aflibercept. SEQ ID NO: 22 is a shortened sequence containing the VEGF-binding portions of human VEGF receptors 1 and 2 without the Fc portion of IgG.

(188) TABLE-US-00012 TABLE 11 SEQ ID NO: Name Amino acid sequence 21 AFL.sub.1

SDTGRPFVEMYSEIPEIIHMTEGRELVIPCRVTSPNITVT
LKKFPLDTLIPDGKRIIWDSRKGFIIISNATYKEIGLLTCE
ATVNGHLYKTNYLTHRQTNTIIDVVLSPSHGIELSVGE
KLVLNCTARTELNVGIDFNWEYPSSKHQHKKLVNRLD
KTQSGSEMKKFLSTLTIDGVTRSDQGLYTCAASSGLMT
KKNSTFVRVHEKDKTHTCPPCPAPPELLGGPSVFLFPPKP
KDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEV
HNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYK
CKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELT
KNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPV

LDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHN HYTQKSLSLSPG 22 AFL.sub.2

SDTGRPFVEMYSEIPEIIHMTEGRELVIPCRVTSPNITVT
LKKFPLDTLIPDGKRIIWDSRKGFIIISNATYKEIGLLTCE
ATVNGHLYKTNYLTHRQTNTIIDVVLSPSHGIELSVGE
KLVLNCTARTELNVGIDFNWEYPSSKHQHKKLVNRLD
KTQSGSEMKKFLSTLTIDGVTRSDQGLYTCAASSGLMT KKNSTFVRVHEK

(189) TABLE 12 provides the sequence of brolocizumab (SEQ ID NO: 23), a humanized single-chain antibody fragment (scFv) inhibitor of VEGF that binds to the receptor binding site of VEGF and thereby interferes with the interaction of VEGF with VEGFR1 and VEGFR2.

(190) TABLE-US-00013 TABLE 12 SEQ ID NO: Name Amino acid sequence 23

BRO.sub.scFv1 EIVMTQSPSTLSASVGDRVIITCQASEIIHSWLAWYQQ
KPGKAPKLLIYLASTLASGVPSRFSGSGSGAEFTLTIS
SLQPDDFATYYCQNVYLASTNGANFGQGTKLTVLGGGG
GSGGGGSGGGGSGGGGSEVQLVESGGGLVQPGGSLRLS
CTASGFSLTDDYYMTWVRQAPGKGLEWVGFDPPDDPY
YATWAKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCA
GGDHNSGWGLDIWGQGT LTVTVSS

(191) TABLE 13 provides sequences related to ranibizumab, a humanized monoclonal antibody fragment (Fab) that binds to and inhibits activity of VEGF. SEQ ID NO: 24 is the heavy chain sequence of ranibizumab. SEQ ID NO: 25 is the light chain sequence of ranibizumab. SEQ ID NO: 26 is a shortened sequence of the ranibizumab heavy chain that can be used in cloning a single-chain antibody fragment (scFv). SEQ ID NO: 27 is a shortened sequence of the ranibizumab light chain that can be used in cloning a single-chain antibody fragment (scFv). SEQ ID NO: 28 is a single-chain antibody fragment (scFv) comprising SEQ ID NO: 26, SEQ ID NO: 27, and SEQ ID NO: 33 (linker peptide, underlined).

(192) TABLE-US-00014 TABLE 13 SEQ ID NO: Name Amino acid sequence 24

RAN.sub.H1 EVQLVESGGGLVQPGGSLRLSCAASGYDFTHYGMNW
VRQAPGKGLEWVGWINTYTGEPTYAADFKRRFTFSL
DTSKSTAYLQMNSLRAEDTAVYYCAKYPYYYGTSHW
YFDVWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGG
TAALGCLVKDYFPEPVTWNSGALTSGLVHTFPAVL
QSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTK VDKKVEPKSCDKTHL 25 RAN.sub.L1
DIQLTQSPSSLSASVGDRVTITCSASQDISNYLNWY

QQKPGKAPKVLIIYFTSSLHSGVPSRFSGSGSGTDFT
LTISSLQPEDFATYYCQQYSTVPWTFGQGTKVEIKR
TVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREA
KVQWKVDNALQSGNSQESVTEQDSKDSSTYSLSSTLT
LSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC 26 RAN.sub.H2
EVQLVESGGGLVQPGGSLRLSCAASGYDFTHYGMNW
VRQAPGKGLEWVGWINTYTGEPTYAADFKRRFTFSL
DTSKSTAYLQMNSLRAEDTAVYYCAKYPYYYGTSHW YFDVWGQGTTLTVSS 27
RAN.sub.L2 DIQLTQSPSSLSASVGDRVTITCSASQDISNYLNWY
QQKPGKAPKVLIIYFTSSLHSGVPSRFSGSGSGTDFT
LTISSLQPEDFATYYCQQYSTVPWTFGQGTKVEIK 28 RAN.sub.scFv1
EVQLVESGGGLVQPGGSLRLSCAASGYDFTHYGMNW
VRQAPGKGLEWVGWINTYTGEPTYAADFKRRFTFSL
DTSKSTAYLQMNSLRAEDTAVYYCAKYPYYYGTSHW
YFDVWGQGTTLTVSSGGGGSGGGGSGGGGSDIQLTQ
SPSSLSASVGDRVTITCSASQDISNYLNWYQQKPGK
APKVLIIYFTSSLHSGVPSRFSGSGSGTDFTLTISSL
QPEDFATYYCQQYSTVPWTFGQGTKVEIK

(193) TABLE 14 provides sequences related to bevacizumab, a humanized monoclonal antibody that binds to and inhibits activity of VEGF. SEQ ID NO: 29 is the heavy chain sequence of bevacizumab. SEQ ID NO: 30 is the light chain sequence of bevacizumab.

(194) TABLE-US-00015 TABLE 14 SEQ ID NO: Name Amino acid sequence 29

BEV.sub.H1 EVQLVESGGGLVQPGGSLRLSCAASGYTFTNYGMNWVRQA
PGKGLEWVGWINTYTGEPTYAADFKRRFTFSLDTSKSTAYL
QMNSLRAEDTAVYYCAKYPHYGSSHWYFDVWGQGTTLVT
VSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
SWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTY
ICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSV
FLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDG
VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYK
CKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQ
VSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSF
FLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLS PGK 30 BEV.sub.L1
DIQMTQSPSSLSASVGDRVTITCSASQDISNYLNWYQQKPGK
APKVLIIYFTSSLHSGVPSRFSGSGSGTDFTLTISSLQPEDFA
TYYCQQYSTVPWTFGQGTKVEIKRTVAAPSVFIFPPSDEQLK
SGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDS
KDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSF NRGEC

(195) TABLE 15 provides sequences related to conbercept, a recombinant protein comprising extracellular domains from VEGF receptors 1 and 2 fused to the Fc portion of human IgG1. SEQ ID NO: 154 is the full amino acid sequence of conbercept. SEQ ID NO: 155 is a shortened sequence containing sequences from VEGF receptors 1 and 2 without the Fc portion of IgG.

(196) TABLE-US-00016 TABLE 15 SEQ ID NO: Name Amino acid sequence 154

CON.sub.1 GRPFVEMYSEIPEIIHMTGRELVIPCRVTSPNITVTLKKFPL
DTLIPDGKRIIWDSRKGFIIISNATYKEIGLLTCEATVNGHLYK
TNYLTHRQTNTIIDVVLSPSHGIELSVGEKLVLNCTARTELNV
GIDFNWEYPSSKHQHKLVNRDLKTQSGSEMKKFLSTLTID
GVTRSDQGLYTCAASSGLMTKKNSTFVRVHEKPFVAFGSG
MESLVEATVGERVRIPAKYLGYPPEIKWYKNGIPLESNHTI
KAGHVLTIMEVSRDTGNYTVILTNPISKEKQSHVVSLLVY

VPPGPGDKTHTCPLCPAELGGPSVFLCPPKPKDITLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKATPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK 155 CON.sub.2
GRPFVEMYSEIPEIIHMTGRELVIPCRVTSPNITVTLKKFPLDTLIPDGKRIIWDSRKGFIIISNATYKEIGLLTCEATVNGHLYKTNYLTHRQTNTIIDVVLSPSHGIELSVGEKLVNCTARTELVN
GIDFNWEYPSSKHQHKLVNRDLKTQSGSEMKKFLSTLTIDGVTRSDQGLYTCAASSGLMTKKNSTFVRVHEKPFVAFGSG
MESLVEATVGERVRIPAKYLGYPPEIKWYKNGIPLESNHTIKAGHVLTIMEVSRDTGNYTVILTNPISKEKQSHVVSLLVVY VPPGPG

(197) TABLE 16 provides sequences related to ramucirumab, a humanized monoclonal antibody that binds to an extracellular domain of VEGFR2 and inhibits VEGFR2 signaling. SEQ ID NO: 156 is the heavy chain sequence of ramucirumab. SEQ ID NO: 157 is the light chain sequence of ramucirumab.

(198) TABLE-US-00017 TABLE 16 SEQ ID NO: Name Amino acid sequence 156
RAM.sub.H1 EVQLVQSGGGLVKPGGSLRLSCAASGFTFSSYSMNWVRQA
PGKGLEWVSSISSSSSYIYYADSVKGRFTISRDNKNSLY
LQMNSLRAEDTAVYYCARVTDADFIDWGQGTMTVTVSSASTK
GPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSG
ALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICN
VNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVF
LFPPKPKDITLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG
VEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSD
GSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK 157 RAM.sub.L1
DIQMTQSPSSVSASIGDRVTITCRASQGIDNWLGWYQQKP
GKAPKLLIYDASNLDTGVPSTRFSGSGSGTYFTLTISLQA
EDFAVYFCQQAQAFPTFGGGTKVDIKGTVAAPSVFIFPP
SDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQ
ESVTEQDSKDSSTLSSTLTLSKADYEKHKVYACEVTHQG LSSPVTKSFNRGEC

(199) TABLE 17 provides DARPIn and DARPIn-derived amino acid sequences that bind VEGF and inhibit VEGFR signaling. SEQ ID NOS: 158-168 comprise designed ankyrin repeats with binding specificity for VEGF. SEQ ID NO: 169 comprises designed ankyrin repeats with a binding specificity for VEGF and designed ankyrin repeats with a binding specificity for serum albumin. SEQ ID NO: 170 comprises designed ankyrin repeats with a binding specificity for VEGF, designed ankyrin repeats with a binding specificity for hepatocyte growth factor, and designed ankyrin repeats with a binding specificity for serum albumin. SEQ ID NOS: 171-177 comprise designed ankyrin repeats with a binding specificity for VEGF. SEQ ID NO: 173 provides the sequence of abicipar, which comprises designed ankyrin repeats with a binding specificity for VEGF. SEQ ID NOS: 178-190 provide individual designed ankyrin repeat sequence motifs with binding specificity for VEGF, wherein X represents any amino acid. SEQ ID NOS: 191-217 comprise designed ankyrin repeats with binding specificity for VEGF.

(200) TABLE-US-00018 TABLE 17 SEQ ID NO: Amino acid sequence 158
DLGKKLLEAARAGQDDEVRLMANGADVNAFDWMGWTPHLAAHE
GHLEIVEVLLKNGADVNDVSGYTPLHLAAADGHLEIVEVLLKHGA
DVNTKDNTGWTPHLHSADLGHLEIVEVLLKNGADVNAQDKFGKTAF

DISIDNGNEDLAEILQKAA 159
DLDKKLLEAARAGQDDEVRIILLKAGADVNAKDYLGWTPHLHLAAHEG
HLEIVEVLLKAGADVNAKDVSGYTPLHLAAADGHLEIVEVLLKAGAD
VNAKDNTGWTPHLHLSADLGHLEIVEVLLKAGADVNAQDKFGKTAFDI
SIDNGNEDLAEILQKAA 160
DLGKKLLEAARAGQDDEVRIILMANGADVNTADSTGWTPHLHLAAPWG
HPEIVEVLLKNGADVNAHDYQGWTPLHLAATLGHLEIVEVLLKHGAD
VNAQDKFGKTAFDISIDNGNEDLAEILQKAA 161
DLGKKLLEAARAGQDDEVRIILMANGADVNTADSTGWTPHLHLAVPWG
HLEIVEVLLKYGADVNAKDFQGWTPHLHLAAAIGHQEIVEVLLKNGAD
VNAQDKFGKTAFDISIDNGNEDLAEILQKAA 162
DLDKKLLEAARAGQDDEVRIILMANGADVNAKDSTGYTPLHLAAPWG
HLEIVEVLLKAGADVNAKDYQGWTPLHLAAAVGHLEIVEVLLKAGA
DVNAQDKSGKTPADLAADAGHEDIAEVLQKAA 163
DLGKKLLEAARAGQDDEVRIILMANGADVNAARDSTGWTPHLHLAAPWG
HPEIVEVLLKNGADVNAADFQGWTPHLHLAAAVGHLEIVEVLLKHGAD
VNAQDKFGKTAFDISIDNGNEDLAEILQKAA 164
DLDKKLLEAARAGQDDEVRIILLKAGADVNAKDSTGWTPHLHLAAPWG
HPEIVEVLLKAGADVNAKDFQGWTPHLHLAAAAGHLEIVEVLLKAGAD
VNAQDKSGKTPADLAADAGHEDIAEVLQKAA 165
DLGKKLLEAARAGQDDEVRIILLKAGADVNAKDSTGWTPHLHLAAPWG
HPEIVEVLLKAGADVNAKDFQGWTPHLHLAAAAGHLEIVEVLLKAGAD
VNAQDKSGKTPADLAADAGHEDIAEVLQKAA 166
DLDKKLLEAARAGQDDEVRIILLKAGADVNAKDSTGWTPHLHLAAPWG
HPEIVEVLLKAGADVNAKDFQGWTPHLHLAAAVGHLEIVEVLLKAGAD
VNAQDKSGKTPADLAADAGHEDIAEVLQKAA 167
DLDKKLLEAARAGQDDEVRIILLKAGADVNAKDSTGWTPHLHLAAPWG
HPEIVEVLLKAGADVNAKDYQGWTPLHLAAAVGHLEIVEVLLKAGA
DVNAQDKSGKTPADLAADAGHEDIAEVLQKAA 168
DLDKKLLEAARAGQDDEVRIILMANGADVNAKDSTGWTPHLHLAAPW
GHLEIVEVLLKAGADVNAKDFQGWTPHLHLAAAVGHLEIVEVLLKAGA
DVNAQDKFGKTAFDISIDNGNEDLAEILQKAA 169
DLGKKLLEAARAGQDDEVRELLKAGADVNAKDYFSHTPLHLAARNG
HLKIVEVLLKAGADVNAKDFAGKTPLHLAANEGHLEIVEVLLKAGAD
VNAQDIFGKTPADIAADAGHEDIAEVLQKAAGSPTPTPTPTPTPTPTPT
TPTGSDLDKKLLEAARAGQDDEVRIILLKAGADVNAKDSTGWTPHLHLA
APWGHPEIVEVLLKAGADVNAKDFQGWTPHLHLAAAAGHLEIVEVLLK
AGADVNAQDKSGKTPADLAADAGHEDIAEVLQKAA 170
GSDLGKKLLEAARAGQDDEVRELLKAGADVNAKDYFSHTPLHLAAR
NGHLKIVEVLLKAGADVNAKDFAGKTPLHLAANEGHLEIVEVLLKAG
ADVNAQDIFGKTPADIAADAGHEDIAEVLQKAAGSPTPTPTPTPTPTPT
PTPTPTGSDLGKKLLEAARAGQDDEVRIILLKAGADVNAKDRYGDTP
HLAADIGHLEIVEVLLKAGADVNAEDYFGNTPLHLAASYGHLEIVEV
LKAGADVNAKDDYGNTPHLAANTGHLEIVEVLLKAGADVNAQDKS
GKTPADLAADAGHEDIAEVLQKAAGSPTPTPTPTPTPTPTPTPTPTGSD
LDKKLLEAARAGQDDEVRIILLKAGADVNAKDSTGWTPHLHLAAPWGH
PEIVEVLLKAGADVNAKDFQGWTPHLHLAAAAGHLEIVEVLLKAGADV
NAQDKSGKTPADLAADAGHEDIAEVLQKAAGSPTPTPTPTPTPTPTPT
TPTGSDLGKKLLEAARAGQDDEVRELLKAGADVNAKDYFSHTPLHLA
ARNGHLKIVEVLLKAGADVNAKDFAGKTPLHLAANEGHLEIVEVLLK

AGADVNAQDIAADPADIGHEDIAEVLQKAA 171
GSDLGKKLLEAARAGQDDEVRLMANGADVNAFDWMGWTPHLHLAA
HEGHLEIVEVLLKNGADVNDVSGYTPLHLAAADGHLEIVEVLLKY
GADVNTKDNTGWTPLHLSADLGRLEIVEVLLKYGADVNAQDKFGKT
AFDISIDNGNEDLAEILQKAASGSPAGSPTSTEEGTSESATPESGPGTSTE
PSEGSAPGSPAGSPTSTEEGTSTEPSEGSAPGTSTEPSEGSAPGTSESATP
ESGPGSEPATSGSETPGSEPATSGSETPGSPAGSPTSTEEGTSESATPESG
PGTSTEPSEGSAPGTSTEPSEGSAPGSPAGSPTSTEEGTSTEPSEGSAPGT
STEPSEGSAPGTSESATPESGPGTSTEPSEGSAPGTSESATPESGPGSEPA
TSGSETPGTSTEPSEGSAPGTSTEPSEGSAPGTSESATPESGPGTSESATP
ESGPGSPAGSPTSTEEGTSESATPESGPGSEPATSGSETPGTSESATPESG
PGTSTEPSEGSAPGTSTEPSEGSAPGTSTEPSEGSAPGTSTEPSEGSAPGT
STEPSEGSAPGTSTEPSEGSAPGSPAGSPTSTEEGTSTEPSEGSAPGTSES
ATPESGPGSEPATSGSETPGTSESATPESGPGSEPATSGSETPGTSESATP
ESGPGTSTEPSEGSAPGTSESATPESGPGSPAGSPTSTEEGTSPAGSPTSTE
EGSPAGSPTSTEEGTSESATPESGPGTSTEPSEGSAPGTSESATPESGPGS
EPATSGSETPGTSESATPESGPGSEPATSGSETPGTSESATPESGPGTSTE
PSEGSAPGSPAGSPTSTEEGTSESATPESGPGSEPATSGSETPGTSESATP
ESGPGSPAGSPTSTEEGTSPAGSPTSTEEGTSTEPSEGSAPGTSESATPESG
PGTSESATPESGPGTSESATPESGPGSEPATSGSETPGSEPATSGSETPGS
PAGSPTSTEEGTSTEPSEGSAPGTSTEPSEGSAPGSEPATSGSETPGTSES
ATPESGPGTSTEPSEGSAPG 172

GSDLGKKLLEAARAGQDDEVRLMANGADVNAFDWMGWTPHLHLAA
HEGHLEIVEVLLKNGADVNDVSGYTPLHLAAADGHLEIVEVLLKH
GADVNTKDNTGWTPLHLSADLGHLEIVEVLLKNGADVNAQDKFGKT
AFDISIDNGNEDLAEILQKAAGGGSGGGSC 173

GSDLDKKLLEAARAGQDDEVRLMANGADVNDSTGWTPHLHLAAP
WGHPEIVEVLLKNGADVNAADFQGWTPHLHLAAVGHLEIVEVLLKY
GADVNAQDKFGKTAFDISIDNGNEDLAEILQKAAGGGSGGGSC 174

GSDLGKKLLEAARAGQDDEVRLMANGADVNTADSTGWTPHLHLAVP
WGHLEIVEVLLKYGADVNAKDFQGWTPHLHLAAAIGHQEIVEVLLKNG
ADVNAQDKFGKTAFDISIDNGNEDLAEILQKAAGSGSASPAAPAPASP
AAPAPSAPAASPAAPAPASPAAPAPSAPAASPAAPAPASPAAPAPSAPA
ASPAAPAPASPAAPAPSAPAASPAAPAPASPAAPAPSAPAASPAAPAPA
SPAAPAPSAPAASPAAPAPASPAAPAPSAPAASPAAPAPASPAAPAPSA
PAASPAAPAPASPAAPAPSAPAASPAAPAPASPAAPAPSAPAASPAAPA
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SAPAASPAAPAPASPAAPAPSAPAASPAAPAPASPAAPAPSAPAASPA
PAPASPAAPAPSAPAASPAAPAPASPAAPAPSAPAASPAAPAPASPAAP
APSAPAASPAAPAPASPAAPAPSAPAASPAAPAPASPAAPAPSAPAASP AAPAPAS 175

GSDLGKKLLEAARVGQDDEVRLMADGADVNASDFKGDTPHLHLAAS
QGHLEIVEVLLKYGADVNDYDMLGWTPHLHLAADLGHLEIVEVLLKY
GADVNAQDRFGKTAFDISIDNGNEDLAEILQKAAGSPSTADGC 176

GSDLGKKLLEAVRAGQDDEVRLMTNGADVNAKDQFGFTPLQLAAY
NGHLEIVEVLLKYGADVNAFDIFGWTPHLHLAADLGHLEIVEVLLKNGA
DVNAQDKFGRTAFDISIDNGNEDLAEILQKAASGSC 177

GSDLGKKLLEAARAGQDDEVRLMANGADVNAVYIGWTPLHLAAA
YGHLEIVEVLLKYSADVNAEDFAGYTPLHLAASNGHLEIVEVLLKYGA
DVNTKDNTGWTPLHLSADLGHLEIVEVLLKYGADVNTQDKFGKTAFD
ISIDNGNEDLAEILQKAAGSPSTADGC 178

XDXXGXTPLHLAAXXGHLEIVEVLLKXGADVNA 179
XDXXGWTPLHLAAXXGHLEIVEVLLKXGADVNA 180
XDXXGXTPLHLAAXXGHLEIVEVLLKXGADVNX 181
XDXXGWTPLHLXADLGXLEIVEVLLKXGADVNX 182
XDXXGXTPLHLAAXXGHXEIVEVLLKXGADVNA 183
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XDXXGWTXLHLAADLGXLEIVEVLLKXGADVNA 191
GSDLGKKLLEAARAGQDDEVRLMANGADVNAFDWMGWTPLHLAA
HEGHLEIVEVLLKNGADVNAATDVSGYTPLHLAAADGHLEIVEVLLKY
GADVNTKDNTGWTPLHLSADLGHEIVEVLLKYGADVNAQDKFGKT
AFDISIDNGNEDLAEILQKAA 192
GSDLGKKLLEAARVGQDDEVRLMANGADVNAFDWMGWTPLHLAA
HEGHLEIVEVLLKNGADVNAATDVSGYTPLHLAAADGHLEIVEVLLKY
GADVNTKDNTGWTPLHLSADLGHEIVEVLLKYGADVNAQDKFGKT
AFDISIDNGNEDLAEILQKAA 193
GSDLGKKLLEAARAGQDDEVRLMANGADVNAFDWMGWTPLHLAA
HEGHLEIVEVLLKNGADVNAATDVSGYTPLHLAAADGHLEIVEVLLKY
GADVNTKDNTGWTPLHLSADLGRLEIVEVLLKYGADVNAQDKFGKT
AFDISIDNGNEDLAEILQKAA 194
GSDLGKKLLEAARAGQDDEVRLMANGADVNAFDWMGWTPLHLAA
HEGHLEIVEVLLKNGTDVNATDVSGYTPLHLAAADGHLEIVEVLLKY
GADVNTKDNTGWTPLHLSADLGHEIVEVLLKHGADVNAQDKFGKT
AFDISIDNGNEDLAEILQKAA 195
GSDLGKKLLEAARAGQDDEVRLMANGADVNAFDWMGWTPLHLAA
HEGHLEIVEVLLKNGADVNAATDVSGYTPLHLAAADGHLEIVEVLLKH
GADVNTKDNTGWTPLHLSADLGHEIVEVLLKNGADVNAQDKFGKT
AFDISIDNGNEDLAEILQKAA 196
GSDLGKKLLEAARAGQDDEVRLMANGADVNAFDWMGWTPLHLAA
HEGHLEIVEVLLKNGADVNAATDVSGYTPLHLAAADGHLEIVEVLLKH
GADVNTKDNTGWTPLHLSADLGHEIVEVLLKNGADINAQDKFGKTA
FDISIDNGNEDLAEILQKAA 197
GSDLGKKLLEAARAGQDDEVRLMANGADVNAFDWMGWTPLHLAA
HEGHLEIVEVLLKNGADVNAATDVSGYTPLHLAAADGHLEIVEVLLKH
GADVNTTDNTGWTPLHLSADLGHEIVEVLLKYGADVNAQDKFGKT
AFDISIDNGNEDLAEILQKAA 198
GSDLGKKLLEAARAGQDDEVRLMANGADVNAFDYMGWTPLHLAA
HNGHMEIVEVLLKYGADVNASDYSGYTPLHLAAADGHLEIVEVLLKY
GADVNTKDNTGWTPLHLSADLGHEIVEVLLKYGADVNAQDKFGKT
AFDISIDNGNEDLAEILQKAA 199
GSDLGKKLLEAARAGQDDEVRLMANGADVNAVDYIGWTPLHLAAA
YGHLEIVEVLLKYSADVNAEDFAGYTPLHLAASNHGHEIVEVLLKYGA
DVNTKDNTGWTPLHLSADLGHEIVEVLLKYGADVNTQDKFGKTA
FDISIDNGNEDLAEILQKAA 200
GSDLGKKLLEAARTGQDDEVRLMANGADVNAATDYMGTPLHLAA

KVGHLEIVEVLLKYGADVNTADSTGWTPLHLAAAMGHLEIAEVLLKY
GADVNTKDNTGWTPLHLSADLGHLEIVEVLLKNGADVNAQDKFGKT
AFDISIDNGNEDLAEILQKAA 201
GSDLGKKLLEAARAGQDDEVIRILMANGADVNTADSTGWTPLHLAAP
WGHPEIVEVLLKNGADVNAADFQGWTPHLHLAAVGHLEIVEVLLKY
GADVNAQDKFGKTAFDISIDNGNEDLAEILQKAA 202
GSDLGKKLLEAARAGQDDEVIRILMANGADVNTADSTGWTPLHLAAP
WGHPEIVEVLLKNGADVNAADFQGWTPHLHLAAVGHLEIVEVLLKH
GADVNAQDKFGKTAFDISIDNGNEDLAEILQKAA 203
GSDLGKKLLEAARAGQDDEVIRILMANGADVNTADSTGWTPLHLAAP
WGHPEIVEVLLKNGADVNAHDYQGWTPHLHLAATLGHLEIVEVLLKY
GADVNAQDKFGKTAFDISIDNGNEDLAEILQKAA 204
GSDLGKKLLEAARAGQDDEVIRILMANGADVNTADSTGWTPLHLVAP
WGHPEIVEVLLKHGADVNTHDYQGWTPHLHLAATLGHLEIVEVLLRYG
ADVNAQDKFGKTAFDISIDNGNEDLAEILQKAA 205
GSDLGKKLLEAARAGQDDEVIRILMANGADVNTADSTGWTPMHLAAP
WGHPEIVEVLLKHGADVNAQDFQGWTPHLHLAAAIGHLEIVEVLLKYG
ADVNAQDKFGKTAFDISIDNGNEDLAEILQKAA 206
GSDLGKKLLEAARAGQDDEVIRILMANGADVNTADSTGWTPLHLAVP
WGHLEIVEVLLKYGADVNAKDFQGWTPHLHLAAAIGHQEIVEVLLKNG
ADVNAQDKFGKTAFDISIDNGNEDLAEILQKAA 207
GSDLGKKLLEAARVGQDDEVIRILMADGADVNASDFKGDTPHLAAS
QGHLEIVEVLLKYGADVNAYDMLGWTPLHLAADLGHLEIVEVLLKY
GADVNAQDRFGKTAFDISIDNGNEDLAEILQKAA 208
GSDLGKKLLEAARVGQDDEVIRILMANGADVNASDFKGDTPHLAAS
QGHLEIVEVLLKNSADVNAFDLLGWTPLHLAADLGHLEIVEVLLKYG
ADVNAQDKFGKTAFDISIDNGNEDLAEILQKAA 209
GSDLGKKLLEAARVGQDDEVIRILMANGADVNALDFKGDTPHLAAA
SGHLEIVEVLLKNGADVNAHDMLSWTPLHLAGDLGHLEIVEVLLKYG
ADVNAQDRFGKTAFDISIDNGNEDLAEILQKAA 210
GSDLGKKLLEAVRAGQDDEVIRILMTNGADVNAKDQFGFTPLQLAAY
NGHLEIVEVLLKYGADVNAFDIFGWTPLHLAADLGHLEIVEVLLKNGA
DVNAQDKFGRTAFDISIDNGNEDLAEILQKAA 211
GSDLGKKLLEAVRAGQDDEVIRILMANGADVNASDNQGTTPHLAAS
HGHLEIVEVLLKYGADVNDAAHDDLGTPLHLSADLGHLEIVEVLLKY
GADVNAQDKFGKTAFDISIDNGNEDLAEILQKAA 212
GSDLGKKLLEATRAGQDDEVIRILMANGADVNASDNQGTTPHLAAS
HGHLEIVEVLLKYGADVNDAAHDDLGTPLHLAADLGHLEIVEVLLKY
GADVNAQDKFGKTAFDISIDNGNEDLAEILQKAA 213
GSDLGKKLLEAARVGQDDEVIRILMADGADVNASDFKGDTPHLAAS
QGHLEIVEVLLKYGADVNAYDMLGWTPLHLAADLGHLEIVEVLLKY
GADVNAQDRFGKTAFDISIDNGNEDLAEILQKAA 214
GSDLGKKLLEAARVGQDDEVIRILMANDADVNASDFKGDTPHLAAS
QGHLEIVEVLLKYGADVNAYDMLGWTPLHLAADLGHLEIVEVLLKH
GADVNAQDKFGKTAFDISIDNGNEDLAEILQKAA 215
GSDLGKKLLEAARAGQDDEVIRILMANGADVNTLDFKSDTPHLAAAAS
GHLEIVEVLLKNGADVNAHDMLSWTPLHLAGDLGHLEIVEVLLKHGA
DVNAQDKFGKTAFDISIDNGNEDLAEILQKAA 216
GSDLGKKLLEAARAGQDDEVIRILMANGADVNAKDIYGRTPLHLAAL
HGHPEIVEVLLKYGADVNaNDYWGTTSLHLVAIWGHLEIVEVLLKYG

ADVNAVDIAGQTPLHLAAAGHLEIVEVLLKHGADVNAQDKFGKTA
FDISIDNGNEDLAEILQKAA 217
GSDLGKKLLEAARAGQDDEVRLMANGADVNaNDDYDGMTPLHLAA
MEGHLEIVEVLLKYGADVNaNDDHYGFTPLHLAWTGRLEIVEVLLKNG
ADVNAADVFGRTPLHLAATSGHLEIVEVLLKYGADVNAQDKFGKTAF
DISIDNGNEDLAEILQKAA

(201) Antibodies or antigen-binding compounds specific for VEGF can be combined with antibodies or antigen-binding compounds specific to HPTP- β (VE-PTP) to form multispecific compounds, such as bispecific compounds that bind VEGF and HPTP- β (VE-PTP). Any compounds in this disclosure specific for VEGF can be combined with any compounds in this disclosure specific to HPTP- β (VE-PTP). Any of the compounds in this disclosure specific for VEGF or HPTP- β (VE-PTP) can be modified as necessary for the generation of a multispecific compound. Non-limiting examples of modifications necessary for the generation of a multispecific compound include the addition of amino acid residues, the removal of amino acid residues, the replacement of amino acid residues, and the use of linkers. In some embodiments, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, or more residues can be added to an N-terminus and/or a C-terminus of a sequence disclosed herein, and the resulting sequence can be used in the generation of a multi-specific construct. In some embodiments, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, or more residues can be removed from an N-terminus and/or a C-terminus of a sequence disclosed herein, and the remaining sequence can be used in the generation of a multi-specific construct. For example, N- and/or C-terminal residues can be removed from SEQ ID NO: 173 (e.g., to generate SEQ ID NO: 244), and the truncated sequence can be used in a multi-specific construct. In some embodiments, the sequences in any of SEQ ID NOS: 13-20 or 246-253 can be modified. For example, one or more C-terminal residues can be removed (e.g., the C-terminal lysine can be removed from any of SEQ ID NOS: 13-16 or 246-249, and the remaining residues (residues 1-467) can be used in a multi-specific construct).

(202) A compound described herein can include a linker between different domains of the compound. A linker can be a chemical bond, for example, a covalent bond or a non-covalent bond. A linker as described herein can include a flexible or rigid linker.

(203) A linker of the disclosure can include a chemical linker. For example, two amino acid sequences of the disclosure can be connected together by a chemical linker. Each chemical linker of the disclosure can be alkylene, alkenylene, alkynylene, heteroalkylene, cycloalkylene, heterocycloalkylene, arylene, or heteroarylene, any of which is optionally substituted. In some embodiments, a chemical linker of the disclosure can be an ester, ether, amide, thioether, or polyethyleneglycol (PEG). In some embodiments, a linker can reverse the order of the amino acids sequence in a compound, for example, so that the amino acid sequences linked by the linker are head-to-head, rather than head-to-tail. Non-limiting examples of such linkers include diesters of dicarboxylic acids, such as oxalyl diester, malonyl diester, succinyl diester, glutaryl diester, adipyl diester, pimetyl diester, fumaryl diester, maleyl diester, phthalyl diester, isophthalyl diester, and terephthalyl diester. Non-limiting examples of such linkers include diamides of dicarboxylic acids, such as oxalyl diamide, malonyl diamide, succinyl diamide, glutaryl diamide, adipyl diamide, pimetyl diamide, fumaryl diamide, maleyl diamide, phthalyl diamide, isophthalyl diamide, and terephthalyl diamide. Non-limiting examples of such linkers include diamides of diamino linkers, such as ethylene diamine, 1,2-di(methylamino)ethane, 1,3-diaminopropane, 1,3-di(methylamino)propane, 1,4-di(methylamino)butane, 1,5-di(methylamino)pentane, 1,6-di(methylamino)hexane, and piperazine.

(204) Non-limiting examples of optional substituents include hydroxyl groups, sulfhydryl groups, halogens, amino groups, nitro groups, nitroso groups, cyano groups, azido groups, sulfoxide groups, sulfone groups, sulfonamide groups, carboxyl groups, carboxaldehyde groups, imine

groups, alkyl groups, halo-alkyl groups, alkenyl groups, halo-alkenyl groups, alkynyl groups, halo-alkynyl groups, alkoxy groups, aryl groups, aryloxy groups, aralkyl groups, arylalkoxy groups, heterocyclyl groups, acyl groups, acyloxy groups, carbamate groups, amide groups, ureido groups, epoxy groups, and ester groups.

(205) A linker can be a peptide. A linker can comprise a linker sequence, for example, a linker peptide sequence. A linker sequence can be, for example, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 51, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, or 70 amino acid residues in length.

(206) A flexible linker can have a sequence containing stretches of glycine and serine residues. The small size of the glycine and serine residues provides flexibility, and allows for mobility of the connected functional domains. The incorporation of serine or threonine can maintain the stability of the linker in aqueous solutions by forming hydrogen bonds with the water molecules, thereby reducing unfavorable interactions between the linker and protein moieties.

(207) Flexible linkers can also contain additional amino acids such as threonine and alanine to maintain flexibility, as well as polar amino acids such as lysine and glutamine to improve solubility.

(208) A flexible linker can comprise repeats of SEQ ID NO: 42 (GGGS), for example, SEQ ID NOS: 42-55. A flexible linker can comprise repeats of SEQ ID NO: 31 (GGGGS), for example, SEQ ID NOS: 31-41. Several other types of flexible linkers, including SEQ ID NO: 59 (KESGSVSSEQLAQFRSLD) and SEQ ID NO: 60 (EGKSSGSGSESKST), can also be used. The SEQ ID NO: 61 (GSAGSAAGSGEF) linker can also be used, in which large hydrophobic residues are minimized to maintain good solubility in aqueous solutions. The length of the flexible linkers can be adjusted to allow for proper folding or to achieve optimal biological activity of the fused proteins.

(209) A rigid linker can have, for example, an alpha helix-structure. An alpha-helical rigid linker can act as a spacer between protein domains. A rigid linker can comprise repeats of SEQ ID NO: 62 (EAAAK), for example, SEQ ID NOS: 62-66. A rigid linker can comprise repeats of SEQ ID NO: 67 (EAAAR), for example, SEQ ID NOS: 67-72. A rigid linker can have a proline-rich sequence, (XP)_n, with X designating alanine, lysine, glutamine, or any amino acid. The presence of proline in non-helical linkers can increase stiffness, and allow for effective separation of protein domains.

(210) A linker can comprise any of the sequences disclosed in TABLE 18, which can be used to link any portion of a compound disclosed herein to any portion of another compound disclosed herein:

(211) TABLE-US-00019

TABLE	18	SEQ	ID	NO:	Sequence
31	GGGGS	32	GGGGSGGGGS		
33	GGGGSGGGGS	34	GGGGSGGGGS		
35	GGGGSGGGGS	36	GGGGSGGGGS		
37	GGGGSGGGGS	38	GGGGSGGGGS		
39	GGGGSGGGGS	40	GGGGSGGGGS		
41	GGGGSGGGGS	42	GGGGSGGGGS		
43	GGGS	44	GGGS		
45	GGGS	46	GGGS		
47	GGGS	48	GGGS		
49	GGGS	50	GGGS		
51	GGGS	52	GGGS		
53	GGGS	54	GGGS		

could be generated as demonstrated for ranibizumab in TABLE 13. The bevacizumab-derived antigen-binding scFv can be appended onto SEQ ID NO: 14 or SEQ ID NO: 247 of antibody HC2:LC1 to generate a tetravalent bispecific antibody wherein the heavy chain of HC2:LC1 is appended with the bevacizumab-derived antigen-binding scFv (FIG. 2). The bevacizumab-derived antigen-binding scFv can be appended onto SEQ ID NO: 17 or SEQ ID NO: 250 of antibody HC2:LC1 to generate a tetravalent bispecific antibody, in which the light chain of HC2:LC1 is appended with the bevacizumab-derived antigen-binding scFv (FIG. 3). The bevacizumab-derived antigen-binding scFv can be appended onto SEQ ID NO: 14 or SEQ ID NO: 247, and the bevacizumab-derived antigen-binding scFv can be appended onto SEQ ID NO: 17 or SEQ ID NO: 250 of antibody HC2:LC1 to generate a hexavalent bispecific antibody, in which the heavy and light chains of HC2:LC1 are appended with the bevacizumab-derived antigen-binding scFv (FIG. 4).

(217) Any of the 16 antibodies described in TABLE 8 can be combined with abicipar or abicipar-related sequences to generate a bispecific antibody. Non-limiting schematic examples are presented in FIG. 2, FIG. 3, and FIG. 4. For example, SEQ ID NO: 173 or SEQ ID NO: 244 can be appended onto SEQ ID NO: 14, SEQ ID NO: 247, residues 1-467 SEQ ID NO: 14, or residues 1-467 SEQ ID NO: 247 of antibody HC2:LC1 to generate a tetravalent bispecific antibody wherein the heavy chain of HC2:LC1 is appended with SEQ ID NO: 173 or SEQ ID NO: 244 (FIG. 2). SEQ ID NO: 173 or SEQ ID NO: 244 can be appended onto SEQ ID NO: 17 or SEQ ID NO: 250 of antibody HC2:LC1 to generate a tetravalent bispecific antibody, in which the light chain of HC2:LC1 is appended with SEQ ID NO: 173 or SEQ ID NO: 244 (FIG. 3). SEQ ID NO: 173 or SEQ ID NO: 244 can be appended onto SEQ ID NO: 14, SEQ ID NO: 247, residues 1-467 SEQ ID NO: 14, or residues 1-467 SEQ ID NO: 247, and SEQ ID NO: 173 or SEQ ID NO: 244 can be appended onto SEQ ID NO: 17 or SEQ ID NO: 250 of antibody HC2:LC1 to generate a hexavalent bispecific antibody wherein the heavy and light chains of HC2:LC1 are appended with SEQ ID NO: 173 or SEQ ID NO: 244 (FIG. 4).

(218) Any of the 16 antibodies described in TABLE 8 can be combined with conbercept or conbercept-related sequences to generate a bispecific antibody. Non-limiting schematic examples are presented in FIG. 2, FIG. 3, and FIG. 4. For example, SEQ ID NO: 155 can be appended onto SEQ ID NO: 14 or SEQ ID NO: 247 of antibody HC2:LC1 to generate a tetravalent bispecific antibody wherein the heavy chain of HC2:LC1 is appended with SEQ ID NO: 155 (FIG. 2). SEQ ID NO: 155 can be appended onto SEQ ID NO: 17 or SEQ ID NO: 250 of antibody HC2:LC1 to generate a tetravalent bispecific antibody, in which the light chain of HC2:LC1 is appended with SEQ ID NO: 155 (FIG. 3). SEQ ID NO: 155 can be appended onto SEQ ID NO: 14 or SEQ ID NO: 247, and SEQ ID NO: 155 can be appended onto SEQ ID NO: 17 or SEQ ID NO: 250 of antibody HC2:LC1 to generate a hexavalent bispecific antibody wherein the heavy and light chains of HC2:LC1 are appended with SEQ ID NO: 155 (FIG. 4).

(219) Any of the 16 antibodies described in TABLE 8 can be combined with ramucirumab or ramucirumab-related sequences to generate a bispecific antibody. Non-limiting schematic examples are presented in FIG. 2, FIG. 3, and FIG. 4. For example, an antigen-binding scFv of ramucirumab could be generated as demonstrated for ranibizumab in TABLE 13. The ramucirumab-derived antigen-binding scFv can be appended onto SEQ ID NO: 14 or SEQ ID NO: 247 of antibody HC2:LC1 to generate a tetravalent bispecific antibody wherein the heavy chain of HC2:LC1 is appended with the ramucirumab-derived antigen-binding scFv (FIG. 2). The ramucirumab-derived antigen-binding scFv can be appended onto SEQ ID NO: 17 or SEQ ID NO: 250 of antibody HC2:LC1 to generate a tetravalent bispecific antibody, in which the light chain of HC2:LC1 is appended with the ramucirumab-derived antigen-binding scFv (FIG. 3). The ramucirumab-derived antigen-binding scFv can be appended onto SEQ ID NO: 14 or SEQ ID NO: 247, and the ramucirumab-derived antigen-binding scFv can be appended onto SEQ ID NO: 17 or SEQ ID NO: 250 of antibody HC2:LC1 to generate a hexavalent bispecific antibody, in which the heavy and

light chains of HC2:LC1 are appended with the ramcirumab-derived antigen-binding scFv (FIG. 4).

(220) Any of the 16 antibodies described in TABLE 8 can be combined with DARPins, DARPin repeats, or sequences therefrom, to generate a multispecific antibody. Non-limiting schematic examples are presented in FIG. 2, FIG. 3, and FIG. 4. For example, an amino acid sequence comprising any of SEQ ID NOS: 158-217 can be appended onto SEQ ID NO: 14 or SEQ ID NO: 247 of antibody HC2:LC1 to generate a tetravalent bispecific antibody wherein the heavy chain of HC2:LC1 is appended with an amino acid sequence comprising any of SEQ ID NOS: 158-217 (FIG. 2). An amino acid sequence comprising any of SEQ ID NOS: 158-217 can be appended onto SEQ ID NO: 17 or SEQ ID NO: 250 of antibody HC2:LC1 to generate a tetravalent bispecific antibody, in which the light chain of HC2:LC1 is appended with an amino acid sequence comprising any of SEQ ID NOS: 158-217 (FIG. 3). An amino acid sequence comprising any of SEQ ID NOS: 158-217 can be appended onto SEQ ID NO: 14 or SEQ ID NO: 247 and onto SEQ ID NO: 17 or SEQ ID NO: 250 of antibody HC2:LC1 to generate a hexavalent bispecific antibody wherein the heavy and light chains of HC2:LC1 are appended with an amino acid sequence comprising any of SEQ ID NOS: 158-217 (FIG. 4).

(221) The Tie2 activating compounds described in TABLE 9 and TABLE 10 can be combined with the HPTP- β (VE-PTP)-binding compounds described in TABLE 2, TABLE 3, TABLE 6, TABLE 7, and TABLE 8.

(222) Any of the 16 antibodies described in TABLE 8 can be combined with collagen IV-derived biomimetic peptide sequences to generate a multispecific antibody. Non-limiting schematic examples are presented in FIG. 2, FIG. 3, and FIG. 4. For example, an amino acid sequence comprising SEQ ID NO: 152 or SEQ ID NO: 153 can be appended onto SEQ ID NO: 14 or SEQ ID NO: 247 of antibody HC2:LC1 to generate a tetravalent bispecific antibody wherein the heavy chain of HC2:LC1 is appended with an amino acid sequence comprising SEQ ID NO: 152 or SEQ ID NO: 153 (FIG. 2). An amino acid sequence comprising SEQ ID NO: 152 or SEQ ID NO: 153 can be appended onto SEQ ID NO: 17 or SEQ ID NO: 250 of antibody HC2:LC1 to generate a tetravalent bispecific antibody, in which the light chain of HC2:LC1 is appended with an amino acid sequence comprising SEQ ID NO: 152 or SEQ ID NO: 153 (FIG. 3). An amino acid sequence comprising SEQ ID NO: 152 or SEQ ID NO: 153 can be appended onto SEQ ID NO: 14 or SEQ ID NO: 247 and onto SEQ ID NO: 17 or SEQ ID NO: 250 of antibody HC2:LC1 to generate a hexavalent bispecific antibody wherein the heavy and light chains of HC2:LC1 are appended with an amino acid sequence comprising SEQ ID NO: 152 or SEQ ID NO: 153 (FIG. 4).

(223) Any of the 16 antibodies described in TABLE 8 can be combined with Ang2 mimetics, or sequences therefrom, to generate a multispecific antibody. Non-limiting schematic examples are presented in FIG. 2, FIG. 3, and FIG. 4. For example, an amino acid sequence comprising any of SEQ ID NOS: 229-230 can be appended onto SEQ ID NO: 14 or SEQ ID NO: 247 of antibody HC2:LC1 to generate a tetravalent bispecific antibody wherein the heavy chain of HC2:LC1 is appended with an amino acid sequence comprising any of SEQ ID NOS: 229-230 (FIG. 2). An amino acid sequence comprising any of SEQ ID NOS: 229-230 can be appended onto SEQ ID NO: 17 or SEQ ID NO: 250 of antibody HC2:LC1 to generate a tetravalent bispecific antibody, in which the light chain of HC2:LC1 is appended with an amino acid sequence comprising any of SEQ ID NOS: 229-230 (FIG. 3). An amino acid sequence comprising any of SEQ ID NOS: 229-230 can be appended onto SEQ ID NO: 14 or SEQ ID NO: 247 and onto SEQ ID NO: 17 or SEQ ID NO: 250 of antibody HC2:LC1 to generate a hexavalent bispecific antibody wherein the heavy and light chains of HC2:LC1 are appended with an amino acid sequence comprising any of SEQ ID NOS: 229-230 (FIG. 4).

(224) The Tie2 activating compounds described in TABLE 9 and TABLE 10 can be combined with the VEGF-binding and VEGFR-binding compounds described in TABLE 11, TABLE 12, TABLE 13, TABLE 14, TABLE 15, TABLE 16, and TABLE 17 to generate multi-specific compounds.

(225) Any of the compounds in this disclosure, e.g., the multi-specific fusion constructs described above, can be modified as necessary to promote desirable protein folding or biological activity.

(226) CDRs

(227) Sequences in this disclosure can comprise complementarity determining regions (CDRs). CDRs can be identified by the Kabat method, the Chothia method, the IMGT method, or the Paratome method. A CDR of a sequence herein can be, for example, between 0 and 91 residues in length, between 0 and 25 residues in length, between 5 and 14 residues in length, about 0 residues in length, about 1 residue in length, about 2 residues in length, about 3 residues in length, about 4 residues in length, about 5 residues in length, about 6 residues in length, about 7 residues in length, about 8 residues in length, about 9 residues in length, about 10 residues in length, about 11 residues in length, about 12 residues in length, about 13 residues in length, about 14 residues in length, about 15 residues in length, about 16 residues in length, about 17 residues in length, about 18 residues in length, about 19 residues in length, about 20 residues in length, about 21 residues in length, about 22 residues in length, about 23 residues in length, about 24 residues in length, or about 25 residues in length.

(228) A compound of this disclosure with a binding specificity for or an ability to modulate HPTP- β (VE-PTP) can comprise, for example, any of the CDRs in TABLE 19, TABLE 20, TABLE 21, TABLE 22, TABLE 23, or TABLE 24.

(229) TABLE 19 provides non-limiting examples of HCDR1 sequences specific for HPTP- β (VE-PTP).

(230) TABLE-US-00020 TABLE 19 SEQ ID NO: Amino acid sequence 76 ANAMN 77 GFTFNAN 78 GFTFNANA 79 FTFNANAMN 80 GFTFNANAMN

(231) TABLE 20 provides non-limiting examples of HCDR2 sequences specific for HPTP- β (VE-PTP).

(232) TABLE-US-00021 TABLE 20 SEQ ID NO: Amino acid sequence 81 RIRTKSNNYATYYAGSVKD 82 RTKSNNYA 83 IRTKSNNYAT 84 WVGRIRTKSNNYATYY 85 WVGRIRTKSNNYATYYAGSVKD 86 WWSRIRTKSNNYATYY 87 WWSRIRTKSNNYATYYAGSVKD

(233) TABLE 21 provides non-limiting examples of HCDR3 sequences specific for HPTP- β (VE-PTP).

(234) TABLE-US-00022 TABLE 21 SEQ ID NO: Amino acid sequence 88 DYYGSSAWITY 89 VRDYYGSSAWITY 90 RDYYGSSAWITY

(235) TABLE 22 provides non-limiting examples of LCDR1 sequences specific for HPTP- β (VE-PTP).

(236) TABLE-US-00023 TABLE 22 SEQ ID NO: Amino acid sequence 91 KASQHVGTAVA 92 QHVGTA 93 QHVGTA

(237) TABLE 23 provides non-limiting examples of LCDR2 sequences specific for HPTP- β (VE-PTP).

(238) TABLE-US-00024 TABLE 23 SEQ ID NO: Amino acid sequence 94 WASTRHT 95 WAS 96 LLIYWASTRHT

(239) TABLE 24 provides non-limiting examples of LCDR3 sequences specific for HPTP- β (VE-PTP).

(240) TABLE-US-00025 TABLE 24 SEQ ID NO: Amino acid sequence 97 QQYSSYPFT 98 QQYSSYPF

(241) Compounds of this disclosure with a binding specificity for or an ability to modulate VEGF can comprise any of the CDRs in TABLE 25, TABLE 26, TABLE 27, TABLE 28, TABLE 29, or TABLE 30.

(242) TABLE 25 provides non-limiting examples of HCDR1 sequences specific for VEGF.

(243) TABLE-US-00026 TABLE 25 SEQ ID NO: Amino acid sequence 99 DYYMYT 100 GFSLTDDYY 101 GFSLTDDYYY 102 FSLTDYYMYT 103 GFSLTDDYYMYT 104 HYGMN

105 GYDFTHY 106 GYDFTHYG 107 YDFTHYGMN 108 GYDFTHYGMN 109 NYGMN 110
GYTFTNY 111 GYTFTNYG 112 YTFTNYGMN 113 GYTFTNYGMN

(244) TABLE 26 provides non-limiting examples of HCDR2 sequences specific for VEGF.

(245) TABLE-US-00027 TABLE 26 SEQ ID NO: Amino acid sequence 114

FIDPDDDDPYYATWAKG 115 DPDDD 116 IDPDDDDP 117 WVGFIIDPDDDDPYYATWA 118

WVGFIIDPDDDDPYYATWAKG 119 WINTYTGEPTYAADFQR 120 NTYTGE 121 INTYTGE

122 WVGWINTYTGEPTY 123 WVGWINTYTGEPTYAADFQR

(246) TABLE 27 provides non-limiting examples of HCDR3 sequences specific for VEGF.

(247) TABLE-US-00028 TABLE 27 SEQ ID NO: Amino acid sequence 124

GDHNSGWGLDI 125 AGGDHNSGWGLDI 126 YPYYYGTSHWYFDV 127

AKYPYYGTSHWYFDV 128 KYPYYGTSHWYFDV 129 YPHYYGSSHWYFDV 130

AKYPHYYGSSHWYFDV 131 KYPHYYGSSHWYFDV

(248) TABLE 28 provides non-limiting examples of LCDR1 sequences specific for VEGF.

(249) TABLE-US-00029 TABLE 28 SEQ ID NO: Amino acid sequence 132

QASEIIHSWLA 133 EIIHSW 134 EIIHSWLA 135 SASQDISNYLN 136 QDISNY 137

QDISNYLN

(250) TABLE 29 provides non-limiting examples of LCDR2 sequences specific for VEGF.

(251) TABLE-US-00030 TABLE 29 SEQ ID NO: Amino acid sequence 138 LASTLAS

139 LAS 140 LLIYLASTLAS 141 FTSSLHS 142 FTS 143 VLIYFTSSLHS

(252) TABLE 30 provides non-limiting examples of LCDR3 sequences specific for VEGF.

(253) TABLE-US-00031 TABLE 30 SEQ ID NO: Amino acid sequence 144

QNVYLASTNGAN 145 QQYSTVPWT 146 QQYSTVPW

(254) TABLE 31 provides aflibercept-derived sequences corresponding to the D2 domain of human VEGF receptor 1 (SEQ ID NO: 147) and the D3 domain of human VEGF receptor 2 (SEQ ID NO: 148).

(255) TABLE-US-00032 TABLE 31 SEQ ID NO: Amino acid sequence 147

SDTGRPFVEMYSEIPEIIHMTEGRELVIPCR VTSPNITVTLKKFPLDTLIPDGKRIIWDNRK

GFIISNATYKEIGLLTCEATVNGHLYKTNYL THRQNTNII 148

DVVLSPSHGIELSVGEKLVNCTARTELVN

IDFNWEYPSSKHQHKLVNRDLKTQSGSEMK

KFLSTLTIDGVTRSDQGLYTCAASSGLMTKK NSTFVRVHEK

(256) TABLE 32 provides abicipar-derived sequences. SEQ ID NOS: 233-236 correspond to

ankyrin repeats within abicipar. SEQ ID NOS: 237-242 provide consensus sequences for VEGF

binding ankyrin repeats. In SEQ ID NO: 237, X1 is K, T, or Y; X2 is N or M; X3 is T or F; X4 is S or A; X5 is H or R; X6 is A, Y, H, or N; and X7 is A or T. In SEQ ID NO: 238, X1 is K, M, N, R, or V; X2 is Y, H, M, or V; X3 is F, L, M, or V; X4 is R, H, V, A, K, or N; X5 is F, D, H, T, Y, M, or K; and X6 is A, H, N, or Y. In SEQ ID NO: 239, X1 is L, S, or T; X2 is G, S, or C; X3 is S or A; X4 is Q, S, M, or N; X5 is L, M, or Q; and X6 is A, H, N, Y, or D. In SEQ ID NO: 240, X1 is K, S, I, N, T, or V; X2 is K, N, W, A, H, M, Q, or S; X3 is F, Q, L, H, or V; X4 is F or T; X5 is Q or H; X6 is Y or S; X7 is N, H, Y, or M; and X8 is A, H, N, or Y. In SEQ ID NO: 241, X1 is A, N, R, V, Y, E, H, I, K, L, Q, S, or T; X2 is S, A, N, R, D, F, L, P, T, or Y; X3 is T, V, S, A, L, or F; X4 is W, F, or H; X5 is P, I, A, L, S, T, V, or Y; X6 is W, F, I, L, T, or V; X7 is L or P; and X8 is A, H, N, or Y. In SEQ ID NO: 242, X1 is H, Q, A, K, R, D, I, L, M, N, V, or Y; X2 is Y, F, or H; X3 is Q, F, or T; X4 is W, M, G, H, N, or T; X5 is T, A, M, L, or V; X6 is I, L, V, D, or T; and X7 is A, H, N, or Y. SEQ ID NO: 244 provides a truncated sequence derived from abicipar that comprises designed ankyrin repeats with a binding specificity for VEGF.

(257) TABLE-US-00033 TABLE 32 SEQ ID NO: Amino acid sequence 233

DLDKKLLEAARAGQDDEVRLMANGADVNRDS 234

TGWTPLHLAAPWGHPEIVEVLLKNGADVNAADF 235

QGWTPLHLAAVGHLEIVEVLLKYGADVNAQDK 236 FGKTAFDISIDNGNEDLAEILQ 237

X1DX2X3GWTPLHLAAX4X5GHLEIVEVLLKX6GADVNA 238
X1DX2X3GWTPLHLAAX4X5GHLEIVEVLLKX6GADVNA 239
X1DFKX2DTPLHLAAX3X4GHX5EIVEVLLKX6GADVNA 240
X1DX2X3GX4TPLX5LAAAX6X7GHLEIVEVLLKX8GADVNA 241
X1DX2X3GX4TPLHLAAX5X6GHX7EIVEVLLKX8GADVNA 242
X1DX2X3GX4TPLHLAAX5X6GHLEIVEVLLKX7GADVNA 244
DLDKKLEAARAGQDDEVRI LMANGADV NARDSTGWTPLHLA
APWGHPEIVEVLLKNGADVNAADFQGWTPHLAAVGHLEI
VEVLLKYGADVNAQDKFGKTAFDISIDNGNEDLAEILQKAA

Homology

(258) A sequence of a compound herein can have at least about 70% homology, at least about 71% homology, at least about 72% homology, at least about 73% homology, at least about 74% homology, at least about 75% homology, at least about 76% homology, at least about 77% homology, at least about 78% homology, at least about 79% homology, at least about 80% homology, at least about 81% homology, at least about 82% homology, at least about 83% homology, at least about 84% homology, at least about 85% homology, at least about 86% homology, at least about 87% homology, at least about 88% homology, at least about 89% homology, at least about 90% homology, at least about 91% homology, at least about 92% homology, at least about 93% homology, at least about 94% homology, at least about 95% homology, at least about 96% homology, at least about 97% homology, at least about 98% homology, at least about 99% homology, at least about 99.1% homology, at least about 99.2% homology, at least about 99.3% homology, at least about 99.4% homology, at least about 99.5% homology, at least about 99.6% homology, at least about 99.7% homology, at least about 99.8% homology, at least about 99.9% homology, at least about 99.91% homology, at least about 99.92% homology, at least about 99.93% homology, at least about 99.94% homology, at least about 99.95% homology, at least about 99.96% homology, at least about 99.97% homology, at least about 99.98% homology, or at least about 99.99% homology to an amino acid sequence provided herein.

(259) Various methods and software programs can be used to determine the homology between two or sequences, such as NCBI BLAST, Clustal W, MAFFT, Clustal Omega, AlignMe, Praline, or another suitable method or algorithm.

(260) Pharmaceutical Compositions

(261) A pharmaceutical composition of the disclosure can be a combination of any pharmaceutical compounds described herein with other chemical components, such as carriers, stabilizers, diluents, dispersing agents, suspending agents, thickening agents, and/or excipients. The pharmaceutical composition facilitates administration of the compound to an organism.

(262) Pharmaceutical formulations for administration can include aqueous solutions of the active compounds in water-soluble form. Suspensions of the active compounds can be prepared as oily injection suspensions. Suitable lipophilic solvents or vehicles include fatty oils such as sesame oil, or synthetic fatty acid esters, such as ethyl oleate or triglycerides, or liposomes. Aqueous injection suspensions can contain substances which increase the viscosity of the suspension, such as sodium carboxymethyl cellulose, sorbitol, or dextran. The suspension can also contain suitable stabilizers or agents which increase the solubility of the compounds to allow for the preparation of highly concentrated solutions. The active ingredient can be in powder form for constitution with a suitable vehicle, for example, sterile pyrogen-free water, before use.

(263) In practicing the methods of treatment or use provided herein, therapeutically-effective amounts of the compounds described herein are administered in pharmaceutical compositions to a subject having a disease or condition to be treated. In some embodiments, the subject is a mammal such as a human. A therapeutically-effective amount can vary widely depending on the severity of the disease, the age and relative health of the subject, the potency of the compounds used, and other factors.

(264) Pharmaceutical compositions can be formulated using one or more physiologically-acceptable carriers comprising excipients and auxiliaries, which facilitate processing of the active compounds into preparations that can be used pharmaceutically. Formulation can be modified depending upon the route of administration chosen. Pharmaceutical compositions comprising compounds described herein can be manufactured, for example, by mixing, dissolving, emulsifying, encapsulating, entrapping, or compression processes.

(265) The pharmaceutical compositions can include at least one pharmaceutically-acceptable carrier, diluent, or excipient and compounds described herein as free-base or pharmaceutically-acceptable salt form. Pharmaceutical compositions can contain solubilizers, stabilizers, tonicity enhancing agents, buffers, and preservatives.

(266) Methods for the preparation of compositions comprising the compounds described herein include formulating the compounds with one or more inert, pharmaceutically-acceptable excipients or carriers to form a solid, semi-solid, or liquid composition. Solid compositions include, for example, powders, tablets, dispersible granules, capsules, and cachets. Liquid compositions include, for example, solutions in which a compound is dissolved, emulsions comprising a compound, or a solution containing liposomes, micelles, or nanoparticles comprising a compound as disclosed herein. Semi-solid compositions include, for example, gels, suspensions and creams. The compositions can be in liquid solutions or suspensions, solid forms suitable for solution or suspension in a liquid prior to use, or as emulsions. These compositions can also contain minor amounts of nontoxic, auxiliary substances, such as wetting or emulsifying agents, pH buffering agents, and other pharmaceutically-acceptable additives.

(267) Non-limiting examples of dosage forms suitable for use in the disclosure include liquid, powder, gel, nanosuspension, nanoparticle, microgel, aqueous or oily suspensions, emulsion, and any combination thereof.

(268) Non-limiting examples of pharmaceutically-acceptable excipients suitable for use in the disclosure include binding agents, disintegrating agents, anti-adherents, anti-static agents, surfactants, anti-oxidants, coating agents, coloring agents, plasticizers, preservatives, suspending agents, emulsifying agents, anti-microbial agents, spheronization agents, and any combination thereof.

(269) A composition of the disclosure can be, for example, an immediate release form or a controlled release formulation. An immediate release formulation can be formulated to allow the compounds to act rapidly. Non-limiting examples of immediate release formulations include readily dissolvable formulations. A controlled release formulation can be a pharmaceutical formulation that has been adapted such that release rates and release profiles of the active agent can be matched to physiological and chronotherapeutic requirements, or has been formulated to effect release of an active agent at a programmed rate. Non-limiting examples of controlled release formulations include granules, delayed release granules, hydrogels (e.g., of synthetic or natural origin), other gelling agents (e.g., gel-forming dietary fibers), matrix-based formulations (e.g., formulations comprising a polymeric material having at least one active ingredient dispersed through), granules within a matrix, polymeric mixtures, and granular masses.

(270) In some, a controlled release formulation is a delayed release form. A delayed release form can be formulated to delay a compound's action for an extended period of time. A delayed release form can be formulated to delay the release of an effective dose of one or more compounds, for example, for about 4, about 8, about 12, about 16, or about 24 hours.

(271) A controlled release formulation can be a sustained release form. A sustained release form can be formulated to sustain, for example, the compound's action over an extended period of time. A sustained release form can be formulated to provide an effective dose of any compound described herein (e.g., provide a physiologically-effective blood profile) over about 4, about 8, about 12, about 16, or about 24 hours.

(272) The disclosed compositions can optionally comprise pharmaceutically-acceptable

preservatives.

(273) Non-limiting examples of pharmaceutically-acceptable excipients can be found, for example, in Remington: The Science and Practice of Pharmacy, Nineteenth Ed (Easton, Pa.: Mack Publishing Company, 1995); Hoover, John E., Remington's Pharmaceutical Sciences, Mack Publishing Co., Easton, Pennsylvania 1975; Liberman, H. A. and Lachman, L., Eds., Pharmaceutical Dosage Forms, Marcel Decker, New York, N.Y., 1980; and Pharmaceutical Dosage Forms and Drug Delivery Systems, Seventh Ed. (Lippincott Williams & Wilkins 1999), each of which is incorporated by reference in its entirety.

(274) A compound described herein can be conveniently formulated into pharmaceutical compositions composed of one or more pharmaceutically-acceptable carriers. See e.g., Remington's Pharmaceutical Sciences, latest edition, by E.W. Martin Mack Pub. Co., Easton, PA, incorporated by reference in its entirety, which discloses typical carriers and conventional methods of preparing pharmaceutical compositions. Such carriers can be carriers for administration of compositions to humans and non-humans, including solutions such as sterile water, saline, and buffered solutions at physiological pH. Pharmaceutical compositions can also include one or more additional active ingredients such as antimicrobial agents, anti-inflammatory agents, and anesthetics.

(275) Non-limiting examples of pharmaceutically-acceptable carriers include saline, Ringer's solution, and dextrose solution. In some embodiments, the pH of the solution can be from about 5 to about 8, and can be from about 7 to about 7.5. Further carriers include sustained release preparations such as semipermeable matrices of solid hydrophobic polymers containing the compound. The matrices can be in the form of shaped articles, for example, films, liposomes, microparticles, or microcapsules.

(276) Compositions suitable for topical administration can be used. In some embodiments, compositions of the disclosure can comprise a liquid comprising an active agent in solution, in suspension, or both. Liquid compositions can include gels. A liquid composition can be, for example, aqueous. A composition is an in situ gellable aqueous composition. In iteration, the composition is an in situ gellable aqueous solution. Such a composition can comprise a gelling agent in a concentration effective to promote gelling upon contact with the eye or lacrimal fluid in the exterior of the eye. Aqueous compositions can have ophthalmically-compatible pH and osmolality. The composition can comprise an ophthalmic depot formulation comprising an active agent for subconjunctival administration. Microparticles comprising an active agent can be embedded in a biocompatible, pharmaceutically-acceptable polymer or a lipid encapsulating agent. The depot formulations can be adapted to release all or substantially all the active material over an extended period of time. The polymer or lipid matrix, if present, can be adapted to degrade sufficiently to be transported from the site of administration after release of all or substantially all the active agent. The depot formulation can be a liquid formulation, comprising a pharmaceutical acceptable polymer and a dissolved or dispersed active agent. Upon injection, the polymer forms a depot at the injections site, for example, by gelifying or precipitating. The composition can comprise a solid article that can be inserted in a suitable location in the eye, such as between the eye and eyelid or in the conjunctival sac, where the article releases the active agent. Solid articles suitable for implantation in the eye in such fashion can comprise polymers and can be bioerodible or non-bioerodible.

(277) Pharmaceutical formulations can include additional carriers, as well as thickeners, diluents, buffers, preservatives, and surface active agents in addition to the agents disclosed herein.

(278) The pH of the disclosed composition can range from about 3 to about 12. The pH of the composition can be, for example, from about 3 to about 4, from about 4 to about 5, from about 5 to about 6, from about 6 to about 7, from about 7 to about 8, from about 8 to about 9, from about 9 to about 10, from about 10 to about 11, or from about 11 to about 12 pH units. The pH of the composition can be, for example, about 3, about 4, about 5, about 6, about 7, about 8, about 9, about 10, about 11, or about 12 pH units. The pH of the composition can be, for example, at least 3,

at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 11 or at least 12 pH units. The pH of the composition can be, for example, at most 3, at most 4, at most 5, at most 6, at most 7, at most 8, at most 9, at most 10, at most 11, or at most 12 pH units. A pharmaceutical formulation disclosed herein can have a pH of from about 5.5 to about 6.5. For example, a formulation of the present disclosure can have a pH of about 5.5, about 5.6, about 5.7, about 5.8, about 5.9, about 6.0, about 6.1, about 6.2, about 6.3, about 6.4, or about 6.5. In some embodiments, the pH is 6.2 ± 0.3 , 6.2 ± 0.2 , 6.2 ± 0.1 , about 6.2, or 6.2.

(279) If the pH is outside the range desired by the formulator, the pH can be adjusted by using sufficient pharmaceutically-acceptable acids and bases.

(280) Depending on the intended mode of administration, the pharmaceutical compositions can be in the form of solid, semi-solid or liquid dosage forms, such as, for example, tablets, suppositories, pills, capsules, powders, liquids, suspensions, lotions, creams, or gels, for example, in unit dosage form suitable for single administration of a precise dosage.

(281) For solid compositions, nontoxic solid carriers include, for example, pharmaceutical grades of mannitol, lactose, starch, magnesium stearate, sodium saccharin, talc, cellulose, glucose, sucrose, and magnesium carbonate.

(282) Non-limiting examples of pharmaceutically active agents suitable for combination with compositions of the disclosure include anti-infectives, i.e., aminoglycosides, antiviral agents, antimicrobials, anti-cholinergics/anti-spasmodics, antidiabetic agents, antihypertensive agents, anti-neoplastics, cardiovascular agents, central nervous system agents, coagulation modifiers, hormones, immunologic agents, immunosuppressive agents, and ophthalmic preparations.

(283) In some embodiments, the pharmaceutical composition provided herein comprises a therapeutically effective amount of a compound in admixture with a pharmaceutically-acceptable carrier and/or excipient, for example, saline, phosphate buffered saline, phosphate and amino acids, polymers, polyols, sugar, buffers, preservatives, and other proteins. Illustrative agents include octylphenoxy polyethoxy ethanol compounds, polyethylene glycol monostearate compounds, polyoxyethylene sorbitan fatty acid esters, sucrose, fructose, dextrose, maltose, glucose, mannitol, dextran, sorbitol, inositol, galactitol, xylitol, lactose, trehalose, bovine or human serum albumin, citrate, acetate, Ringer's and Hank's solutions, cysteine, arginine, carnitine, alanine, glycine, lysine, valine, leucine, polyvinylpyrrolidone, polyethylene, and glycol.

(284) In some embodiments, a pharmaceutical formulation disclosed herein can comprise: (i) a compound or antibody disclosed herein; (ii) a buffer; (iii) a non-ionic detergent; (iv) a tonicity agent; and (v) a stabilizer. In some embodiments, the pharmaceutical formulation disclosed herein is a stable liquid pharmaceutical formulation.

(285) In some embodiments, an ophthalmic formulation disclosed herein can comprise: (i) a compound or antibody disclosed herein; (ii) a buffer; (iii) a non-ionic detergent; (iv) a tonicity agent; and (v) a stabilizer. In some embodiments, the ophthalmic formulation disclosed herein is a stable liquid pharmaceutical formulation or a stable liquid ophthalmic formulation.

(286) In some embodiments, a pharmaceutical formulation or ophthalmic formulation disclosed herein is a liquid formulation that can comprise about 5 mg/mL to about 150 mg/mL of antibody or compound, about 7.5 mg/mL to about 140 mg/mL of antibody or compound, about 10 mg/mL to about 130 mg/mL of antibody or compound, about 10 mg/mL to about 100 mg/mL of antibody or compound, about 20 mg/mL to about 80 mg/mL of antibody or compound, or about 30 mg/mL to about 70 mg/mL of antibody or compound. For example, a formulation of the present disclosure can comprise about 5 mg/mL, about 10 mg/mL, about 15 mg/mL, about 20 mg/mL, about 25 mg/mL, about 30 mg/mL, about 35 mg/mL, about 40 mg/mL, about 50 mg/mL, about 60 mg/mL, about 70 mg/mL, about 80 mg/mL, about 90 mg/mL, about 100 mg/mL, about 120 mg/mL, about 140 mg/mL, or about 150 mg/mL of a compound, antibody, or antigen-binding fragment thereof described herein.

(287) In some embodiments, a pharmaceutical formulation or ophthalmic formulation disclosed

herein can comprise a buffer. In some embodiments, the buffer serves to maintain a stable pH and to help stabilize a compound or antibody disclosed herein. In some embodiments, the buffer or buffer system comprises at least one buffer that has a buffering range that overlaps fully or in part the range of pH 5.5-7.4. In some embodiments, the buffer has a pKa of about 6.2 ± 0.5 . In some embodiments, the buffer comprises a sodium phosphate buffer. In some embodiments, the sodium phosphate is present at a concentration of about 5 mM to about 15 mM, about 6 mM to about 14 mM, about 7 mM to about 13 mM, about 8 mM to about 12 mM, about 9 mM to about 11 mM, or about 10 mM. In certain embodiments, the buffer system comprises sodium phosphate at 10 mM, at a pH of 6.2 ± 0.3 or 6.1 ± 0.3 .

(288) In some embodiments, a pharmaceutical formulation or ophthalmic formulation disclosed herein can comprise a non-ionic detergent. In some embodiments, the non-ionic detergent is a nonionic polymer containing a polyoxyethylene moiety. In some embodiments, the non-ionic detergent is any one or more of polysorbate 20, poloxamer 188 or polyethylene glycol 3350. In some embodiments, the non-ionic detergent is polysorbate 20. In some embodiments, the non-ionic detergent is polysorbate 80. In some embodiments, a pharmaceutical formulation or ophthalmic formulation disclosed herein can contain about 0.01% to about 1% non-ionic detergent. For example, a formulation of the present disclosure can comprise about 0.0085%, about 0.01%, about 0.02%, about 0.03%, about 0.04%, about 0.05%, about 0.06%, about 0.07%, about 0.08%, about 0.09%, about 0.1%, about 0.11%, about 0.12%, about 0.13%, about 0.14%, about 0.15%, about 0.16%, about 0.17%, about 0.18%, about 0.19%, about 0.20%, about 0.21%, about 0.22%, about 0.23%, about 0.24%, about 0.25%, about 0.3%, about 0.4%, about 0.5%, about 0.6%, about 0.7%, about 0.8%, about 0.9%, about 1%, about 1.1%, about 1.15%, about 1.2%, about 1.25%, about 1.3%, about 1.35%, about 1.4%, about 1.45%, about 1.5%, about 1.55%, about 1.6%, about 1.65%, about 1.7%, about 1.75%, about 1.8%, about 1.85%, about 1.9%, about 1.95%, or about 2% polysorbate 20, polysorbate 80 or poloxamer 188.

(289) In some embodiments, a pharmaceutical formulation or ophthalmic formulation disclosed herein can comprise a tonicity agent. In some embodiments, the tonicity agent is sodium chloride or potassium chloride. In some embodiments, the tonicity agent is sodium chloride. In some embodiments, the sodium chloride is present at a concentration of about 5 mM to about 100 mM, about 10 mM to about 50 mM, or about 40 mM.

(290) In some embodiments, a pharmaceutical formulation or ophthalmic formulation disclosed herein can comprise a stabilizer. In some embodiments, the stabilizer is a thermal stabilizer that can stabilize an antibody or compound disclosed herein under conditions of thermal stress. In some embodiments, the stabilizer maintains greater than about 93% of the compound or antibody in a native conformation when the solution containing the compound or antibody and the thermal stabilizer is kept at about 45° C. for up to about 28 days. In some embodiments, the stabilizer prevents aggregation of the compound or antibody and less than 4% of the compound or antibody is aggregated when the solution containing the compound or antibody and the thermal stabilizer is kept at about 45° C. for up to about 28 days. In some embodiments, the stabilizer maintains greater than about 96% of the compound or antibody in a native conformation when the solution containing the compound or antibody and the thermal stabilizer is kept at about 37° C. for up to about 28 days. In some embodiments, the stabilizer prevents aggregation of the compound or antibody and less than about 2% of the compound or antibody is aggregated when the solution containing the compound or antibody and the thermal stabilizer is kept at about 37° C. for up to about 28 days.

(291) In some embodiments, the thermal stabilizer is a sugar or sugar alcohol, for example, sucrose, sorbitol, glycerol, trehalose, or mannitol, or any combination thereof. In some embodiments, the stabilizer is a sugar. In some embodiments, the sugar is sucrose, mannitol or trehalose. In some embodiments, the stabilizer is sucrose. In some embodiments, a pharmaceutical formulation or ophthalmic formulation disclosed herein can comprise about 1% to about 20% sugar

or sugar alcohol, about 2% to about 18% sugar or sugar alcohol, about 3% to about 15% sugar or sugar alcohol, about 4% to about 10% sugar or sugar alcohol, or about 5% sugar or sugar alcohol. For example, a pharmaceutical formulation or ophthalmic formulation of the present disclosure can comprise about 4%, about 5%, about 6%, about 7%, about 8%, about 9%, about 10%, about 11%, about 12%, about 13%, or about 14% sugar or sugar alcohol (e.g., sucrose, trehalose or mannitol). In some embodiments, the stabilizer is at a concentration of from about 1% w/v to about 20% w/v. In some embodiments, the stabilizer is sucrose at a concentration of from about 1% w/v to about 15% w/v, or from about 1% w/v to about 10% w/v. In some embodiments, the stabilizer is sucrose at a concentration of 5% w/v or about 5% w/v. In some embodiments, the stabilizer is sucrose at a concentration of 7.5% w/v or about 7.5% w/v. In some embodiments, the stabilizer is sucrose at a concentration of 10% w/v or about 10% w/v. In some embodiments, the stabilizer is sucrose at a concentration of 12.5% w/v or about 12.5% w/v. In some embodiments, the stabilizer is sucrose at a concentration of 15% w/v or about 15% w/v. In some embodiments, the stabilizer is sucrose at a concentration of 20% w/v or about 20% w/v.

(292) Administration of Pharmaceutical Compositions

(293) A pharmaceutical composition disclosed herein can be administered in a therapeutically-effective amount by various forms and routes including, for example, oral, topical, parenteral, intravenous injection, intravenous infusion, subcutaneous injection, subcutaneous infusion, intramuscular injection, intramuscular infusion, intradermal injection, intradermal infusion, intraperitoneal injection, intraperitoneal infusion, intracerebral injection, intracerebral infusion, subarachnoid injection, subarachnoid infusion, intraocular injection, intraspinal injection, intrasternal injection, ophthalmic administration, endothelial administration, local administration, intranasal administration, intrapulmonary administration, rectal administration, intraarterial administration, intrathecal administration, inhalation, intralesional administration, intradermal administration, epidural administration, absorption through epithelial or mucocutaneous linings (e.g., oral mucosa, rectal and intestinal mucosa), intracapsular administration, subcapsular administration, intracardiac administration, transtracheal administration, subcuticular administration, subarachnoid administration, subcapsular administration, intraspinal administration, or intrasternal administration.

(294) A pharmaceutical composition can be administered in a local manner, for example, via injection of the compound directly into an organ, optionally in a depot or sustained release formulation or implant. A pharmaceutical composition can be provided in the form of a rapid release formulation, in the form of an extended release formulation, or in the form of an intermediate release formulation. A rapid release form can provide an immediate release. An extended release formulation can provide a controlled release or a sustained delayed release.

(295) In some embodiments, a pump can be used for delivery of the pharmaceutical composition. In some embodiments, a pen delivery device can be used, for example, for subcutaneous delivery of a composition of the disclosure. Such a pen delivery device can be reusable or disposable. A reusable pen delivery device can use a replaceable cartridge that contains a pharmaceutical composition disclosed herein. Once all of the pharmaceutical composition within the cartridge has been administered and the cartridge is empty, the empty cartridge can readily be discarded and replaced with a new cartridge that contains the pharmaceutical composition. The pen delivery device can then be reused. A disposable pen has no replaceable cartridge. Rather, the disposable pen delivery device comes prefilled with the pharmaceutical composition held in a reservoir within the device. Once the reservoir is emptied of the pharmaceutical composition, the entire device is discarded.

(296) A pharmaceutical composition disclosed herein can be administered, for example, to the eye via any suitable form or route including, for example, topical, oral, systemic, intravitreal, intracameral, intracanalicular, subconjunctival, subtenon, retrobulbar, intraocular, intrastromal, intracorneal, posterior juxtasclear, periocular, subretinal, or suprachoroidal administration. The

delivery method can include an invasive method for direct delivery of the composition to ocular cells. In some embodiments, a liquid pharmaceutical composition comprising an antibody or compound can be delivered via a subretinal injection, intravitreal injection (e.g., front, mid or back vitreal injection), intravitreal implant, intraorbital injection, intraorbital administration, subcutaneous injection, intracameral injection, intracranial injection, subconjunctival injection, subconjunctival implant, injection into the anterior chamber via the temporal limbus, intrastromal injection, intracorneal injection, aqueous humor injection, subtenon injection, or subtenon implant. The compositions can be administered by injecting the formulation in any part of the eye including anterior chamber, posterior chamber, vitreous chamber (intravitreal), retina proper, and/or subretinal space.

(297) A pharmaceutical composition disclosed herein can be delivered via a non-invasive method. Examples of non-invasive modes of administering the formulation can include using a needleless injection device, and topical administration, for example, eye drops to the cornea. Multiple administration routes can be employed for efficient delivery of the pharmaceutical composition. In some embodiments, the composition is delivered via multiple administration routes, for example, subretinal and intravitreal, to increase the efficiency of antibody delivery. In some embodiments, the subretinal and/or intravitreal injection is preceded by a vitrectomy.

(298) In some embodiments, a liquid formulation comprising from 10 mg/mL to 120 mg/mL of antibody or compound is in a prefilled syringe and is administered intravitreally in a volume of up to about 500 μ L. In some embodiments, a liquid formulation comprising from 10 mg/mL to 120 mg/mL of antibody or compound is in a prefilled syringe and is administered intravitreally in a volume of up to about 100 μ L. In some embodiments, a liquid formulation comprising from 10 mg/mL to 120 mg/mL of antibody or compound is in a prefilled syringe and is administered intravitreally in a volume of about 50 μ L.

(299) A pharmaceutical composition disclosed herein can be targeted to any suitable ocular cell including, for example, endothelial cells such as vascular endothelial cells, cells of the retina such as retinal pigment epithelium (RPE), corneal cells, fibroblasts, astrocytes, glial cells, pericytes, iris epithelial cells, cells of neural origin, ciliary epithelial cells, Müller cells, muscle cells surrounding and attached to the eye such as cells of the lateral rectus muscle, orbital fat cells, cells of the sclera and episclera, cells of the trabecular meshwork, or connective tissue cells.

(300) Dosing

(301) A compound, antibody, or therapeutic agent described herein can be administered before, during, or after the occurrence of a disease or condition, and the timing of administering a composition containing the compound, antibody, or therapeutic agent can vary. For example, the composition can be used as a prophylactic and can be administered continuously to subjects with a propensity to conditions or diseases in order to lessen a likelihood of the occurrence of the disease or condition. The composition can be administered to a subject already suffering from a disease or condition, in an amount sufficient to cure or at least partially arrest the symptoms of the disease or condition, or to cure, heal, improve, or ameliorate the condition. The composition can be administered to a subject during or as soon as possible after the onset of the symptoms. The administration of the compound, antibody, or therapeutic agent can be initiated within the first 48 hours of the onset of the symptoms, within the first 24 hours of the onset of the symptoms, within the first 6 hours of the onset of the symptoms, or within 3 hours of the onset of the symptoms. The initial administration can be via any practical route, such as by any route described herein using any formulation described herein. The compound, antibody, or therapeutic agent can be administered as soon as is practicable after the onset of a disease or condition is detected or suspected, and for a length of time necessary for the treatment of the disease, such as, for example, from about 1 month to about 3 months. The length of treatment can vary for each subject. Amounts effective for this use can vary based on the severity and course of the disease or condition, previous therapy, the subject's health status, weight, and response to the drugs, and the judgment of the

treating physician. Improvement of clinical symptoms can be monitored, for example, by indirect ophthalmoscopy, fundus photography, fluorescein angiography, electroretinography, external eye examination, slit lamp biomicroscopy, applanation tonometry, pachymetry, optical coherence tomography, or autorefractometry.

(302) A pharmaceutical composition described herein can be in a unit dosage form suitable for a single administration of a precise dosage. In unit dosage form, the formulation can be divided into unit doses containing appropriate quantities of one or more compounds, antibodies or therapeutic agents. The unit dosage can be in the form of a package containing discrete quantities of the formulation. Non-limiting examples are packaged injectables, vials, and ampoules. An aqueous suspension composition disclosed herein can be packaged in a single-dose non-reclosable container. Multiple-dose reclosable containers can be used, for example, in combination with or without a preservative. A formulation for injection disclosed herein can be present in a unit dosage form, for example, in ampoules, or in multi-dose containers with a preservative.

(303) Multiple compounds, antibodies or therapeutic agents disclosed herein can be administered in any order or simultaneously. If simultaneously, the multiple compounds, antibodies or therapeutic agents can be provided in a single, unified form, or in multiple forms, for example, as multiple separate injections or infusions. The compounds, antibodies or therapeutic agents can be packed together or separately, in a single package or in a plurality of packages. One or all of the compounds, antibodies or therapeutic agents can be given in multiple doses. If not simultaneous, the timing between the multiple doses can vary to as much as about a month.

(304) An intraocular injection can be performed between any interval of time to improve efficiency of delivery and/or to minimize or avoid damage to surrounding tissue. The interval of time between two or more intraocular injections can be from, for example, about 1 minute to about 60 minutes, about 1 minute to about 5 minutes, about 5 minutes to about 10 minutes, about 10 minutes to about 15 minutes, about 15 minutes to about 20 minutes, about 20 minutes to about 25 minutes, about 25 minutes to about 30 minutes, about 30 minutes to about 35 minutes, about 35 minutes to about 40 minutes, about 40 minutes to about 45 minutes, about 45 minutes to about 50 minutes, about 50 minutes to about 55 minutes, or about 55 minutes to about 60 minutes.

(305) An intraocular injection can be performed at any rate. The rate of intraocular injection can be from, for example, about 1 $\mu\text{L}/\text{sec}$ to about 500 $\mu\text{L}/\text{sec}$, about 1 $\mu\text{L}/\text{sec}$ to about 10 $\mu\text{L}/\text{sec}$, about 10 $\mu\text{L}/\text{sec}$ to about 20 $\mu\text{L}/\text{sec}$, about 20 $\mu\text{L}/\text{sec}$ to about 30 $\mu\text{L}/\text{sec}$, about 30 $\mu\text{L}/\text{sec}$ to about 40 $\mu\text{L}/\text{sec}$, about 40 $\mu\text{L}/\text{sec}$ to about 50 $\mu\text{L}/\text{sec}$, about 50 $\mu\text{L}/\text{sec}$ to about 60 $\mu\text{L}/\text{sec}$, about 60 $\mu\text{L}/\text{sec}$ to about 70 $\mu\text{L}/\text{sec}$, about 70 $\mu\text{L}/\text{sec}$ to about 80 $\mu\text{L}/\text{sec}$, about 80 $\mu\text{L}/\text{sec}$ to about 90 $\mu\text{L}/\text{sec}$, about 90 $\mu\text{L}/\text{sec}$ to about 100 $\mu\text{L}/\text{sec}$, about 100 $\mu\text{L}/\text{sec}$ to about 110 $\mu\text{L}/\text{sec}$, about 110 $\mu\text{L}/\text{sec}$ to about 120 $\mu\text{L}/\text{sec}$, about 120 $\mu\text{L}/\text{sec}$ to about 130 $\mu\text{L}/\text{sec}$, about 130 $\mu\text{L}/\text{sec}$ to about 140 $\mu\text{L}/\text{sec}$, about 140 $\mu\text{L}/\text{sec}$ to about 150 $\mu\text{L}/\text{sec}$, about 150 $\mu\text{L}/\text{sec}$ to about 160 $\mu\text{L}/\text{sec}$, about 160 $\mu\text{L}/\text{sec}$ to about 170 $\mu\text{L}/\text{sec}$, about 170 $\mu\text{L}/\text{sec}$ to about 180 $\mu\text{L}/\text{sec}$, about 180 $\mu\text{L}/\text{sec}$ to about 190 $\mu\text{L}/\text{sec}$, about 190 $\mu\text{L}/\text{sec}$ to about 200 $\mu\text{L}/\text{sec}$, about 200 $\mu\text{L}/\text{sec}$ to about 300 $\mu\text{L}/\text{sec}$, about 300 $\mu\text{L}/\text{sec}$ to about 400 $\mu\text{L}/\text{sec}$, or about 400 $\mu\text{L}/\text{sec}$ to about 500 $\mu\text{L}/\text{sec}$.

(306) A compound, antibody, or therapeutic agent disclosed herein can be administered at a dosage of about 0.0001 mg/kg to about 1000 mg/kg, about 0.001 mg/kg to about 100 mg/kg, about 0.01 mg/kg to about 100 mg/kg, about 0.01 mg/kg to about 20 mg/kg, about 0.02 mg/kg to about 7 mg/kg, about 0.03 mg/kg to about 5 mg/kg, about 0.05 mg/kg to about 3 mg/kg, about 0.1 mg/kg to about 50 mg/kg, about 0.1 mg/kg to about 0.5 mg/kg, about 0.2 mg/kg to about 0.6 mg/kg, about 0.3 mg/kg to about 0.7 mg/kg, about 0.4 mg/kg to about 0.8 mg/kg, about 0.1 mg/kg to about 0.9 mg/kg, about 0.01 mg/kg to about 50 mg/kg, about 0.1 mg/kg to about 10 mg/kg, about 1 mg/kg to about 10 mg/kg, about 5 mg/kg to about 10 mg/kg, about 1 mg/kg to about 5 mg/kg, or about 3 mg/kg to about 7 mg/kg by mass of the subject.

(307) A compound, antibody, or therapeutic agent described herein can be administered at any interval desired. The administration of the compound, antibody, or therapeutic agent can have

regular or irregular dosing schedules to accommodate either the person administering the compound, antibody, or therapeutic agent or the subject receiving the compound, antibody, or therapeutic agent. For example, the compound, antibody, or therapeutic agent can be administered twice a day, once a day, five times a week, four times a week, three times a week, two times a week, once a week, once every two weeks, once every three weeks, once every four weeks, once a month, once every five weeks, once every six weeks, once every eight weeks, once every two months, once every twelve weeks, once every three months, once every four months, once every six months, once a year, or less frequently. In some embodiments, administration is every other week.

(308) The amount administered can be of the same amount in each dose or the dosage can vary between doses. For example, a first amount can be administered in the morning and a second amount can be administered in the evening.

(309) A compound, antibody, or therapeutic agent described herein can be administered in any amount necessary or convenient. For example, a compound described herein can be administered in an amount from about 0.05 mg to about 300 mg, about 0.1 mg to about 300 mg, about 0.1 mg to about 200 mg, about 0.1 mg to about 100 mg, about 0.05 mg to about 1.5 mg, about 0.1 mg to about 1.5 mg, about 0.05 mg to about 1 mg, about 1 mg to about 1.5 mg, about 0.5 mg to about 6 mg, about 1 mg to about 4 mg, about 2 mg to about 10 mg, about 10 mg to about 30 mg, about 30 mg to about 50 mg, about 50 mg to about 70 mg, about 70 mg to about 100 mg, or about 0.1 mg to about 1 mg, about 0.05 mg, about 0.06 mg, about 0.07 mg, about 0.08 mg, about 0.09 mg, about 0.1 mg, about 0.11 mg, about 0.12 mg, about 0.13 mg, about 0.14 mg, about 0.15 mg, about 0.16 mg, about 0.17 mg, about 0.18 mg, about 0.19 mg, about 0.2 mg, about 0.21 mg, about 0.22 mg, about 0.23 mg, about 0.24 mg, about 0.25 mg, about 0.26 mg, about 0.27 mg, about 0.28 mg, about 0.29 mg, about 0.3 mg, about 0.31 mg, about 0.32 mg, about 0.33 mg, about 0.34 mg, about 0.35 mg, about 0.36 mg, about 0.37 mg, about 0.38 mg, about 0.39 mg, about 0.4 mg, about 0.41 mg, about 0.42 mg, about 0.43 mg, about 0.44 mg, about 0.45 mg, about 0.46 mg, about 0.47 mg, about 0.48 mg, about 0.49 mg, about 0.5 mg, about 0.51 mg, about 0.52 mg, about 0.53 mg, about 0.54 mg, about 0.55 mg, about 0.56 mg, about 0.57 mg, about 0.58 mg, about 0.59 mg, about 0.6 mg, about 0.61 mg, about 0.62 mg, about 0.63 mg, about 0.64 mg, about 0.65 mg, about 0.66 mg, about 0.67 mg, about 0.68 mg, about 0.69 mg, about 0.7 mg, about 0.71 mg, about 0.72 mg, about 0.73 mg, about 0.74 mg, about 0.75 mg, about 0.76 mg, about 0.77 mg, about 0.78 mg, about 0.79 mg, about 0.8 mg, about 0.81 mg, about 0.82 mg, about 0.83 mg, about 0.84 mg, about 0.85 mg, about 0.86 mg, about 0.87 mg, about 0.88 mg, about 0.89 mg, about 0.9 mg, about 0.91 mg, about 0.92 mg, about 0.93 mg, about 0.94 mg, about 0.95 mg, about 0.96 mg, about 0.97 mg, about 0.98 mg, about 0.99 mg, about 1 mg, about 1.5 mg, about 2 mg, about 2.5 mg, about 3 mg, about 3.5 mg, about 4 mg, about 4.5 mg, about 5 mg, about 5.5 mg, about 6 mg, about 6.5 mg, about 7 mg, about 7.5 mg, about 8 mg, about 8.5 mg, about 9 mg, about 9.5 mg, about 10 mg, about 15 mg, about 20 mg, about 25 mg, about 30 mg, about 35 mg, about 40 mg, about 45 mg, about 50 mg, about 55 mg, about 60 mg, about 65 mg, about 70 mg, about 75 mg, about 80 mg, about 85 mg, about 90 mg, about 95 mg, about 100 mg, about 110 mg, about 120 mg, about 130 mg, about 140 mg, about 150 mg, about 160 mg, about 170 mg, about 180 mg, about 190 mg, about 200 mg, about 210 mg, about 220 mg, about 230 mg, about 240 mg, about 250 mg, about 260 mg, about 270 mg, about 280 mg, about 290 mg, or about 300 mg per dose for a subject by any route of administration.

(310) Combination Therapies

(311) A pharmaceutical composition provided herein can be administered in conjunction with other therapies, for example, chemotherapy, radiation, surgery, anti-inflammatory agents, or vitamins. The other agents can be administered prior to, after, or concomitantly with the pharmaceutical composition.

(312) In some embodiments, a compound or antibody described herein can be used singly or in

combination with one or more therapeutic agents as a component of mixtures.

(313) In some embodiments, the disclosure provides co-administration of a multispecific compound or antibody targeting, for example, HPTP- β (VE-PTP) and VEGF, with one or more additional anti-VEGF agents, which can stabilize the vasculature against neovascularization. In some embodiments, co-administration of the multispecific compound or antibody with one or more additional anti-VEGF agents can stabilize the vasculature against leakage. An anti-VEGF agent can be a compound, a recombinant protein, an antibody, an antigen-binding fragment, variant, or derivative thereof (e.g., a scFv), a protein comprising one or more designed ankyrin repeats, a designed ankyrin repeat protein (DARPin), an ankyrin protein, an ankyrin repeat protein, an affibody, an avimer, an adnectin, an anticalin, a Fynomer, a Kunitz domain, a knottin, a β -hairpin mimetic, or a peptide derived from one or more receptors, e.g. VEGF receptors, or the VEGF-binding portions of human VEGF receptors 1 and 2.

(314) Non-limiting examples of anti-VEGF agents include bevacizumab (Avastin®), ranibizumab (Lucentis®), aflibercept (Eylea®), conbercept, brotuzumab, RTH258, VEGF receptor tyrosine kinase inhibitors such as sorafenib, sunitinib, axitinib, pazopanib, vandetanib, cabozantinib, regorafenib, and 4-(4-bromo-2-fluoroanilino)-6-methoxy-7-(1-methylpiperidin-4-ylmethoxy)quinazoline (ZD6474), VEGF variants, soluble VEGF receptor fragments or traps, aptamers capable of blocking VEGF or VEGFR (e.g. Pegaptanib), neutralizing anti-VEGFR antibodies or fragments thereof (e.g., ramucirumab, p1C11, 1121, 1121B), anti-KDR antibodies, anti-flt1 antibodies, low molecular weight inhibitors of VEGFR tyrosine kinases, DARPins that bind VEGF (e.g., abicipar, MP0112, MP0250), proteins comprising one or more designed ankyrin repeats that bind VEGF, adnectins (e.g., CT-322), anticalins (e.g., PRS-050), collagen IV-derived biomimetic peptides (e.g., AXT-107)

(315) Further non-limiting examples of VEGF-modulating agents include anti-inflammatory agents, for example, dexamethasone, fluocinolone, and triamcinolone. The multispecific compound or antibody targeting, for example, HPTP- β (VE-PTP) and VEGF, can be administered in combination with any additional anti-VEGF agent in any combination, for example, at the beginning of treatment, at any time during treatment, or at any time after treatment with the additional anti-VEGF agent has concluded. In addition, the dosage of the multispecific compound or antibody can be adjusted during treatment. Also, the dosage of the additional anti-VEGF agent can be adjusted during treatment. The multispecific compound or antibody can be administered, for example, monthly, once every 3 months, once every 6 months, or yearly, wherein the additional anti-VEGF agent is administered at any frequency between treatments. Also disclosed herein are methods for treating a disease or condition as disclosed herein. The method comprises administering to a subject: a) a therapeutically-effective amount of a multispecific compound or antibody targeting, for example, HPTP- β (VE-PTP) and VEGF; and b) a therapeutically-effective amount of an additional anti-VEGF agent;

wherein the administration of the multispecific compound or antibody and the additional anti-VEGF agent can be conducted as described herein.

(316) In some embodiments, the disclosure provides co-administration of a multispecific compound or antibody targeting, for example, HPTP- β (VE-PTP) and VEGF, with one or more additional anti-HPTP- β (VE-PTP) agents, which can stabilize the vasculature against neovascularization. In some embodiments, co-administration of the multispecific compound or antibody with one or more additional anti-HPTP- β (VE-PTP) agents can stabilize the vasculature against leakage. An anti-HPTP- β (VE-PTP) agent can be a compound, a recombinant protein, an antibody, an antigen-binding fragment, variant, or derivative thereof (e.g., a scFv), a protein comprising one or more designed ankyrin repeats, a designed ankyrin repeat protein (DARPin), an ankyrin protein, an ankyrin repeat protein, an affibody, an avimer, an adnectin, an anticalin, a Fynomer, a Kunitz domain, a knottin, a β -hairpin mimetic, or a peptide derived from one or more receptors. In some embodiments, the additional anti-HPTP- β (VE-PTP) agent(s) can activate Tie2

signaling by promoting protein phosphorylation, such as phosphorylation of the Tie2 protein. In some embodiments, the additional anti-HPTP- β (VE-PTP) agent(s) can bind to HPTP- β (VE-PTP). (317) The multispecific compound or antibody targeting, for example, HPTP- β (VE-PTP) and VEGF, can be administered in combination with any additional anti-HPTP- β (VE-PTP) agent in any combination, for example, at the beginning of treatment, at any time during treatment, or at any time after treatment with the additional anti-HPTP- β (VE-PTP) agent has concluded. The dosage of the multispecific compound or antibody can be adjusted during treatment. The dosage of the additional anti-HPTP- β (VE-PTP) agent can be adjusted during treatment. The multispecific compound or antibody can be administered, for example, monthly, once every 3 months, once every 6 months, or yearly, wherein the additional anti-HPTP- β (VE-PTP) agent is administered at any frequency between treatments. Also disclosed herein are methods for treating a disease or condition as disclosed herein. The method comprises administering to a subject: a) a therapeutically-effective amount of a multispecific compound or antibody targeting, for example, HPTP- β (VE-PTP) and VEGF; and b) a therapeutically-effective amount of an additional anti-HPTP- β (VE-PTP) agent;

wherein the administration of the multispecific compound or antibody and the additional anti-HPTP- β (VE-PTP) agent can be conducted as described herein.

(318) In some embodiments, the disclosure provides co-administration of a multispecific compound or antibody targeting, for example, HPTP- β (VE-PTP) and VEGF, with one or more additional Tie2 receptor activating compounds. An additional Tie2 receptor activating compound can be, for example, an angiopoietin 1 recombinant protein, an Ang1 mimetic, a Tie2 agonist, a peptide, a HPTP- β (VE-PTP) phosphatase inhibitor, a Tie2-peptomimetic, a tetrameric polyethylene oxide clustered peptide, a collagen IV-biomimetic peptide, a compound, a recombinant protein, an antibody, an antigen-binding fragment, variant, or derivative thereof (e.g., an scFv), an affibody, an avimer, an adnectin, a protein comprising one or more designed ankyrin repeats, a designed ankyrin repeat protein (DARPin), an ankyrin protein, an ankyrin repeat protein, an affibody, an avimer, an adnectin, an anticalin, a Fynomer, a Kunitz domain, a knottin, a β -hairpin mimetic, or a peptide derived from one or more receptors. In some embodiments, the one or more additional Tie2 receptor activating compounds are small molecules. In some embodiments, the one or more additional Tie2 receptor activating compounds improve drainage through ocular lymphatics, Schlemm's canal, or corneal limbal lymphatics. In some embodiments, the one or more additional Tie2 receptor activating compounds are administered as eye drops. In some embodiments, the one or more additional Tie2 receptor activating compounds are administered to treat primary open angle glaucoma, age-related macular degeneration, cardiovascular disease, or cystic kidney disease. In some embodiments, the one or more additional Tie2 receptor activating compounds can be, for example, MAN-01, AXT-107, or vasculotide. In some embodiments, a compound disclosed herein can be co-administered with, for example, MAN-01, AXT-107, or vasculotide.

(319) The multispecific compound or antibody targeting, for example, HPTP- β (VE-PTP) and VEGF, can be administered in combination with any additional Tie2 receptor activating compound in any combination, for example, at the beginning of treatment, at any time during treatment, or at any time after treatment with the additional Tie2 receptor activating compound has concluded. The dosage of the multispecific compound or antibody can be adjusted during treatment. The dosage of the additional Tie2 receptor activating compound can be adjusted during treatment. The multispecific compound or antibody can be administered, for example, monthly, once every 3 months, once every 6 months, or yearly, wherein the additional Tie2 receptor activating compound is administered at any frequency between treatments with the multispecific compound or antibody. Also disclosed herein are methods for treating a disease or condition as disclosed herein. The method comprises administering to a subject: a) a therapeutically-effective amount of a multispecific compound or antibody targeting, for example, HPTP- β (VE-PTP) and VEGF; and b)

a therapeutically-effective amount of a Tie2 activator;

wherein the administration of the multispecific compound or antibody and the additional Tie2 receptor activating compound can be conducted as described herein.

Mouse Models

Oxygen-Induced Ischemic Retinopathy Model

(320) The oxygen-induced ischemic retinopathy model can be considered to mimic aspects of proliferative retinal neovascularization and proliferative diabetic retinopathy. One week old mice can be placed in an airtight chamber and exposed to hyperoxia ($75\pm 3\%$ oxygen) for five days, resulting in hyperoxia-induced neovascularization, for example, at the junction between the vascularized and avascular retina between postnatal day 17 and 21. Mice can be dosed with a compound of interest, for example, an antibody or compound of the disclosure, to determine the effect on neovascularization and/or vascular leakage. Neovascularization and/or vascular leakage can be assessed as described below.

(321) Rho/VEGF Mouse Model

(322) The Rho/VEGF mouse model can mimic aspects of neovascular age-related macular degeneration. Transgenic mice with vascular endothelial growth factor (VEGF) expression driven by the rhodopsin promoter (rho/VEGF mice) can develop retinal neovascularization, retinal angiomatous proliferation, and retinal vascular leakage. In rho/VEGF mice, VEGF expression in photoreceptors can begin between postnatal days 5 and 10, the period when the deep capillary bed is developing. Neovascularization can originate from the deep capillary bed of the retina and grow into the subretinal space. Mice can be dosed with a compound of interest, for example, an antibody or compound of the disclosure, to determine the effect on neovascularization and/or vascular leakage. Neovascularization and/or vascular leakage can be assessed as described below.

(323) Tet/Opsin/VEGF Mouse Model

(324) Mice with a VEGF under the control of a reverse tetracycline transactivator (rtTA) inducible promoter coupled to the rhodopsin promoter (Tet/opsin/VEGF mice) can be used as an inducible model of neovascularization, retinal vascular leakage, and retinal detachment. In these mice, VEGF transgene expression in the retina can be induced by administering doxycycline.

Neovascularization can be evident by three to four days after VEGF induction. Neovascularization can be more extensive, and can cause outer retinal folds followed by total retinal detachment within about five days. Mice can be dosed with a compound of interest, for example, an antibody or compound of the disclosure, to determine the effect on neovascularization, vascular leakage, and/or retinal detachment. Neovascularization and/or vascular leakage can be assessed as described below. To assess retinal detachment, eyes can be frozen in cutting temperature embedding solution, ten micron sections cut through the entire eye, and sections stained with Hoechst. Sections can be examined by light microscopy, the mean length of the retinal detachment per section measured by image analysis, and the detached percentage of the retina calculated.

(325) Tet/Opsin/Ang2 Mouse Model

(326) Tet/opsin/Ang2 mice have inducible expression of Ang2 in the retina. These mice can be used to study the impact of Ang2 expression in various experimental conditions. For example, Tet/opsin/Ang2 mice can be used to determine the effect of Ang2 expression on neovascularization when VEGF levels are high versus low, or the effect of Ang2 expression in the oxygen induced ischemic retinopathy model. Mice can be dosed with a compound of interest, for example, an antibody or compound of the disclosure, to determine the impact of Ang2 expression on therapeutic efficacy.

(327) Laser-Induced Choroidal Neovascularization Model

(328) The laser-induced choroidal neovascularization model can be considered to mimic aspects of neovascular age-related macular degeneration. Anesthetized mice can have their pupils dilated, and burns can be delivered, for example, to the retina by a krypton laser using a slit lamp system and a cover glass as a contact lens. Multiple burns can be produced in a single eye, for example, burns in

three locations per eye. Burns can cause rupture of Bruch's membrane. Choroidal neovascularization can be assessed at various time points after laser treatment, for example, one week, two weeks, or four weeks after laser treatment. Mice can be dosed with a compound of interest, for example, an antibody or compound of the disclosure, to determine the effect on neovascularization. Eyecups can be stained with FITC-labeled GSA, choroids flat mounted, and the area of choroidal neovascularization at each Bruch's membrane rupture site measured with fluorescence microscopy and image analysis. Neovascularization and/or vascular leakage can also be assessed as described below.

(329) Assessment of Neovascularization

(330) Neovascularization can be assessed using a range of techniques.

(331) Fluorescein angiograms can be done by taking serial fundus photographs after injection of a dye to reveal blood vessels, allowing identification of the presence, location, and size of a neovascular complex. For example, an intraperitoneal injection of 0.3 mL of 1% fluorescein sodium can be given, serial fundus photographs taken, and the choroidal neovascularization area, total lesion area, and leakage area can be measured.

(332) Eyes can be processed for evaluation by fluorescent, light, or electron microscopy. For example, mice can be perfused with fluorescently labelled dextran, or eyes can be injected with GSA or antibodies targeting PECAM. Eyes can be processed for observation under a fluorescent microscope, and the extent of neovascularization can be quantified, for example, by quantifying the area of neovascularization per retina, by quantifying the area of neovascularization at each Bruch's membrane rupture site, or by quantifying the number of nuclei of new vessels extending from the retina into the vitreous.

(333) The area of retinal neovascularization can be determined, for example, using FITC-labeled GSA lectin and fluorescence microscopy. Eyes can be fixed in 10% formalin, dissected intact, washed with PBS, blocked in 8% swine serum, and stained with FITC-labeled GSA lectin for a time appropriate to stain retinal neovascularization and hyaloid vessels, but not normal retinal vessels (e.g., 40-50 minutes). Retinas can be flat mounted, digital images can be obtained with a fluorescent microscope and merged into a single image of the entire retina. Software can be used to measure the area of neovascularization per retina.

(334) The area of subretinal neovascularization can be determined, for example, using FITC-labeled GSA lectin and fluorescence microscopy. Eyes can be fixed in 10% formalin, and retinas can be dissected, blocked with 5% swine serum, stained with FITC-conjugated GSA for two hours to stain vascular cells, and flat mounted with the photoreceptor side up. The area of subretinal neovascularization can be measured with fluorescence microscopy and image analysis.

(335) Assessment of Retinal Vascular Leakage

(336) Retinal vascular leakage can be assessed by measuring extravasated serum albumin using an immunofluorescent technique. For example, eyes can be fixed in 10% formalin, retinas can be dissected, washed, blocked, and stained using an anti-albumin antibody and a fluorescently-conjugated secondary antibody. The vessels can be labeled by counterstaining with GSA lectin, and retinas flat mounted. Retinas can be examined by fluorescence microscopy, and the area of albumin staining determined by image analysis. Retinal vascular leakage is relevant to, for example, diabetic macular edema and macular edema due to retinal vein occlusion.

(337) Miles Assay for Vascular Leakage

(338) The Miles assay can be used to assess vascular leakage in dermal subcutaneous blood vessels. Evans blue is a dye that binds albumin. Under physiologic conditions the endothelium is impermeable to albumin, so Evans blue bound albumin remains restricted within blood vessels. In pathologic conditions that promote increased vascular permeability, endothelial cells partially lose close contacts. The endothelium becomes permeable to small proteins such as albumin. Mice can be injected intravenously with 1% Evans blue dye in PBS, and injected intradermally with VEGF. Thirty minutes after intradermal injections, tissue at the intradermal injection sites can be excised

and extracted in formamide to assess Evans blue dye extravasation. Vascular leakage can be quantified by measurement of the dye incorporated per milligram of tissue, for example, using optical density measurements and a standard curve. Mice can be dosed with a compound of interest, for example, an antibody or compound of the disclosure, to determine the effect on vascular leakage.

(339) Cancer Models

(340) The efficacy of a compound or antibody of the disclosure as a treatment for cancer can be tested in a range of cancer models. In some cancer models, cancer cells from a cell line can be implanted in a recipient animal, for example, a mouse. Non-limiting examples of suitable mouse strains include C57BL6, BALB/C, C3H, FVB/N, and FVB/N-Tg(MMTV-PyVT)634Mul. Non-limiting examples of suitable cancer cell lines include 4T1, E0771, and P0008. Cells of the 4T1 or E0771 cell lines can, for example, be implanted into a mammary pad as a model of breast cancer. 4T1 or E0771 cells can be implanted, for example, into the third mammary fat pad of female nude mice.

(341) The size of solid tumors can be measured with calipers and tumor volume calculated. Tumors can be processed for histopathological evaluation or fluorescence microscopy, to assess, for example, tumor area, tumor grade, metastases count, metastases area, the number of cell nuclei per tumor focus, intratumoral vessel diameter, intratumoral vessel density, tumor vascular maturity, perivascular cell coverage, proximity between perivascular cells and endothelial cells, or to grade tumors foci as intravascular or extravascular.

(342) Spontaneous metastasis models can be used to study the effects of a compound or antibody of the disclosure on metastasis. After tumor implant, a primary tumor can be resected (e.g., upon reaching 5 mm in size), and the animal later evaluated macroscopically or histopathologically for metastases, for example, to determine the number and size of metastases in the lungs, liver, lymph nodes, and bones.

(343) To evaluate the impact of vessel stability on metastasis, a model can be used wherein tumor cells can be injected intravenously, and the animal subsequently evaluated for intravascular versus extravasated metastases. For example, 4T-1 cells can be injected intravenously, and the lungs processed for histopathologic evaluation to quantify intravascular versus extravascular metastases.

(344) Metastases can be counted and measured macroscopically, for example, by immersing lungs in Bouin's solution, then examining them with a stereomicroscope.

(345) To measure tumor vessel permeability in vivo, intravital multiphoton microscopy can be used. Animals can be injected with fluorescently labeled bovine serum albumin, for example, pre-treatment and post-treatment time points. At each time point, two distinct regions within the tumor can be selected and a 200 micron image stack recording BSA fluorescence taken through each region every ten minutes for an hour, using a multiphoton laser scanning microscope. The analysis approach can involve three-dimensional vessel tracing to create vessel metrics and a three-dimensional map of voxel intensity versus distance to the nearest vessel over time. Images can be corrected for sample movement over time with three-dimensional image registration. The normalized transvascular flux can be calculated.

(346) Illustrative CDR Combinations

(347) In some embodiments, an antibody or compound of this disclosure comprises a HCDR1. In some embodiments, an antibody or compound of this disclosure comprises a HCDR2. In some embodiments, an antibody or compound of this disclosure comprises a HCDR3. In some embodiments, an antibody or compound of this disclosure comprises a LCDR1. In some embodiments, an antibody or compound of this disclosure comprises a LCDR2. In some embodiments, an antibody or compound of this disclosure comprises a LCDR3.

(348) In some embodiments, an antibody or compound of this disclosure comprises a HCDR1 and a HCDR2. In some embodiments, an antibody or compound of this disclosure comprises a HCDR1 and a HCDR3. In some embodiments, an antibody or compound of this disclosure comprises a

113, SEQ ID NO: 123, SEQ ID NO: 130, SEQ ID NO: 135, SEQ ID NO: 143, and SEQ ID NO: 145.

(390) In some embodiments, an antibody or compound of this disclosure comprises SEQ ID NO: 80, SEQ ID NO: 85, SEQ ID NO: 89, SEQ ID NO: 91, SEQ ID NO: 96, SEQ ID NO: 97, SEQ ID NO: 103, SEQ ID NO: 118, SEQ ID NO: 125, SEQ ID NO: 132, SEQ ID NO: 140, and SEQ ID NO: 144. In some embodiments, an antibody or compound of this disclosure comprises SEQ ID NO: 80, SEQ ID NO: 87, SEQ ID NO: 89, SEQ ID NO: 91, SEQ ID NO: 96, SEQ ID NO: 97, SEQ ID NO: 103, SEQ ID NO: 118, SEQ ID NO: 125, SEQ ID NO: 132, SEQ ID NO: 140, and SEQ ID NO: 144. In some embodiments, an antibody or compound of this disclosure comprises SEQ ID NO: 80, SEQ ID NO: 85, SEQ ID NO: 89, SEQ ID NO: 91, SEQ ID NO: 96, SEQ ID NO: 97, SEQ ID NO: 108, SEQ ID NO: 123, SEQ ID NO: 127, SEQ ID NO: 135, SEQ ID NO: 143, and SEQ ID NO: 145. In some embodiments, an antibody or compound of this disclosure comprises SEQ ID NO: 80, SEQ ID NO: 87, SEQ ID NO: 89, SEQ ID NO: 91, SEQ ID NO: 96, SEQ ID NO: 97, SEQ ID NO: 108, SEQ ID NO: 123, SEQ ID NO: 127, SEQ ID NO: 135, SEQ ID NO: 143, and SEQ ID NO: 145. In some embodiments, an antibody or compound of this disclosure comprises SEQ ID NO: 80, SEQ ID NO: 85, SEQ ID NO: 89, SEQ ID NO: 91, SEQ ID NO: 96, SEQ ID NO: 97, SEQ ID NO: 113, SEQ ID NO: 123, SEQ ID NO: 130, SEQ ID NO: 135, SEQ ID NO: 143, and SEQ ID NO: 145. In some embodiments, an antibody or compound of this disclosure comprises SEQ ID NO: 80, SEQ ID NO: 87, SEQ ID NO: 89, SEQ ID NO: 91, SEQ ID NO: 96, SEQ ID NO: 97, SEQ ID NO: 113, SEQ ID NO: 123, SEQ ID NO: 130, SEQ ID NO: 135, SEQ ID NO: 143, and SEQ ID NO: 145.

EXAMPLES

Example 1: A Tetravalent Bispecific Antibody Comprising Antibody HC2:LC1 and Aflibercept-Derived Sequences

(391) To generate a tetravalent bispecific antibody in which the heavy chain of antibody HC2:LC1 was fused to an aflibercept-derived VEGF-binding domain, an amino acid sequence was generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) SEQ ID NO: 14 (heavy chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 22 (aflibercept-derived sequence).

(392) The resulting polypeptide (SEQ ID NO: 149) was co-expressed with SEQ ID NO: 17, to provide tetravalent, bispecific antibody HC2-AFL:LC1, comprising the sequences shown in TABLE 33. Amino acids 1-19 of SEQ ID NO: 149 are the heavy chain signal peptide (SEQ ID NO: 11). Amino acids 1-20 of SEQ ID NO: 17 are the light chain signal peptide (SEQ ID NO: 12). In some embodiments, a mature HC2-AFL:LC1 does not comprise the signal peptides. For example, a mature HC2-AFL:LC1 of the disclosure can comprise SEQ ID NO: 254 and SEQ ID NO: 250.

(393) TABLE-US-00034

TABLE	33	SEQ ID NO:	Name	Amino acid sequence
149	signal		MGWTLVFLFLLSVTAGVHSEVQLVESGGGLVQPGGSLRLSC peptide-	
			AASGFTFNANAMNWVRQAPGKGLEWVGRIRTKSNNYATY HC2-	
			YAGSVKDRFTISRDDSKNSLYLQMNSLKTEDTAVYYCVRD AFL	
			YYGSSAWITYWGQGT LVTVSS ASTKGPSVFPLAPCSRSTSES	
			TAALGCLVKDYFPEPVT TVSWNS GALTSGVHTFPAVLQSSGL	
			YSLSSVVTVPSSSLG TKTYTCNV DHKPSNTKVDKRVESKYG	
			PPCPPCPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDV	
			SQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVL	
			TVLHQDWLNGKEYKCKVSNKGLPSSIEKTISKAKGQPREPQ	
			VYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPE	
			NNYKTTTPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMH	
			EALHNHYTQKSLSL SLG GGGGSSDTGRPFVEMYSEIPEIIH	
			MTEGRELVIPCRVTS PNITVTL KKFPLDTLIPDGKRIIWDSRK	
			GFIISNATYKEIGLLTCEATVNGHLYKTNLYLTHRQNTNIIDVV	

LSPSHGIELSVGIDLFNWEYPSSKHQH
KKLVNRDLKTQSGSEMKKFLSTLTIDGVTRSDQGLYTCAAS SGLMTKKNSTFVRVHEK 17
Signal MVSSAQFLGLLLLCFQGTRCDVVMTQSPSFLSASVGDRVTIT peptide-
CKASQHVGTAVAWYQQRPGKAPKLLIYWASTRHTGVPSRF LC1
SGSGSGTEFTLTISLQPEDFATYFCQQYSSYPFTFGGGGTKLEI
KRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQ
WKVDNALQSGNSQESVTEQDSKDYSLSTLTLSKADYEK
HKVYACEVTHQGLSSPVTKSFNRGEC 254 HC2-
EVQLVESGGGLVQPGGSLRLSCAASGFTFNANAMNWVRQA AFL
PGKGLEWVGRIRTKSNNYATYYAGSVKDRFTISRDDSKNSL
YLQMNSLKTEDTAVYYCVRDYYGSSAWITYWGQGTLVTV
SSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVS
WNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTITYT
CNVDHKPSNTKVDKRVESKYGPPCPPCPAPEFLGGPSVFLFP
PKPKDTLMISRTPEVTCVVVDVSQEDPEVQFNWYVDGVEV
HNAKTKPREEQFNSTYRVVSVLTVTLHQLDNLNGKEYKCKVS
NKGLPSSIEKTISKAKGQPREPQVYTLPPSQEEMTKNQVSLT
CLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYS
RLTVDKSRWQEGNVFSCSVMHEALHNHYTQKSLSLGLGK
GGGSSDTGRPFVEMYSEIPEIIHMTEGRELVIPCRVTSPNITVT
LKKFPLDTLIPDGKRIIWDSRKGFIISNATYKEIGLLTCEATVN
GHLYKTNYLTHRQTNTIIDVVLSPSHGIELSVGEKLVNCTA
RTELNVGIDFNWEYPSSKHQHKKLVNRDLKTQSGSEMKKF
LSTLTIDGVTRSDQGLYTCAASSGLMTKKNSTFVRVHEK

Example 2: A Tetravalent Bispecific Antibody Comprising Antibody HC2:LC1 and Brolucizumab-Derived Sequences

(394) To generate a tetravalent bispecific antibody in which the heavy chain of antibody HC2:LC1 was fused to a brolucizumab-derived VEGF-binding domain, an amino acid sequence was generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) SEQ ID NO: 14 (heavy chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 23 (brolucizumab-derived sequence).

(395) The resulting polypeptide (SEQ ID NO: 150) was co-expressed with SEQ ID NO: 17, to provide tetravalent, bispecific antibody HC2-BRO:LC1, comprising the sequences shown in TABLE 34. Amino acids 1-19 of SEQ ID NO: 150 are the heavy chain signal peptide (SEQ ID NO: 11). Amino acids 1-20 of SEQ ID NO: 17 are the light chain signal peptide (SEQ ID NO: 12). In some embodiments, a mature HC2-BRO:LC1 does not comprise the signal peptides. For example, a mature HC2-BRO:LC1 of the disclosure can comprise SEQ ID NO: 255 and SEQ ID NO: 250.

(396) TABLE-US-00035 TABLE 34 SEQ ID NO: Name Amino acid sequence 150 signal
MGWTLVFLFLLSVTAGVHSEVQLVESGGGLVQPGGSLRLSC peptide-

AASGFTFNANAMNWVRQAPGKGLEWVGRIRTKSNNYATY HC2-
YAGSVKDRFTISRDDSKNSLYLQMNSLKTEDTAVYYCVRD BRO
YYGSSAWITYWGQGTLVTVSSASTKGPSVFPLAPCSRSTSES
TAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGL
YSLSSVVTVPSSSLGTITYTCNVDPKPSNTKVDKRVESKYG
PPCPPCPAPEFLGGPSVFLFPPPKPKDTLMISRTPEVTCVVVDV
SQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVL
TVLHQLDNLNGKEYKCKVSNKGLPSSIEKTISKAKGQPREPQ
VYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPE
NNYKTTPPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMH
EALHNHYTQKSLSLGLGKGGGGSEIVMTQSPSTLSASVGDR

VIITCQAEIHSWLAQYQQKPGKAPKLLIYLASTLASGVPSR
 FSGSGSGAEFTLTISLQPDDEFATYYCQNVYLASTNGANFGQ
 GTKLTVLGGGGGSGGGGSGGGGSGGGGSEVQLVESGGGLV
 QPGGSLRLSCTASGFSLTDDYYMTWVRQAPGKGLEWVGFI
 DPDDDPYYATWAKGRFTISRDN SKNTLYLQMNSLRAEDTA
 VYYCAGGDHNSGWGLDIWGQGTLVTVSS 17 Signal
 MVSSAQFLGLLLLCFQGTRCDVVM TQSPSFLSASVGDRVTIT peptide-
 CKASQHVGTAWAWYQQRPGKAPKLLIYWASTRHTGVPSRF LC1
 SSGSGSGTEFTLTISLQPEDFATYFCQQYSSYPFTFGGGTKLEI
 KRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQ
 WKVDNALQSGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEK
 HKVYACEVTHQGLSSPVTKSFNRGEC 255 HC2-
 EVQLVESGGGLVQPGGSLRLSCAASGFTFNANAMNWVRQA BRO
 PGKGLEWVGRIRTKSN NYATYYAGSVKDRFTISRDDSKNSL
 YLQMNSLKTEDTAVYYCVRDYYGSSAWITYWGQGTLVTV
 SSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVS
 WNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTKTYT
 CNVDHKPSNTKVDKRVESKYGPPCPPCPAPEFLGGPSVFLFP
 PKPKDTLMISRTPETCVVVDVVSQEDPEVQFNWYVDGVEV
 HNAKTKPREEQFNSTYRVVSVLTVLHQDWLNGKEYKCKV
 NKGLPSSIEKTIKAKGQPREPQVYTLPPS QEEMTKNQVSLT
 CLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYS
 RLTVDKSRWQEGNVFSCSVMHEALHNHYTQKSLSLGLGK
GGGSEIVMTQSPSTLSASVGDRVIITCQASEIIHSWLAWYQQ
 KPGKAPKLLIYLASTLASGVPSRFSGSGSGAEFTLTISLQPD
 DEFATYYCQNVYLASTNGANFGQGGTKLTVLGGGGGSGGGG
 SGGGSGGGGSEVQLVESGGGLVQPGGSLRLSCTASGFSLT
 DYYMTWVRQAPGKGLEWVGFI DPDDDPYYATWAKGRFTIS
 RDN SKNTLYLQMNSLRAEDTAVYYCAGGDHNSGWGLDIW GQGTLVTVSS

Example 3: A Tetravalent Bispecific Antibody Comprising Antibody HC2:LC1 and Ranibizumab-Derived Sequences

(397) To generate a tetravalent bispecific antibody in which the heavy chain of antibody HC2:LC1 was fused to a ranibizumab-derived VEGF-binding domain, an amino acid sequence was generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) SEQ ID NO: 14 (heavy chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 28 (ranibizumab-derived sequence).

(398) The resulting polypeptide (SEQ ID NO: 151) was co-expressed with SEQ ID NO: 17, to provide tetravalent, bispecific antibody HC2-RAN:LC1, comprising the sequences shown in TABLE 35. Amino acids 1-19 of SEQ ID NO: 151 are the heavy chain signal peptide (SEQ ID NO: 11). Amino acids 1-20 of SEQ ID NO: 17 are the light chain signal peptide (SEQ ID NO: 12). In some embodiments, a mature HC2-RAN:LC1 does not comprise the signal peptides. For example, a mature HC2-RAN:LC1 of the disclosure can comprise SEQ ID NO: 256 and SEQ ID NO: 250.

(399) TABLE-US-00036 TABLE 35 SEQ ID NO: Name Amino acid sequence 151 Signal
 MGWTLVFLFLLSVTAGVHSEVQLVESGGGLVQPGGSLRLS peptide-
 AASGFTFNANAMNWVRQAPGKGLEWVGRIRTKSN NYATY HC2-
 YAGSVKDRFTISRDDSKNSLYLQMNSLKTEDTAVYYCVRD RAN
 YYGSSAWITYWGQGTLVTVSSASTKGPSVFPLAPCSRSTSES
 TAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGL
 YSLSSVVTVPSSSLGTKTYTCNVDHKPSNTKVDKRVESKYG
 PPCPPCPAPEFLGGPSVFLFPPKPKDTLMISRTPETCVVVDV

SQEDPEVQFNVDGVEVHNAKTKPREEQFNSTYRVVSVL
TVLHQDWLNGKEYKCKVSNKGLPSSIEKTISKAKGQPREPQ
VYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPE
NNYKTTTPPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMH
EALHNHYTQKSLSLGLKGGGGSEVQLVESGGGLVQPGGSL
RLSCAASGYDFTHYGMNWVRQAPGKGLEWVGWINTYTGE
PTYAADFKRRFTFSLDTSKSTAYLQMNSLRAEDTAVYYCAK
YPYYYGTSHWYFDVWGQGTTLVTVSSGGGGSGGGGSGGGG
SDIQLTQSPSSLSASVGDRVTITCSASQDISNYLNWYQQKPG
KAPKVLIIYFTSSLHSGVPSRFSGSGSGTDFTLTISLQPEDFAT
YYCQQYSTVTPWTFGQGTKVEIK 17 Signal
MVSSAQFLGLLLLCFQGTRCDVVMTQSPSFLSASVGDRVTIT peptide-
CKASQHVGTAWAWYQQRPGKAPKLLIYWASTRHTGVPSRF LC1
SGSGSGTEFTLTISLQPEDFATYFCQQYSSYPFTFGGGGTKLEI
KRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQ
WKVDNALQSGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEK
HKVYACEVTHQGLSPVTKSFNRGEC 256 HC2-
EVQLVESGGGLVQPGGSLRLSCAASGFTFNANAMNWVRQA RAN
PGKGLEWVGRIRTKSNNYATYYAGSVKDRFTISRDDSKNSL
YLQMNSLKTEDTAVYYCVRDYYGSSAWITYWGQGTTLVTV
SSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVS
WNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTKEYT
CNVDHKPSNTKVDKRVESKYGPPCPPCPAPEFLGGPSVFLFP
PKPKDTLMISRTPEVTCVVVDVSQEDPEVQFNWYVDGVEV
HNAKTKPREEQFNSTYRVVSVLTVLHQDWLNGKEYKCKVS
NKGLPSSIEKTISKAKGQPREPQVYTLPPSQEEMTKNQVSLT
CLVKGFYPSDIAVEWESNGQPENNYKTTTPPVLDSDGSFFLYS
RLTVDKSRWQEGNVFSCSVMHEALHNHYTQKSLSLGLKGG
GGGSEVQLVESGGGLVQPGGSLRLSCAASGYDFTHYGMNW
VRQAPGKGLEWVGWINTYTGEPTYAADFKRRFTFSLDTSKS
TAYLQMNSLRAEDTAVYYCAKYPYYYGTSHWYFDVWGQ
GTLVTVSSGGGGSGGGGSGGGGSDIQLTQSPSSLSASVGDR
VTITCSASQDISNYLNWYQQKPGKAPKVLIIYFTSSLHSGVPS
RFSGSGSGTDFTLTISLQPEDFATYYCQQYSTVTPWTFGQGT KVEIK

Example 4: Characterization of Bispecific Compounds

(400) The following multi-specific antibodies of the disclosure were generated: (i) HC2-RAN:LC1, (ii) HC2-AFL:LC1, (iii) HC2-BRO:LC1, (iv) HC2-ABI:LC1, (v) HC2:LC1-AFL, (vi) HC2:LC1-BRO, (vii) HC2:LC1-ABI, (viii) HC2-AFL:LC1-AFL, (ix) HC2-BRO:LC1-BRO, and (x) HC2-ABI:LC1-ABI. Enzyme-linked immunosorbent assays (ELISAs) were performed to determine binding of these antibodies to VEGF and HPTP-03 (VE-PTP). Binding to HPTP-03 (VE-PTP) was confirmed for all constructs, and binding to VEGF was confirmed for all constructs except for HC2-RAN:LC1.

(401) The tetravalent, bispecific antibodies described in EXAMPLES 1-3 were produced and characterized as described in TABLE 36, FIG. 16, FIG. 17, and FIG. 18.

(402) TABLE 36 provides results from small scale production and characterization of bispecific candidates.

(403) TABLE-US-00037

TABLE 36	Candidate	Yield	HC MW	LC MW	Intact	Name	(mg)	(Da)**
	HC2-RAN:LC1	0.84	76113/76109	23400/23399	198981	HC2-AFL:LC1	1.32	
	72756/72748	23400/23399	192270	HC2-BRO:LC1	0.73	75882/75877	23400/23399	198532

**m/c = measured/calculated

(404) FIG. 16 provides ELISA data demonstrating binding of a tetravalent bispecific antibody HC2-AFL:LC1 to HPTP- β (top panel) and VEGF (bottom panel). Binding is compared to controls (HC2:LC1 in top panel, VEGF-R2-Fc chimera in bottom panel).

(405) FIG. 17 provides ELISA data demonstrating binding of a tetravalent bispecific antibody HC2-BRO:LC1 to HPTP- β (top panel) and VEGF (bottom panel). Binding is compared to controls (HC2:LC1 in top panel, VEGF-R2-Fc chimera in bottom panel).

(406) FIG. 18 provides ELISA data evaluating binding of a tetravalent bispecific antibody HC2-RAN:LC1 to HPTP- β (top panel) and VEGF (bottom panel). Binding is compared to controls (HC2:LC1 in top panel, VEGF-R2-Fc chimera in bottom panel).

Example 5: A Tetravalent Bispecific Antibody Comprising Antibody HC2:LC1 and Abicipar-Derived Sequences

(407) To generate a tetravalent bispecific antibody in which the heavy chain of antibody HC2:LC1 was fused to an abicipar-derived VEGF-binding domain, an amino acid sequence was generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) Residues 1-467 of SEQ ID NO: 14 (heavy chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 244 (abicipar-derived sequence).

(408) The resulting polypeptide (SEQ ID NO: 231) was co-expressed with SEQ ID NO: 17, to provide tetravalent, bispecific antibody HC2-ABI:LC1, comprising the sequences shown in TABLE 37. Amino acids 1-19 of SEQ ID NO: 231 are the heavy chain signal peptide (SEQ ID NO: 11). Amino acids 1-20 of SEQ ID NO: 17 are the light chain signal peptide (SEQ ID NO: 12). In some embodiments, a mature HC2-ABI:LC1 does not comprise the signal peptides. For example, a mature HC2-ABI:LC1 of the disclosure can comprise SEQ ID NO: 257 and SEQ ID NO: 250.

(409) TABLE-US-00038

TABLE	37	SEQ	ID	NO:	Name	Amino acid sequence	231	Signal
MGWTLVFLFLLSVTAGVHSEVQLVESGGGLVQPGGSLRLSC	peptide-							

AASGFTFNANAMNWVRQAPGKGLEWVGRIRTKSNNYATY HC2-
YAGSVKDRFTISRDDSKNSLYLQMNSLKTEDTAVYYCVRD ABI

YYGSSAWITYWGQGT~~LVTVSSASTKGPSVFPLAPCSRSTSE~~

TAALGCLVKDYFPEPVT~~VS~~WNSGALTSGVHTFPAVLQSSGL

YSLSSVVTVPSSSLG~~TKTYTCNV~~DHKPSNTKVDKR~~VESKYG~~

PPCPPCPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVDV

SQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVL

TVLHQDWLNGKEYKCKVSNKGLPSSIEKTISKAKGQPREPQ

VYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPE

NNYKTTTPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMH

EALHNHYTQKSLSL~~SLG~~GGGSDLDK~~KLLEA~~RAGQDDEV

RILMANGADV~~NARD~~STGWTPHLAAPWGHPEIVEVLLKNG

ADVNAADFQGWTPHLAAAVGHLEIVEVLLKYGADVNAQ

DKFGKTAFDISIDNGNEDLAEILQKAA 17 Signal

MVSSAQFLGLLLLCFQGTRCDVVM~~TQSP~~SFLSASVGDRVTIT peptide-

CKASQHVGTAWAWYQQRPGKAPKLLIYWASTRHTGVPSRF LC1

SGSGSGTEFTLTIS~~SLQ~~PEDFATYFCQQYSSYPFTFGGGTKLEI

KRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQ

WKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEK

HKVYACEVTHQGLSSPVTKSFNRGEC 257 HC2-

EVQLVESGGGLVQPGGSLRLSCAASGFTFNANAMNWVRQA ABI

PGKGLEWVGRIRTKSNNYATYYAGSVKDRFTISRDDSKNSL

YLQMNSLKTEDTAVYYCVRDYYGSSAWITYWGQGT~~LVTV~~

SSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVS

WNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG~~TKTYT~~

CNV~~DHKPSNTKVDKR~~VESKYGPPCPPCPAPEFLGGPSVFLFP

PKPKDTLMISRTPEVTCVVVDVSDQEDPEVQFNWYVDGVEV
HNAKTKPREEQFNSTYRVVSVLTVLHQDWLNGKEYKCKVVS
NKGLPSSIEKTISKAKGQPREPQVYTLPPSQEEMTKNQVSLT
CLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYS
RLTVDKSRWQEGNVFSCSVMHEALHNHYTQKSLSLGLGGG
GGSDLDKLLLEAARAGQDDEVRLMANGADVNAARDSTGW
TPLHLAAPWGHPEIVEVLLKNGADVNAADFQGWTPHLHLAA
AVGHLEIVEVLLKYGADVNAQDKFGKTAFDISIDNGNEDLA EILQKAA

Example 6: A Hexavalent Bispecific Antibody Comprising Antibody HC2:LC1 and Brolucizumab-Derived Sequences

(410) To generate a heavy chain with brolucizumab-derived VEGF-binding domains added at the C-terminus, an amino acid sequence was generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) SEQ ID NO: 14 (heavy chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 23 (brolucizumab-derived sequence).

(411) To generate a light chain with brolucizumab-derived VEGF-binding domains added at the C-terminus, an amino acid sequence was generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) SEQ ID NO: 17 (light chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 23 (brolucizumab-derived sequence).

(412) The resulting polypeptides, SEQ ID NO: 150 and SEQ ID NO: 218 were co-expressed to provide a hexavalent, bispecific antibody HC2-BRO:LC1-BRO shown in TABLE 38. Amino acids 1-19 of SEQ ID NO: 150 are the heavy chain signal peptide (SEQ ID NO: 11). Amino acids 1-20 of SEQ ID NO: 218 are the light chain signal peptide (SEQ ID NO: 12). In some embodiments, a mature HC2-BRO:LC1-BRO does not comprise the signal peptides. For example, a mature HC2-BRO:LC1-BRO of the disclosure can comprise SEQ ID NO: 255 and SEQ ID NO: 258.

(413) TABLE-US-00039 TABLE 38 SEQ ID NO: Name Amino acid sequence 150 Signal peptide-

AASGFTFNANAMNWVRQAPGKGLEWVGRIRTKSNNYATY HC2-
YAGSVKDRFTISRDDSKNSLYLQMNSLKTEDTAVYYCVRD BRO
YYGSSAWITYWGQGTLVTVSSASTKGPSVFPLAPCSRSTSES
TAALGCLVKDYFPEPTVSWNSGALTSGVHTFPAVLQSSGL
YSLSSVVTVPSSSLGTQTYTCNVDPHKPSNTKVDKRVESKYG
PPCPPCPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDV
SQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVL
TVLHQDWLNGKEYKCKVSNKGLPSSIEKTISKAKGQPREPQ
VYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPE
NNYKTTTPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMH
EALHNHYTQKSLSLGLGKGGGGSEIVMTQSPSTLSASVGDR
VIITCQASEIIHSWLAWYQQKPGKAPKLLIYLASTLASGVPSR
FSGSGSGAEFTLTISLQPDDEATYYCQNVYLASTNGANFGQ
GTKLTVLGGGGGSGGGGSGGGGSGGGGSEVQLVESGGGLV
QPGGSLRLSCTASGFSITDYYMTWVRQAPGKGLEWVGFI
DPDDDPYYATWAKGRFTISRDNSTNTLYLQMNSLRAEDTA
VYYCAGGDHNSGWGLDIWGQGTLVTVSS 218 Signal
MVSSAQFLGLLLLCFQGTRCDVVMTQSPSFLSASVGDRVTIT peptide-
CKASQHVGTAVAWYQQRPGKAPKLLIYWASTRHTGVPSRF LC1-
SGSGSGTEFTLTISLQPEDFATYFCQQYSSYPFTFGGGTKLEI BRO
KRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQ
WKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEK

HKVYACEVTHQGLSSPVTKSFNRGECGGGGSEIVMTQSPST
 LSASVGDRVIITCQASEIIHSWLAWYQQKPGKAPKLLIYLAS
 TLASGVPSRFSGSGSGAEFTLTISLQPDDEFATYYCQNVYLA
 STNGANFGQGTKLTVLGGGGGSGGGGSGGGGSGGGGSEVQ
 LVESGGGLVQPGGSLRLSCTASGFSLTDYYYMTWVRQAPG
 KGLEWVGFIDPDDDPYYATWAKGRFTISRDN SKNTLYLQM
 NSLRAEDTAVYYCAGGDHNSGWGLDIWGQGTLVTVSS 258 LC1-
 DVVMTQSPSFLSASVGDRVTITCKASQHVGTAVAWYQQRP BRO
 GKAPKLLIYWASTRHTGVPSRFSGSGSGTEFTLTISLQPEDF
 ATYFCQQYSSYPFTFGGGTKLEIKRTVAAPSVFIFPPSDEQLK
 SGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQ
 DSKDSTYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTK
 SFNRGECGGGGSEIVMTQSPSTLSASVGDRVIITCQASEIIHS
 WLAWYQQKPGKAPKLLIYLASTLASGVPSRFSGSGSGAEFT
 LTISLQPDDEFATYYCQNVYLASTNGANFGQGTKLTVLGGG
 GSGGGGSGGGGSGGGGSEVQLVESGGGLVQPGGSLRLSC
 TASGFSLTDYYYMTWVRQAPGKGLEWVGFIDPDDDPYYAT
 WAKGRFTISRDN SKNTLYLQMNSLRAEDTAVYYCAGGDHN SGWGLDIWGQGTLVTVSS

Example 7: A Hexavalent Bispecific Antibody Comprising Antibody HC2:LC1 and Aflibercept-Derived Sequences

(414) To generate a heavy chain with aflibercept-derived VEGF-binding domains added at the C-terminus, an amino acid sequence was generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) SEQ ID NO: 14 (heavy chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 22 (aflibercept-derived sequence).

(415) To generate a light chain with aflibercept-derived VEGF-binding domains added at the C-terminus, an amino acid sequence was generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) SEQ ID NO: 17 (light chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 22 (aflibercept-derived sequence).

(416) The resulting polypeptides, SEQ ID NO: 149 and SEQ ID NO: 219 were co-expressed to provide a hexavalent, bispecific antibody HC2-AFL:LC1-AFL shown in TABLE 39. Amino acids 1-19 of SEQ ID NO: 149 are the heavy chain signal peptide (SEQ ID NO: 11). Amino acids 1-20 of SEQ ID NO: 219 are the light chain signal peptide (SEQ ID NO: 12). In some embodiments, a mature HC2-AFL:LC1-AFL does not comprise the signal peptides. For example, a mature HC2-AFL:LC1-AFL of the disclosure can comprise SEQ ID NO: 254 and SEQ ID NO: 259.

(417) TABLE-US-00040 TABLE 39 SEQ ID NO: Name Amino acid sequence 149 Signal peptide-
 MGWTLVFLFLLSVTAGVHSEVQLVESGGGLVQPGGSLRLSC
 AASGFTFNANAMNWVRQAPGKGLEWVGRIRTKSN NYATY HC2-
 YAGSVKDRFTISRDDSKNSLYLQMNSLKTEDTAVYYCVRD AFL
 YYGSSAWITYWGQGTLVTVSSASTKGPSVFPLAPCSRSTSES
 TAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGL
 YSLSSVVTVPSSSLGTKTYTCNVDHKPSNTKVKDKRVESKYG
 PPCPPCPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDV
 SQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVL
 TVLHQDWLNGKEYKCKVSNKGLPSSIEKTISKAKGQPREPQ
 VYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPE
 NNYKTTTPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMH
 EALHNHYTQKSLSLGLKGGGGSSDTGRPFVEMYSEIPEIIH
 MTEGRELVIPCRVTSPNITVTLKKFPLDTLIPDGKRIIWDSRK

GFIISNATYKEIGLLTCEATVNGHLYKTNYLTHRQTNTIIDVV
 LSPSHGIELSVGEKLVNCTARTELVNGIDFNWEYPSSKHQH
 KKLVNRLDKTQSGSEMKKFLSTLTIDGVTRSDQGLYTCAAS SGLMTKKNSTFVRVHEK
 219 Signal MVSSAQFLGLLLLCFQGTRCDVVMTQSPSFLSASVGDRVTTIT peptide-
 CKASQHVGTAVAWYQQRPGKAPKLLIYWASTRHTGVPSRF LC1-
 SGSGSGTEFTLTISLQPEDFATYFCQQYSSYPFTFGGGGTKLEI AFL
 KRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQ
 WKVDNALQSGNSQESVTEQDSKDYSLSTLTLSKADYEK
 HKVYACEVTHQGLSSPVTKSFNRGECGGGGSSDTGRPFVEM
 YSEIPEIIHMTEGRELVIPCRVTSPNITVTLKKFPLDTLIPDGK
 RIIWDSRKGFIIISNATYKEIGLLTCEATVNGHLYKTNYLTHR
 QTNTIIDVVLSPSHGIELSVGEKLVNCTARTELVNGIDFNW
 EYPSSKHQHKKLVNRLDKTQSGSEMKKFLSTLTIDGVTRSD
 QGLYTCAASSGLMTKKNSTFVRVHEK 259 LC1-
 DVVMTQSPSFLSASVGDRVTTITCKASQHVGTAVAWYQQRP AFL
 GKAPKLLIYWASTRHTGVPSRFSGSGSGTEFTLTISLQPEDF
 ATYFCQQYSSYPFTFGGGGTKLEIKRTVAAPSVFIFPPSDEQLK
 SGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQ
 DSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTK
 SFNRGECGGGGSSDTGRPFVEMYSEIPEIMMTEGRELVIPCR
 VTSPNITVTLKKFPLDTLIPDGKRRIWDSRKGFIIISNATYKEIG
 LLTCEATVNGHLYKTNYLTHRQTNTIIDVVLSPSHGIELSVG
 EKLVLNCTARTELVNGIDFNWEYPSSKHQHKKLVNRLDKT
 QSGSEMKKFLSTLTIDGVTRSDQGLYTCAASSGLMTKKNST FVRVHEK

Example 8: A Hexavalent Bispecific Antibody Comprising Antibody HC2:LC1 and Ranibizumab-Derived Sequences

(418) To generate a heavy chain with ranibizumab-derived VEGF-binding domains added at the C-terminus, an amino acid sequence is generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) SEQ ID NO: 14 (heavy chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 28 (ranibizumab-derived sequence).

(419) To generate a light chain with ranibizumab-derived VEGF-binding domains added at the C-terminus, an amino acid sequence is generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) SEQ ID NO: 17 (light chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 28 (ranibizumab-derived sequence).

(420) The resulting polypeptides, SEQ ID NO: 151 and SEQ ID NO: 243 are co-expressed to provide a hexavalent, bispecific antibody HC2-RAN:LC1-RAN shown in TABLE 40. Amino acids 1-19 of SEQ ID NO: 151 are the heavy chain signal peptide (SEQ ID NO: 11). Amino acids 1-20 of SEQ ID NO: 243 are the light chain signal peptide (SEQ ID NO: 12). In some embodiments, a mature HC2-RAN:LC1-RAN does not comprise the signal peptides. For example, a mature HC2-RAN:LC1-RAN of the disclosure can comprise SEQ ID NO: 256 and SEQ ID NO: 260.

(421) TABLE-US-00041 TABLE 40 SEQ ID NO: Name Amino acid sequence 151 Signal
 MGWTLVFLFLLSVTAGVHSEVQLVESGGGLVQPGGSLRLSC peptide-
 AASGFTFNANAMNWVRQAPGKGLEWVGRIITKSNNYATY HC2-
 YAGSVKDRFTISRDDSKNSLYLQMNSLKTEDTAVYYCVRD RAN
 YYGSSAWITYWGQGTLVTVSSASTKGPSVFPLAPCSRSTSES
 TAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGL
 YSLSSVVTVPSSSLGTQTYTCNVDHKPSNTKVDKRVESKYG
 PPCPPCPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDV

SQEDPEVQFNVDGVEVHNAAKTKPREEQFNSTYRVVSVL
 TVLHQDWLNGKEYKCKVSNKGLPSSIEKTISKAKGQPREPQ
 VYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPE
 NNYKTTTPVLDSGSSFLYSRLTVDKSRWQEGNVFSCSVMH
 EALHNHYTQKSLSLGLKGGGGSEVQLVESGGGLVQPGGSL
 RLSCAASGYDFTHYGMNWVRQAPGKGLEWVGWINTYTGE
 PTYAADFKRRFTFSLDTSKSTAYLQMNSLRAEDTAVYYCAK
 YPYYYGTSHWYFDVWGQGTLVTVSSGGGGSGGGGSGGGG
 SDIQLTQSPSSLSASVGDRVTITCSASQDISNYLNWYQQKPG
 KAPKVLIIYFTSSLHSGVPSRFSGSGSGTDFTLTISLQPEDFAT
 YYCQQYSTVTPWTFGGQGTKVEIK 243 Signal
 MVSSAQFLGLLLLCFQGTRCDVVMTQSPSFLSASVGDRVTIT peptide-
 CKASQHVGTAVAWYQQRPGKAPKLLIYWASTRHTGVPSRF LC1-
 SGSGSGTEFTLTISLQPEDFATYFCQQYSSYPFTFGGGGTKLEI RAN
 KRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQ
 WKVDNALQSGNSQESVTEQDSKDYSLSTLTLSKADYEK
 HKVYACEVTHQGLSSPVTKSFNRGECGGGGSEVQLVESGG
 GLVQPGGSLRLSCAASGYDFTHYGMNWVRQAPGKGLEWV
 GWINTYTGEPTYAADFKRRFTFSLDTSKSTAYLQMNSLRAE
 DTAVYYCAKYPYYYGTSHWYFDVWGQGTLVTVSSGGGGSG
 GGGSGGGGSDIQLTQSPSSLSASVGDRVTITCSASQDISNYL
 NWYQQKPGKAPKVLIIYFTSSLHSGVPSRFSGSGSGTDFTLT
 ISLQPEDFATYYCQQYSTVTPWTFGGQGTKVEIK 260 LC1-
 DVVMTQSPSFLSASVGDRVTITCKASQHVGTAVAWYQQRP RAN
 GKAPKLLIYWASTRHTGVPSRFSGSGSGTEFTLTISLQPEDF
 ATYFCQQYSSYPFTFGGGGTKLEIKRTVAAPSVFIFPPSDEQLK
 SGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQ
 DSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTK
 SFNRGECGGGGSEVQLVESGGGLVQPGGSLRLSCAASGYDF
 THYGMNWVRQAPGKGLEWVGWINTYTGEPTYAADFKRRF
 TFLDTSKSTAYLQMNSLRAEDTAVYYCAKYPYYYGTSHW
 YFDVWGQGTLVTVSSGGGGSGGGGSGGGGSDIQLTQSPSSL
 SASVGDRVTITCSASQDISNYLNWYQQKPGKAPKVLIIYFTSS
 LHSVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQYSTVTP WTFGGQGTKVEIK

Example 9: A Hexavalent Bispecific Antibody Comprising Antibody HC2:LC1 and Abicipar-Derived Sequences

(422) To generate a heavy chain with abicipar-derived VEGF-binding domains added at the C-terminus, an amino acid sequence was generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) Residues 1-467 of SEQ ID NO: 14 (heavy chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 244 (abicipar-derived sequence).

(423) To generate a light chain with abicipar-derived VEGF-binding domains added at the C-terminus, an amino acid sequence was generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) SEQ ID NO: 17 (light chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 244 (abicipar-derived sequence).

(424) The resulting polypeptides, SEQ ID NO: 231 and SEQ ID NO: 232, were co-expressed to provide a hexavalent, bispecific antibody HC2-ABI:LC1-ABI shown in TABLE 41. Amino acids 1-19 of SEQ ID NO: 231 are the heavy chain signal peptide (SEQ ID NO: 11). Amino acids 1-20 of SEQ ID NO: 232 are the light chain signal peptide (SEQ ID NO: 12). In some embodiments, a

mature HC2:LC1-ABI does not comprise the signal peptides. For example, a mature HC2-ABI:LC1-ABI of the disclosure can comprise SEQ ID NO: 257 and SEQ ID NO: 261.

(425) TABLE-US-00042 TABLE 41 SEQ ID NO: Name Amino acid sequence 231 Signal MGWTLVFLFLLSVTAGVHSEVQLVESGGGLVQPGGSLRLSC peptide-AASGFTFNANAMNWVRQAPGKGLEWVGRIKTSNNYATY HC2-YAGSVKDRFTISRDDSKNSLYLQMNSLKTEDTAVYYCVRD ABI YYGSSAWITYWGQGTLVTVSSASTKGPSVFPLAPCSRSTSESTAAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYTCNVDHKPSNTKVDKRVESKYGPPCPPCPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSDQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVTLHQDWLNGKEYKCKVSNKGLPSSIEKTISKAKGQPREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMHEALHNHYTQKSLSLGGGGGGSDLDKLLLEAARAGQDDEV RILMANGADV NARDSTGWTPLHLAAPWGHPEIVEVLLKNG ADVNAADFQGWTPHLAAAVGHLEIVEVLLKYGADVNAQ DKFGKTAFDISIDNGNEDLAEILQKAA 232 Signal MVSSAQFLGLLLLCFQGTRCDVVMQTQSPSFLSASVGDRVTIT peptide-CKASQHVGTAVAWYQQRPGKAPKLLIYWASTRHTGVPSRF LC1-SGSGSGTEFTLTISLQPEDFATYFCQQYSSYPFTFGGGGTKLEI ABI KRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGECGGGGSDLDKLLLEAARAGQDDEV RILMANGADV NARDSTGWTPLHLAAPWGHPEIVEVLLKNGADVNAADFQGWTPHLAAAVGHLEIVEVLLKYGADVNAQDKFGKTAFDISIDNGNEDLAEILQKAA 261 LC1-DVVMQTQSPSFLSASVGDRVTITCKASQHVGTAVAWYQQRPGKAPKLLIYWASTRHTGVPSRFSGSGSGTEFTLTISLQPEDFATYFCQQYSSYPFTFGGGGTKLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGECGGGGSDLDKLLLEAARAGQDDEV RILMANGADV NARDSTGWTPLHLAAPWGHPEIVEVLLKNGADVNAADFQGWTPHLAAAVGHLEIVEVLLKYGADVNAQDKFGKTAFDISIDNGNEDLAEILQKAA

Example 10: A Tetravalent Bispecific Antibody Comprising Antibody HC2:LC1 and Brolucizumab-Derived Sequences

(426) To generate a light chain with brolucizumab-derived VEGF-binding domains added at the C-terminus, an amino acid sequence was generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) SEQ ID NO: 17 (light chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 23 (brolucizumab-derived sequence).

(427) The resulting polypeptide (SEQ ID NO: 218) was co-expressed with SEQ ID NO: 14, to provide tetravalent, bispecific antibody HC2:LC1-BRO, comprising the sequences shown in TABLE 42. Amino acids 1-19 of SEQ ID NO: 14 are the heavy chain signal peptide (SEQ ID NO: 11). Amino acids 1-20 of SEQ ID NO: 218 are the light chain signal peptide (SEQ ID NO: 12). In some embodiments, a mature HC2:LC1-BRO does not comprise the signal peptides. For example, a mature HC2:LC1-BRO of the disclosure can comprise SEQ ID NO: 247 and SEQ ID NO: 258.

(428) TABLE-US-00043 TABLE 42 SEQ ID NO: Name Amino acid sequence 14 Signal MGWTLVFLFLLSVTAGVHSEVQLVESGGGLVQPGGSLRLSC peptide-

AASGFTFNANAMNWRQAPGKLEWVGRIRTKSNNYATY HC2
 YAGSVKDRFTISRDDSKNSLYLQMNSLKTEDTAVYYCVRD
 YYGSSAWITYWGQGTTLVTVSSASTKGPSVFPLAPCSRSTSES
 TAALGCLVKDYFPEPTVSWNSGALTSGVHTFPAVLQSSGL
 YSLSSVVTVPSSSLGKTKYTCNVDPKPSNTKVDKRVESKYG
 PPCPPCPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDV
 SQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVL
 TVLHQDWLNGKEYKCKVSNKGLPSSIEKTISKAKGQPREPQ
 VYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPE
 NNYKTTTPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMH EALHNHYTQKSLSLGLGK
 218 Signal MVSSAQFLGLLLLCFQGTRCDVVMTQSPSFLSASVGDRVITIT peptide-
 CKASQHVGTAVAWYQQRPGKAPKLLIYWASTRHTGVPSRF LC1-
 SGSGSGTEFTLTISLQPEDFATYFCQQYSSYPFTFGGGGTKLEI BRO
 KRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQ
 WKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEK
 HKVYACEVTHQGLSPVTKSFNRGECGGGGSEIVMTQSPST
 LSASVGDRVITCQASEIIHSWLAWYQQKPGKAPKWYLAS
 TLASGVPSRFSGSGSGAEFTLTISLQPDDEFATYYCQNVYLA
 STNGANFGQGTKLTVLGGGGGSGGGGSGGGGSGGGGSEVQ
 LVESGGGLVQPGGSLRLSCTASGFSLTDYYYMTWVRQAPG
 KGLEWVGFIDPDDDPYYATWAKGRFTISRDN SKNTLYLQM
 NSLRAEDTAVYYCAGGDHNSGWGLDIWGQGTTLVTVSS

Example 11: A Tetravalent Bispecific Antibody Comprising Antibody HC2:LC1 and Aflibercept-Derived Sequences

(429) To generate a light chain with aflibercept-derived VEGF-binding domains added at the C-terminus, an amino acid sequence was generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) SEQ ID NO: 17 (light chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 22 (aflibercept-derived sequence).

(430) The resulting polypeptide (SEQ ID NO: 219) was co-expressed with SEQ ID NO: 14, to provide tetravalent, bispecific antibody HC2:LC1-AFL, comprising the sequences shown in TABLE 43. Amino acids 1-19 of SEQ ID NO: 14 are the heavy chain signal peptide (SEQ ID NO: 11). Amino acids 1-20 of SEQ ID NO: 219 are the light chain signal peptide (SEQ ID NO: 12). In some embodiments, a mature HC2:LC1-AFL does not comprise the signal peptides. For example, a mature HC2:LC1-AFL of the disclosure can comprise SEQ ID NO: 247 and SEQ ID NO: 259.

(431) TABLE-US-00044 TABLE 43 SEQ ID NO: Name Amino acid sequence 14 Signal MGWTLVFLFLLSVTAGVHSEVQLVESGGGLVQPGGSLRLSC peptide-

AASGFTFNANAMNWRQAPGKLEWVGRIRTKSNNYATY HC2
 YAGSVKDRFTISRDDSKNSLYLQMNSLKTEDTAVYYCVRD
 YYGSSAWITYWGQGTTLVTVSSASTKGPSVFPLAPCSRSTSES
 TAALGCLVKDYFPEPTVSWNSGALTSGVHTFPAVLQSSGL
 YSLSSVVTVPSSSLGKTKYTCNVDPKPSNTKVDKRVESKYG
 PPCPPCPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDV
 SQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVL
 TVLHQDWLNGKEYKCKVSNKGLPSSIEKTISKAKGQPREPQ
 VYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPE
 NNYKTTTPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMH EALHNHYTQKSLSLGLGK
 219 Signal MVSSAQFLGLLLLCFQGTRCDVVMTQSPSFLSASVGDRVITIT peptide-
 CKASQHVGTAVAWYQQRPGKAPKLLIYWASTRHTGVPSRF LC1-
 SGSGSGTEFTLTISLQPEDFATYFCQQYSSYPFTFGGGGTKLEI AFL

KRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQ
WKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEK
HKVYACEVTHQGLSSPVTKSFNRGECGGGGSSDTGRPFVEM
YSEIPEIIHMTEGRELVIPCRVTSPNITVTLKKFPLDTLIPDGK
RIIWDSRKGFIISNATYKEIGLLTCEATVNGHLYKTNYLTHR
QTNTIIDVVLSPSHGIELSVGEKLVNCTARTELNVGIDFNW
EYPSSKHQHKLVNRDLKTQSGSEMKKFLSTLTIDGVTRSD
QGLYTCAASSGLMTKKNSTFVRVHEK

Example 12: A Tetravalent Bispecific Antibody Comprising Antibody HC2:LC1 and Ranibizumab-Derived Sequences

(432) To generate a light chain with ranibizumab-derived VEGF-binding domains added at the C-terminus, an amino acid sequence is generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) SEQ ID NO: 17 (light chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 28 (ranibizumab-derived sequence).

(433) The resulting polypeptide (SEQ ID NO: 243) is co-expressed with SEQ ID NO: 14, to provide tetravalent, bispecific antibody HC2:LC1-RAN, comprising the sequences shown in TABLE 44. Amino acids 1-19 of SEQ ID NO: 14 are the heavy chain signal peptide (SEQ ID NO: 11). Amino acids 1-20 of SEQ ID NO: 243 are the light chain signal peptide (SEQ ID NO: 12). In some embodiments, a mature HC2:LC1-RAN does not comprise the signal peptides. For example, a mature HC2:LC1-RAN of the disclosure can comprise SEQ ID NO: 247 and SEQ ID NO: 260.

(434) TABLE-US-00045 TABLE 44 SEQ ID NO: Name Amino acid sequence 14 Signal MGWTLVFLFLLSVTAGVHSEVQLVESGGGLVQPGGSLRLSC peptide-

AASGFTFNANAMNWVRQAPGKGLEWVGRIRTKSNNYATY HC2
YAGSVKDRFTISRDDSKNSLYLQMNSLKTEDTAVYYCVRD
YYGSSAWITYWGQGTLLVTVSSASTKGPSVFPLAPCSRSTSES
TAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGL
YSLSSVVTVPSSSLGKTYTCNVDHKPSNTKVDKRVESKYG
PPCPPCPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDV
SQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVL
TVLHQDWLNGKEYKCKVSNKGLPSSIEKTISKAKGQPREPQ
VYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPE
NNYKTTTPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMH EALHNHYTQKSLSLGLK
243 Signal MVSSAQFLGLLLLCFQGTRCDVVMQTSPSFLSASVGDRVTTIT peptide-
CKASQHVGTAVAWYQQRPGKAPKLLIYWASTRHTGVPSRF LC1-
SGSGSGTEFTLTISSLQPEDFATYFCQQYSSYPFTFGGGGTKLEI RAN
KRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQ
WKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEK
HKVYACEVTHQGLSSPVTKSFNRGECGGGGSEVQLVESGG
GLVQPGGSLRLSCAASGYDFTHYGMNWVRQAPGKGLEWV
GWINTYTGEPTYAADFKRRFTFSLDTSKSTAYLQMNSLRAE
DTAVYYCAKYPYYYGTSHWYFDVWGQGTLLVTVSSGGGGS
GGGGSGGGGSDIQLTQSPSSLSASVGDRVTTITCSASQDISNYL
NWYQQKPGKAPKVLIIYFTSSLHSGVPSRFSGSGSGTDFTLTI
SSLQPEDFATYYCQQYSTVPWTFGQGTKVEIK

Example 13: A Tetravalent Bispecific Antibody Comprising Antibody HC2:LC1 and Abicipar-Derived Sequences

(435) To generate a light chain with abicipar-derived VEGF-binding domains added at the C-terminus, an amino acid sequence was generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) SEQ ID NO: 17 (light chain of antibody HC2:LC1);

2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 244 (abicipar-derived sequence).

(436) The resulting polypeptide (SEQ ID NO: 232) was co-expressed with SEQ ID NO: 245 (residues 1-467 of SEQ ID NO: 14), to provide tetravalent, bispecific antibody HC2:LC1-ABI, comprising the sequences shown in TABLE 45. Amino acids 1-19 of SEQ ID NO: 245 are the heavy chain signal peptide (SEQ ID NO: 11). Amino acids 1-20 of SEQ ID NO: 232 are the light chain signal peptide (SEQ ID NO: 12). In some embodiments, a mature HC2:LC1-ABI does not comprise the signal peptides. For example, a mature HC2:LC1-ABI of the disclosure can comprise SEQ ID NO: 247 and SEQ ID NO: 261.

(437) TABLE-US-00046 TABLE 45 SEQ ID NO: Name Amino acid sequence 245 Signal peptide-
MGWTLVFLFLLSVTAGVHSEVQLVESGGGLVQPGGSLRLSC
AASGFTFNANAMNWVRQAPGKGLEWVGRIRTKSNNYATY HC2

YAGSVKDRFTISRDDSKNSLYLQMNSLKTEDTAVYYCVRD
YYGSSAWITYWGQGTLVTVSSASTKGPSVFPLAPCSRSTSES
TAALGCLVKDYFPEPVTWNSGALTSGVHTFPAVLQSSGL
YSLSSVVTVPSSSLGTKTYTCNVDHKPSNTKVDKRVESKYG
PPCPPCPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDV
SQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVL
TVLHQDWLNGKEYKCKVSNKGLPSSIEKTISKAKGQPREPQ
VYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPE
NNYKTTTPVLDSGDSFFLYSRLTVDKSRWQEGNVFSCSVMH EALHNHYTQKSLSLSLG

232 Signal MVSSAQFLGLLLLCFQGTRCDVVMTQSPSFLSASVGDRVTIT peptide-

CKASQHVGTAWAWYQQRPGKAPKLLIYWASTRHTGVPSRF LC1-
SGSGSGTEFTLTISLQPEDFATYFCQQYSSYPFTFGGGTKLEI ABI

KRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQ
WKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEK
HKVYACEVTHQGLSSPVTKSFNRGECGGGGSDLDKLLLEA
ARAGQDDEVRLMANGADVNAARDSTGWTPHLAAPWGHP
EIVEVLLKNGADVNAADFQGWTPHLAAAVGHLEIVEVLL
KYGADVNAQDKFGKTAFDISIDNGNEDLAEILQKAA 262 HC2
EVQLVESGGGLVQPGGSLRLSCAASGFTFNANAMNWVRQA
PGKGLEWVGRIRTKSNNYATYYAGSVKDRFTISRDDSKNSL
YLQMNSLKTEDTAVYYCVRDYYGSSAWITYWGQGTLVTV
SSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTW
NSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTKTYT
CNVDHKPSNTKVDKRVESKYGPPCPPCPAPEFLGGPSVFLF
PKPKDTLMISRTPEVTCVVVDVSQEDPEVQFNWYVDGVEV
HNAKTKPREEQFNSTYRVVSVLTVLHQDWLNGKEYKCKV
SNKGLPSSIEKTISKAKGQPREPQVYTLPPSQEEMTKNQVSL
TCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSGDSFFLY
SRLTVDKSRWQEGNVFSCSVMHEALHNHYTQKSLSLSLG

Example 14: Bispecific Antibodies that Activate Tie2

(438) Human Umbilical Cord Endothelial Cells (HUVECs) were seeded onto T75 flasks coated with porcine gelatin. Cell maintenance was performed using complete medium (EGM® or EGM®-2) and sub-cultured using Trypsin/EDTA into 100 mm dishes. After 3 days, the 100 mm dishes were rinsed, and treated in basal medium (EBM®, EBM®-2, OptiMEM™ I) for 30 minutes at 37° C./5% CO.sub.2 with one of the following antibodies at 5 or 50 nM as indicated in FIG. 19:

(i) HC2:LC1, a humanized monoclonal antibody specific for HPTP-β; (ii) HC2-BRO:LC1, a tetravalent bispecific antibody comprising brolicizumab-derived VEGF-binding domains fused to the C-termini of the heavy chains of HC2:LC1; (iii) HC2-AFL:LC1, a tetravalent bispecific

antibody comprising aflibercept-derived VEGF-binding domains fused to the C-termini of the heavy chains of HC2:LC1; (iv) HC2-BRO:LC1-BRO, a hexavalent antibody comprising brolucizumab-derived VEGF-binding domains fused to the C-termini of the heavy chains and the light chains of HC2:LC1; or (v) HC2-AFL:LC1-AFL, a hexavalent antibody comprising aflibercept-derived VEGF-binding domains fused to the C-termini of the heavy chains and the light chains of HC2:LC1. After treatment, the cells were rinsed with ice cold PBS containing 1 mM NaOV and lysed in Complete Triton™ X Lysis Buffer (20 mM Tris-HCl, 137 mM NaCl, 10% Glycerol, 1% Triton™ X-100, 2 mM EDTA, 1 mM NaOV, 1 mM NaF, 1 mM PMSF, 1 µg/mL leupeptin, 1 g/mL pepstatin).

(439) The lysates were immunoprecipitated with anti-Tie2 and anti-VEGFR2 antibodies. 1-10 µg of VEGFR2 antibody (MAB3573, #89109), TIE-2 antibody (Ab33), and 25 µL of Protein A/G agarose beads were added to 1 mL of HUVEC lysate, and the tubes were placed on a rotating platform for 1-3 days at 4° C. The IP reaction tubes were rinsed with 1 mL complete Triton™-X lysis buffer and resuspended in 2× loading dye containing DTT. The tubes were incubated at 95° C. for five minutes, spun down, and 25 µL loaded per well into a Tris Glycine gel. The samples were resolved on a gel, transferred to a PVDF membrane, and subjected to serial western blot to detect phosphotyrosine (phospho-Tie2 and phospho-VEGFR2), followed by re-probing to blot for total Tie2, and total VEGFR2. The gel was run at 125V for 75 minutes, before transfer to a PVDF membrane. The membranes were blocked in 5% BSA/0.05% Tween® Tris Wash Buffer for 1 hour at room temperature (RT) on a rotating platform. Primary antibodies (PY99; VEGFR2-A3; TIE-2 Ab33) were added for 1 hour at a 1:1000 dilution in wash buffer. Secondary antibody (anti-mouse hrp) was added for 1 hour at a 1:1000 dilution in wash buffer. The membranes were rinsed three times with 0.05-0.1% Tween® 20+Tris Buffered Saline (TBS) between steps. An ECL detection system was used to visualize bands. Re-probe was performed on lots treated with 200 mM Glycine for 24-48 h.

(440) As shown in FIG. 19, all of the tested bispecific antibodies increased basal Tie2 activation in HUVECS, while basal VEGFR2 phosphorylation was not affected. The top panel provides a western blot showing Tie2 activation and VEGFR2 activation as shown through detection of phospho-Tie2 and phospho-VEGFR2, respectively (top half), and total Tie2 and VEGFR2 (bottom half). The bottom panel provides the densitometric ratios of phosphorylated Tie2 to total Tie2.

(441) The assay was repeated with the following antibodies of the disclosure, which also enhanced Tie2 activation in the absence of exogenous Ang1: (i) HC2-RAN:LC1, (ii) HC2-ABI:LC1, (iii) HC2:LC1-AFL, (iv) HC2:LC1-BRO, (v) HC2:LC1-ABI, and (vi) HC2-ABI:LC1-ABI.

Example 15: Bispecific Antibodies Enhance Ang1-Mediated Tie2 Activation and Block VEGF-Mediated VEGFR2 Activation

(442) Human Umbilical Cord Endothelial Cells (HUVECs) were seeded onto T75 flasks coated with porcine gelatin. Cell maintenance was performed using complete medium (EGM® or EGM®-2) and sub-cultured using Trypsin/EDTA into 100 mm dishes. After 3 days, the 100 mm dishes were rinsed, and mock-pre-treated or pre-treated in basal medium (EBM®, EBM®-2, OptiMEM™ I) for 30 minutes at 37° C./5% CO.sub.2 with one of the following antibodies at 5 or 50 nM as indicated in FIG. 20: (i) HC2:LC1, a humanized monoclonal antibody specific for HPTP-β; (ii) HC2-BRO:LC1, a tetravalent bispecific antibody comprising brolucizumab-derived VEGF-binding domains fused to the C-termini of the heavy chains of HC2:LC1; (iii) HC2-AFL:LC1, a tetravalent bispecific antibody comprising aflibercept-derived VEGF-binding domains fused to the C-termini of the heavy chains of HC2:LC1; (iv) HC2-BRO:LC1-BRO, a hexavalent antibody comprising brolucizumab-derived VEGF-binding domains fused to the C-termini of the heavy chains and the light chains of HC2:LC1; or (v) HC2-AFL:LC1-AFL, a hexavalent antibody comprising aflibercept-derived VEGF-binding domains fused to the C-termini of the heavy chains and the light chains of HC2:LC1. After pre-treatment, cells were treated with VEGF (5 ng/mL) and Ang1 (50 ng/mL) for 6 minutes at 37° C./5% CO.sub.2 in basal medium (Phosphate buffered

saline, PBS+0.2% Bovine Serum Albumin, or OptiMEM™ I). After treatment, the cells were rinsed with ice cold PBS containing 1 mM NaOV and lysed in Complete Triton™ X Lysis Buffer (20 mM Tris-HCl, 137 mM NaCl, 10% Glycerol, 1% Triton™ X-100, 2 mM EDTA, 1 mM NaOV, 1 mM NaF, 1 mM PMSF, 1 µg/mL leupeptin, 1 µg/mL pepstatin).

(443) The lysates were immunoprecipitated with anti-Tie2 and anti-VEGFR2 antibodies. 1-10 µg of VEGFR2 antibody (MAB3573, #89109), TIE-2 antibody (Ab33), and 25 µL of Protein A/G agarose beads were added to 1 mL of HUVEC lysate, and the tubes were placed on a rotating platform for 1-3 days at 4° C. The IP reaction tubes were rinsed with 1 mL complete Triton™-X lysis buffer and resuspended in 2× loading dye containing DTT. The tubes were incubated at 95° C. for five minutes, spun down, and 25 µL loaded per well into a Tris Glycine gel. The samples were resolved on a gel, transferred to a PVDF membrane, and subjected to serial western blot to detect phosphotyrosine (phospho-Tie2 and phospho-VEGFR2), followed by re-probing to blot for total Tie2, and total VEGFR2. The gel was run at 125V for 75 minutes, before transfer to a PVDF membrane. The membranes were blocked in 5% BSA/0.05% Tween® Tris Wash Buffer for 1 hour at room temperature (RT) on a rotating platform. Primary antibodies (PY99; VEGFR2-A3; TIE-2 Ab33) were added for 1 hour at a 1:1000 dilution in wash buffer. Secondary antibody (anti-mouse hrp) was added for 1 hour at a 1:1000 dilution in wash buffer. The membranes were rinsed three times with 0.05-0.1% Tween® 20+Tris Buffered Saline (TBS) between steps. An ECL detection system was used to visualize bands. Re-probe was performed on lots treated with 200 mM Glycine for 24-48 h.

(444) Treatment with VEGF and Ang1 resulted in increased phosphorylation of VEGFR2 and Tie2 (FIG. 20). Treatment with the HC2:LC1 antibody, specific for HPTP-β, enhanced Ang1-mediated Tie2 activation in cells treated with Ang1 and VEGF. Treatment with the bispecific antibodies enhanced Ang1-mediated Tie2 activation and blocked VEGF-mediated VEGFR2 activation in cells treated with Ang1 and VEGF. The top panel provides a western blot showing Tie2 activation and VEGFR2 activation as shown through detection of phospho-Tie2 and phospho-VEGFR2, respectively (top half), and total Tie2 and VEGFR2 (bottom half). The lower panel provides the densitometric ratios of phosphorylated Tie2 to total Tie2 and phosphorylated VEGFR2 to total VEGFR2.

Example 16: Bispecific Antibodies Enhance Ang1-Mediated Tie2 Activation and Block VEGF-Mediated VEGFR2 Activation (Immunoprecipitation and Western Blot Assays)

(445) Human Umbilical Cord Endothelial Cells (HUVECs) were seeded onto T75 flasks coated with porcine gelatin. Cell maintenance was performed using complete medium (EGM® or EGM®-2) and sub-cultured using Trypsin/EDTA into 100 mm dishes. After 3 days, the 100 mm dishes were rinsed, and pre-treated in basal medium (EBM®, EBM®-2, OptiMEM™ I) for 30 minutes at 37° C./5% CO.sub.2 with one of multi-specific antibodies of the disclosure as indicated for each figure. After pre-treatment, the cells were mock-treated or treated with VEGF and Ang1 for 6 minutes at 37° C./5% CO.sub.2 in basal medium (Phosphate buffered saline, PBS+0.2% Bovine Serum Albumin, or OptiMEM™ I). After treatment, the cells were rinsed with ice cold PBS containing 1 mM NaOV and lysed in Complete Triton™ X Lysis Buffer (20 mM Tris-HCl, 137 mM NaCl, 10% Glycerol, 1% Triton™ X-100, 2 mM EDTA, 1 mM NaOV, 1 mM NaF, 1 mM PMSF, 1 µg/mL leupeptin, 1 g/mL pepstatin).

(446) The lysates were immunoprecipitated with anti-Tie2 and anti-VEGFR2 antibodies. 1-10 g of VEGFR2 antibody (MAB3573, #89109), TIE-2 antibody (Ab33), and 25 µL of Protein A/G agarose beads were added to 1 mL of HUVEC lysate, and the tubes were placed on a rotating platform for 1-3 days at 4° C. The IP reaction tubes were rinsed with 1 mL complete Triton™-X lysis buffer and resuspended in 2× loading dye containing DTT. The tubes were incubated at 95° C. for five minutes, spun down, and 25 µL loaded per well into a Tris Glycine gel. The samples were resolved on a gel, transferred to a PVDF membrane, and subjected to serial western blot to detect phosphotyrosine (phospho-Tie2 and phospho-VEGFR2), followed by re-probing to blot for total

Tie2, and total VEGFR2. The gel was run at 125V for 75 minutes, before transfer to a PVDF membrane. The membranes were blocked in 5% BSA/0.05% Tween® Tris Wash Buffer for 1 hour at room temperature (RT) on a rotating platform. Primary antibodies (PY99; VEGFR2-A3; TIE-2 Ab33) were added for 1 hour at a 1:1000 dilution in wash buffer. Secondary antibody (anti-mouse hrp) was added for 1 hour at a 1:1000 dilution in wash buffer. The membranes were rinsed three times with 0.05-0.1% Tween® 20+Tris Buffered Saline (TBS) between steps. An ECL detection system was used to visualize bands. Re-probe was performed on lots treated with 200 mM Glycine for 24-48 h. The assay was repeated using different multi-specific antibodies of the disclosure, and different concentrations of VEGF and Ang1 as indicated for the following figures.

(447) For FIG. 21A, FIG. 21B, and FIG. 21C, the cells were mock-pre-treated or pre-treated at 5 nM or 50 nM with one of: (i) HC2-ABI:LC1, a tetravalent bispecific antibody comprising abicipar-derived VEGF-binding domains fused to the C-termini of the heavy chains of HC2:LC1 (a humanized monoclonal antibody specific for HPTP-0); (ii) HC2-ABI:LC1-ABI, a hexavalent antibody comprising abicipar-derived VEGF-binding domains fused to the C-termini of the heavy chains and the light chains of HC2:LC1; or (iii) HC2-AFL:LC1-AFL, a hexavalent antibody comprising aflibercept-derived VEGF-binding domains fused to the C-termini of the heavy chains and the light chains of HC2:LC1. After pre-treatment, the cells were mock-treated (-) or treated (+) with VEGF (5 ng/mL) and Ang1 (50 ng/mL). Treatment with VEGF and Ang1 resulted in increased phosphorylation of VEGFR2 and Tie2 (FIG. 21A, FIG. 21B, and FIG. 21C). Treatment with the bispecific antibodies enhanced Tie2 activation and blocked VEGFR2 activation, including in cells treated with Ang1 and VEGF (FIG. 21A, FIG. 21B, and FIG. 21C). FIG. 21A provides a western blot showing Tie2 activation and VEGFR2 activation as shown through detection of phospho-Tie2 and phospho-VEGFR2, respectively (top half), and total Tie2 and VEGFR2 (bottom half). FIG. 21B provides the densitometric ratio of phosphorylated Tie2 to total Tie2, normalized to untreated cells. FIG. 21C provides the densitometric ratio of phosphorylated VEGFR2 to total VEGFR2, normalized to untreated cells.

(448) For FIG. 22A, FIG. 22B, and FIG. 22C, the cells were mock-pre-treated or pre-treated with one of the following antibodies at 5 nM: (i) HC2-BRO:LC1-BRO, a hexavalent antibody comprising brolocizumab-derived VEGF-binding domains fused to the C-termini of the heavy chains and the light chains of HC2:LC1 (a humanized monoclonal antibody specific for HPTP-β); (ii) HC2-AFL:LC1-AFL, a hexavalent antibody comprising aflibercept-derived VEGF-binding domains fused to the C-termini of the heavy chains and the light chains of HC2:LC1; or (iii) HC2-ABI:LC1:ABI, a hexavalent bispecific antibody comprising abicipar-derived VEGF-binding domains fused to the C-termini of the heavy chains and the light chains of HC2:LC1. After pre-treatment, the cells were mock-treated (-), treated with 5 ng/mL VEGF and 50 ng/mL Ang1 (+), or treated with 50 ng/mL VEGF and 250 ng/mL of Ang1 (+++). Treatment with VEGF and Ang1 resulted in increased phosphorylation of VEGFR2 and Tie2 (FIG. 22A, FIG. 22B, and FIG. 22C). Treatment with the bispecific antibodies enhanced Tie2 activation and blocked VEGFR2 activation, including in cells treated with Ang1 and VEGF. (FIG. 22A, FIG. 22B, and FIG. 22C). FIG. 22A provides a western blot showing Tie2 activation and VEGFR2 activation as shown through detection of phospho-Tie2 and phospho-VEGFR2, respectively (top half), and total Tie2 and VEGFR2 (bottom half). FIG. 22B provides the densitometric ratio of phosphorylated Tie2 to total Tie2, normalized to untreated cells. FIG. 22C provides the densitometric ratio of phosphorylated VEGFR2 to total VEGFR2, normalized to untreated cells.

Example 17: Bispecific Antibodies Enhance Ang1-Mediated Tie2 Activation and Block VEGF-Mediated VEGFR2 Activation (Electrochemiluminescence Assays)

(449) Human Umbilical Cord Endothelial Cells (HUVECs) were seeded onto T75 flasks coated with porcine gelatin. Cell maintenance was performed using complete medium (EGM® or EGM®-2) and sub-cultured using Trypsin/EDTA into 100 mm dishes. After 3 days, the 100 mm dishes were rinsed, and mock-pre-treated or pre-treated in basal medium (EBM®, EBM®-2,

OptiMEM™ I) for 30 minutes at 37° C./5% CO₂ with one the antibodies indicated below for each figure.

(450) After pre-treatment, the cells were mock-treated or treated with VEGF and Ang1 for 6 minutes at 37° C./5% CO₂ in basal medium (Phosphate buffered saline, PBS+0.2% Bovine Serum Albumin, or OptiMEM™ I). After treatment, the cells were rinsed with ice cold PBS containing 1 mM NaOH and lysed in Complete Triton™ X Lysis Buffer (20 mM Tris-HCl, 137 mM NaCl, 10% Glycerol, 1% Triton™ X-100, 2 mM EDTA, 1 mM NaOH, 1 mM NaF, 1 mM PMSF, 1 µg/mL leupeptin, 1 g/mL pepstatin).

(451) Phosphorylated Tie2 and phosphorylated VEGFR2 were quantified by electrochemiluminescence. Primary or capture Antibodies were spotted onto 96 well Sector® Imager plates. 5 µL of Tie2 antibody (30 g/mL, AF313) or VEGFR2 antibody (30 g/mL, 89109) were coated for 1 hour (at room temperature) or overnight (4° C.). Each plate was washed three times with TBS+0.02% Tween® 20 (this was performed between each step). Wells were blocked with MSD® Blocker A-3% in wash buffer for 1 hour on a rotating platform at room temperature. 25 µL of HUVEC lysates were added to each well directly and incubated for 1 hour on a rotating platform at room temperature. Detection antibody (1) was diluted to 2 µg/mL in 1% blocker/wash buffer (Ab33; AF2720/Y992; pTyr 1214; NB100 530), and 25 µL/well incubated for 1 hour on a rotating platform at room temperature. Detection antibody (2) was diluted to 1 µg/mL in 1% blocker/wash buffer (Goat anti-mouse, GAM; Goat anti-rabbit, GAR; both Sulfo Tag®-labeled) and 25 µL/well incubated for 1 hour on a rotating platform at room temperature. Signal was captured using 150 µL of Meso Scale Discovery (MSD®) read buffer in each well on an MSD® imager instrument. The assay was repeated using different multi-specific antibodies of the disclosure, and different concentrations of VEGF and Ang1 as indicated for the following figures.

(452) For FIG. 23A and FIG. 23B, the cells were mock-treated or pre-treated with one of the following antibodies at 5 nM: (i) HC2-BRO:LC1, a tetravalent bispecific antibody comprising brolicizumab-derived VEGF-binding domains fused to the C termini of the heavy chains of HC2:LC1 (a humanized monoclonal antibody specific for HTP-β); (ii) HC2:LC1-BRO, a tetravalent bispecific antibody comprising brolicizumab-derived VEGF-binding domains fused to the C termini of the light chains of HC2:LC1; (iii) HC2-BRO:LC1-BRO, a hexavalent bispecific antibody comprising brolicizumab-derived VEGF-binding domains fused to the C termini of the heavy chains and the light chains of HC2:LC1; (iv) HC2-AFL:LC1, a tetravalent bispecific antibody comprising aflibercept-derived VEGF-binding domains fused to the C termini of the heavy chains of HC2:LC1; (v) HC2:LC1-AFL, a tetravalent bispecific antibody comprising aflibercept-derived VEGF-binding domains fused to the C termini of the light chains of HC2:LC1; (vi) HC2-AFL:LC1-AFL, a hexavalent bispecific antibody comprising aflibercept-derived VEGF-binding domains fused to the C termini of the heavy chains and the light chains of HC2:LC1; (vii) HC2-ABI:LC1, a tetravalent bispecific antibody comprising abicipar-derived VEGF-binding domains fused to the C termini of the heavy chains of HC2:LC1; (viii) HC2:LC1-ABI, a tetravalent bispecific antibody comprising abicipar-derived VEGF-binding domains fused to the C termini of the light chains of HC2:LC1; or (ix) HC2-ABI:LC1-ABI, a hexavalent bispecific antibody comprising abicipar-derived VEGF-binding domains fused to the C termini of the heavy chains and the light chains of HC2:LC1. FIG. 23A and FIG. 23B provide the ratio of phosphorylated Tie2 to total Tie2 and phosphorylated VEGFR2 to total VEGFR2, respectively, normalized to untreated cells. Treatments were as follows: mock-treated (-), treated with 5 ng/mL VEGF and 50 ng/mL Ang1 (+), or treated with 25 ng/mL VEGF and 250 ng/mL of Ang1 (+++). Treatment with VEGF and Ang1 resulted in increased phosphorylation of Tie2 (FIG. 23A) and VEGFR2 (FIG. 23B). Pre-treatment with the multi-specific antibodies enhanced Tie2 activation (FIG. 23A) and inhibited VEGFR2 activation (FIG. 23B) in cells treated with Ang1 and VEGF.

(453) For FIG. 24A and FIG. 24B, the cells were mock-treated or pre-treated with one of the following antibodies at 5 nM: (i) HC2-BRO:LC1-BRO, a hexavalent bispecific antibody

comprising brotuzumab-derived VEGF-binding domains fused to the C termini of the heavy chains and the light chains of HC2:LC1 (a humanized monoclonal antibody specific for HPTP- β); (ii) HC2-AFL:LC1-AFL, a hexavalent bispecific antibody comprising aflibercept-derived VEGF-binding domains fused to the C termini of the heavy chains and the light chains of HC2:LC1; or (iii) HC2-ABI:LC1-ABI, a hexavalent bispecific antibody comprising abicipar-derived VEGF-binding domains fused to the C termini of the heavy chains and the light chains of HC2:LC1. FIG. 24A and FIG. 24B provide the ratio of phosphorylated Tie2 to total Tie2 and phosphorylated VEGFR2 to total VEGFR2, respectively, normalized to untreated cells. Treatments were as follows: mock-treated (-), treated with 5 ng/mL VEGF and 50 ng/mL Ang1 (+), or treated with 25 ng/mL VEGF and 250 ng/mL of Ang1 (+++). Treatment with VEGF and Ang1 resulted in increased phosphorylation of Tie2 (FIG. 24A) and VEGFR2 (FIG. 24B). Pre-treatment with the multi-specific antibodies enhanced Tie2 activation (FIG. 24A) and inhibited VEGFR2 activation (FIG. 24B) in cells treated with Ang1 and VEGF.

(454) For FIG. 25A and FIG. 25B, the cells were mock-treated or pre-treated with one of the following antibodies at 5 or 50 nM: (i) HC2-BRO:LC1, a tetravalent bispecific antibody comprising brotuzumab-derived VEGF-binding domains fused to the C termini of the heavy chains of HC2:LC1 (a humanized monoclonal antibody specific for HPTP-0); or (ii) HC2-BRO:LC1-BRO, a hexavalent bispecific antibody comprising brotuzumab-derived VEGF-binding domains fused to the C termini of the heavy chains and the light chains of HC2:LC1. FIG. 25A and FIG. 25B provide the ratio of phosphorylated Tie2 to total Tie2 and phosphorylated VEGFR2 to total VEGFR2, respectively, normalized to untreated cells. Treatments were as follows: mock-treated (-), treated with 5 ng/mL VEGF and 50 ng/mL Ang1 (+), or treated with 25 ng/mL VEGF and 250 ng/mL of Ang1 (+++). Treatment with VEGF and Ang1 resulted in increased phosphorylation of Tie2 (FIG. 25A) and VEGFR2 (FIG. 25B). Pre-treatment with the multi-specific antibodies enhanced Tie2 activation (FIG. 25A) and inhibited VEGFR2 activation (FIG. 25B) in cells treated with Ang1 and VEGF.

(455) For FIG. 26A and FIG. 26B, the cells were mock-treated or pre-treated with one of the following antibodies at 5 nM or 50 nM: (i) HC2-ABI:LC1, a tetravalent bispecific antibody comprising abicipar-derived VEGF-binding domains fused to the C termini of the heavy chains of HC2:LC1 (a humanized monoclonal antibody specific for HPTP- β); (ii) HC2-ABI:LC1-ABI, a hexavalent bispecific antibody comprising abicipar-derived VEGF-binding domains fused to the C termini of the heavy chains and the light chains of HC2:LC1; or (iii) HC2-AFL:LC1-AFL, a hexavalent bispecific antibody comprising aflibercept-derived VEGF-binding domains fused to the C termini of the heavy chains and the light chains of HC2:LC1. FIG. 26A and FIG. 26B provide the ratio of phosphorylated Tie2 to total Tie2 and phosphorylated VEGFR2 to total VEGFR2, respectively, normalized to untreated cells. Treatments were as follows: mock-treated (-), or treated with 5 ng/mL VEGF and 50 ng/mL Ang1 (+). Treatment with VEGF and Ang1 resulted in increased phosphorylation of Tie2 (FIG. 26A) and VEGFR2 (FIG. 26B). Pre-treatment with the multi-specific antibodies enhanced Tie2 activation (FIG. 26A) and inhibited VEGFR2 activation (FIG. 26B) in cells treated with Ang1 and VEGF.

(456) For FIG. 27A and FIG. 27B, the cells were mock-treated or pre-treated with one of the following antibodies at 5 nM or 50 nM: (i) HC2-ABI:LC1, a tetravalent bispecific antibody comprising abicipar-derived VEGF-binding domains fused to the C termini of the heavy chains of HC2:LC1 (a humanized monoclonal antibody specific for HPTP- β); (ii) HC2-ABI:LC1-ABI, a hexavalent bispecific antibody comprising abicipar-derived VEGF-binding domains fused to the C termini of the heavy chains and the light chains of HC2:LC1; or (iii) HC2-AFL:LC1, a tetravalent bispecific antibody comprising aflibercept-derived VEGF-binding domains fused to the C termini of the heavy chains of HC2:LC1. FIG. 27A and FIG. 27B provide the ratio of phosphorylated Tie2 to total Tie2 and phosphorylated VEGFR2 to total VEGFR2, respectively, normalized to untreated cells. Treatments were as follows: mock-treated (-), or treated with 25 ng/mL VEGF and 250

ng/mL Ang1 (+). Treatment with VEGF and Ang1 resulted in increased phosphorylation of Tie2 (FIG. 27A) and VEGFR2 (FIG. 27B). Pre-treatment with the multi-specific antibodies enhanced Tie2 activation (FIG. 27A) and inhibited VEGFR2 activation (FIG. 27B) in cells treated with Ang1 and VEGF.

Example 18: Multi-Specific Antibodies Bind HPTP- β and VEGF at High Affinity (Biacore™ Surface Plasmon Resonance Assays)

(457) Biacore™ (a label-free, real-time biomolecular interaction analysis system using surface plasmon resonance) surface plasmon resonance assays were performed to determine the equilibrium dissociation constant ($K_{sub.D}$) of antibodies of the disclosure for VEGF and HPTP- β (VE-PTP). Binding experiments were performed on Biacore™ 3000/Biacore™ T-200 instruments at 25° C.

(458) The assay buffer contained 10 mM HEPES buffer (pH 7.4), 150 mM NaCl, 3 mM EDTA, and 0.05% P20 (polyoxyethylenesorbitan). The regeneration buffer contained 10 mM Glycine buffer (pH 1.75). The conjugation buffer contained 10 mM sodium acetate buffer (pH 5). A flow rate of 5 L/minute was used for capturing ligand. A flow rate of 30 L/minute was used for kinetic analysis.

(459) For analysis of binding of antibodies to HPTP-0, poly-histidine tagged HPTP- β extracellular domain (ECD) $\frac{1}{2}$ was immobilized on a chip surface via anti-His antibodies. Goat anti-His antibody was first immobilized on the surface of the chip by direct immobilization using EDC/NHS (N-ethyl-N'-(3-dimethyl aminopropyl carbodiimide)/N-hydroxy succinamide) coupling chemistry on flow cell 2 of the Biacore™ CM5 (Carboxymethylated dextran coated) chip. Unoccupied sites were blocked with 1M ethanolamine. The His-tagged ligand HPTP- β $\frac{1}{2}$ ECD was captured at a response unit (RU) of 50. The analyte (antibody) was flowed over the chip at a single analyte concentration at a time. The binding of analyte to the ligand was monitored in real time to obtain on ($k_{sub.a}$) and off ($k_{sub.d}$) rates. The equilibrium constant ($K_{sub.D}$) was calculated from the observed $k_{sub.a}$ and $k_{sub.d}$.

(460) For analysis of binding of the antibodies to VEGF, the antibodies were immobilized on the surface of the chip by direct immobilization using EDC/NHS coupling chemistry on flow cell 2 of the Biacore™ CM5 chip. Unoccupied sites were blocked with 1M ethanolamine. The analyte (VEGF) was flowed over the chip at a single analyte concentration at a time. The binding of analyte to the ligand was monitored in real time to obtain on ($k_{sub.a}$) and off ($k_{sub.d}$) rates. The equilibrium constant ($K_{sub.D}$) was calculated from the observed $k_{sub.a}$ and $k_{sub.d}$.

(461) Scouting analysis was performed using 10 nM of analyte to determine approximate $K_{sub.D}$. Chi square analysis was carried out between the actual sensorgram and the sensorgram generated from the BIAAnalysis software to confirm the accuracy of the analysis; values of 1-2 were considered accurate and below 1 highly accurate.

(462) TABLE 46 provides equilibrium dissociation constants based on scouting experiments performed at a single ligand concentration. NB=no significant binding detected.

(463) TABLE-US-00047 TABLE 46 VEGF VEGF HPTP- β HPTP- β Construct $K_{sub.D}$ chi sq
 $K_{sub.D}$ chi sq HC2:LC1 NB NB 7.32E-11M 0.0439 (73.2 pM) Aflibercept 1.60E-10M 0.278 NB
NB (160 pM) HC2-AFL:LC1 3.50E-10M 0.13 3.20E-10M 0.336 (350 pM) (320 pM) HC2-BRO:LC1 1.12E-13M 0.148 3.25E-10M 0.323 (112 fM) (325 pM) HC2-ABI:LC1 1.10E-13M 1.68 1.22E-10M 1.38 (110 fM) (122 pM) HC2-RBZ:LC1 NB 5.86E-13M 0.16 (586 fM)
HC2:LC1-AFL 8.25E-10M 0.189 4.67E-10M 1.48 (825 pM) (467 pM) HC2:LC1-BRO 3.51E-14M 0.185 3.09E-10M 0.691 (35.1 fM) (309 pM) HC2:LC1-ABI 5.19E-10M 3.09 6.10E-11M 1.47 (519 pM) (61 pM) HC2-AFL:LC1-AFL 5.57E-10M 2.53 6.94E-11M 3.14 (557 pM) (69.4 pM) HC2-BRO:LC1-BRO 1.79E-12M 0.809 3.50E-12M 2.15 (1.79 pM) (3.5 pM) HC2-ABI:LC1-ABI 6.80E-10M 8.02 2.05E-10M 1.18 (680 pM) (205 pM)

(464) Full kinetic analysis was performed using 0 nM, 0.625 nM, 1.25 nM, 5 nM, and 10 nM of the analyte to determine $K_{sub.D}$. Chi square analysis was carried out between the actual sensorgram

and the sensorgram generated from the BIAanalysis software to confirm the accuracy of the analysis; values of 1-2 were considered accurate and below 1 highly accurate.

(465) TABLE 47 provides equilibrium dissociation constants based on full kinetic experiments performed at multiple ligand concentrations. NB=no significant binding detected.

(466) TABLE-US-00048 TABLE 47 VEGF VEGF HPTP- β HPTP- β Construct K.sub.D chi sq
K.sub.D chi sq HC2:LC1 NB NB 1.21E-10M 0.123 (121 pM) Aflibercept 3.75E-11M 0.106 NB
NB (37.5 pM) HC2-AFL:LC1 4.63E-11M 0.0875 2.19E-10M 0.0295 (46.3 pM) (219 pM) HC2-
BRO:LC1 2.18E-13M 0.452 1.54E-10M 0.0817 (218 fM) (154 pM) HC2-ABI:LC1 4.00E-14M
0.223 1.03E-12M 0.113 (40 fM) (1.03 pM)

(467) These results demonstrate that the multi-specific antibodies HC2-AFL:LC1, HC2-BRO:LC1, HC2-ABI:LC1, HC2:LC1-AFL, HC2:LC1-BRO, HC2:LC1-ABI, HC2-AFL:LC1-AFL, HC2-BRO:LC1-BRO, and HC2-ABI:LC1-ABI bind to HPTP- β and VEGF with high affinity.

Example 19: A Hexavalent Antibody Comprising Antibody HC2:LC1, Abicipar-Derived Sequences, and Brolucizumab-Derived Sequences

(468) To generate a heavy chain with abicipar-derived VEGF-binding domains added at the C-terminus, an amino acid sequence is generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) Residues 1-467 of SEQ ID NO: 14 (heavy chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 244 (abicipar-derived sequence).

(469) To generate a light chain with brolucizumab-derived VEGF-binding domains added at the C-terminus, an amino acid sequence is generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) SEQ ID NO: 17 (light chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 23 (brolucizumab-derived sequence).

(470) The resulting polypeptides, SEQ ID NO: 231 and SEQ ID NO: 218, are co-expressed to provide a hexavalent antibody HC2-ABI:LC1-BRO shown in TABLE 48. Amino acids 1-19 of SEQ ID NO: 231 are the heavy chain signal peptide (SEQ ID NO: 11). Amino acids 1-20 of SEQ ID NO: 218 are the light chain signal peptide (SEQ ID NO: 12). In some embodiments, a mature HC2-ABI:LC1-BRO does not comprise the signal peptides. For example, a mature HC2-ABI:LC1-BRO of the disclosure can comprise SEQ ID NO: 257 and SEQ ID NO: 258. The antibody is bispecific for target molecules, comprising a specificity for HPTP- β (VE-PTP) and VEGF. The antibody is trispecific for target epitopes, comprising a specificity for one HPTP- β (VE-PTP) epitope and for two VEGF epitopes.

(471) TABLE-US-00049 TABLE 48 SEQ ID NO: Name Amino acid sequence 231
Signal MGWTLVFLFLLSVTAGVHSEVQLVESGGGLVQPGGSLRLSC peptide-
AASGFTFNANAMNWVRQAPGKGLEWVGRIRTKSNNYATY HC2-
YAGSVKDRFTISRDDSKNSLYLQMNSLKTEDTAVYYCVRD ABI
YYGSSAWITYWGQGTLVTVSSASTKGPSVFPLAPCSRSTSES
TAALGCLVKDYFPEPTVSWNSGALTSGVHTFPAVLQSSGL
YSLSSVVTVPSSSLGTKTYTCNVDHKPSNTKVDKRVESKYG
PPCPPCPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDV
SQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVL
TVLHQDWLNGKEYKCKVSNKGLPSSIEKTISKAKGQPREPQ
VYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPE
NNYKTTTPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMH
EALHNHYTQKSLSLSLGGGGGSDLDKLLLEAARAGQDDEV
RILMANGADV NARDSTGWTPLHLAAPWGHPEIVEVLLKNG
ADVNAADFQGWTPHLAAAVGHLEIVEVLLKYGADVNAQ
DKFGKTAFDISIDNGNEDLAEILQKAA 218 Signal
MVSSAQFLGLLLLCFQGTRCDVVMQTQSPSFLSASVGDRVTIT peptide-

CKASQHVGTAVWYQQRP GKAPKLLIYWASTRHTGVPSRF LC1-
SGSGSGTEFTLTISSLQPEDFATYFCQQYSSYPFTFGGGGTKLEI BRO
KRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQ
WKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEK
HKVYACEVTHQGLSSPVTKSFNRGECGGGGSEIVMTQSPST
LSASVGDRVIITCQASEIIHSWLAWYQQKPGKAPKLLIYLAS
TLASGVPSRFSGSGSGAEFTLTISSLQPDDEFATYYCQNVYLA
STNGANFGQGTKLTVLGGGGGSGGGGSGGGGSGGGGSEVQ
LVESGGGLVQPGGSLRLSCTASGFSLT DYYYMTWVRQAPG
KGLEWVGFI DPDDDPYYATWAKGRFTISRDN SKNTLYLQM
NSLRAEDTAVYYCAGGDHNSGWGLDIWGQGTLVTVSS

Example 20: A Hexavalent Antibody Comprising Antibody HC2:LC1, Aflibercept-Derived Sequences, and BroLucizumab-Derived Sequences

(472) To generate a heavy chain with broLucizumab-derived VEGF-binding domains added at the C-terminus, an amino acid sequence is generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) SEQ ID NO: 14 (heavy chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 23 (broLucizumab-derived sequence).

(473) To generate a light chain with aflibercept-derived VEGF-binding domains added at the C-terminus, an amino acid sequence is generated comprising the following appended amino acid sequences, from N-terminus to C-terminus: 1) SEQ ID NO: 17 (light chain of antibody HC2:LC1); 2) SEQ ID NO: 31 (linker peptide, underlined); and 3) SEQ ID NO: 22 (aflibercept-derived sequence).

(474) The resulting polypeptides, SEQ ID NO: 150 and SEQ ID NO: 219, are co-expressed to provide a hexavalent antibody HC2-BRO:LC1-AFL shown in TABLE 49. Amino acids 1-19 of SEQ ID NO: 150 are the heavy chain signal peptide (SEQ ID NO: 11). Amino acids 1-20 of SEQ ID NO: 219 are the light chain signal peptide (SEQ ID NO: 12). In some embodiments, a mature HC2-BRO:LC1-AFL does not comprise the signal peptides. For example, a mature HC2-BRO:LC1-AFL of the disclosure can comprise SEQ ID NO: 255 and SEQ ID NO: 259. The antibody is bispecific for target molecules, comprising a specificity for HPTP- β (VE-PTP) and VEGF. The antibody is trispecific for target epitopes, comprising a specificity for one HPTP- β (VE-PTP) epitope and for two VEGF epitopes.

(475) TABLE-US-00050 TABLE 49 SEQ ID NO: Name Amino acid sequence 150 Signal peptide-
MGWTLVFLFLLSVTAGVHSEVQLVESGGGLVQPGGSLRLSC
AASGFTFNANAMNWVRQAPGKGLEWVGRI RTKSN NYATY HC2-
YAGSVKDRFTISRDDSKNSLYLQMNSLKTEDTAVYYCVRD BRO
YYGSSAWITYWGQGTLVTVSSASTKGPSVFPLAPCSRSTSES
TAALGCLVKDYFPEPVT VSWNSGALTSGVHTFPAVLQSSGL
YSLSSVVTVPSSSLG TKTYTCNV D HKPSNTKVDKR VESKYG
PPCPPCPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDV
SQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVL
TVLHQDWLNGKEYKCKVSNKGLPSSIEKTISKAKGQPREPQ
VYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPE
NNYKTTTPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMH
EALHNHYTQKSLSLSLGKGGGGSEIVMTQSPSTLSASVGDR
VIITCQASEIIHSWLAWYQQKPGKAPKLLIYLASTLASGVPSR
FSGSGSGAEFTLTISSLQPDDEFATYYCQNVYLASTNGANFGQ
GTKLTVLGGGGGSGGGGSGGGGSGGGGSEVQLVESGGGLV
QPGGSLRLSCTASGFSLT DYYYMTWVRQAPGKGLEWVGFI
DPDDDPYYATWAKGRFTISRDN SKNTLYLQMNSLRAEDTA

VYYCAGDHNSGWGLDIWGQGTLTVSS 219 Signal
MVSSAQFLGLLLLCFQGTRCDVVMQTQSPSFLSASVGDRVTIT peptide-
CKASQHVGTAVAWYQQRPGKAPKLLIYWASTRHTGVPSRF LC1-
SGSGSGTEFTLTISLQPEDFATYFCQQYSSYPFTFGGGTKLEI
AFLKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQ
WKVDNALQSGNSQESVTEQDSKDYSLSTLTLSKADYEK
HKVYACEVTHQGLSSPVTKSFNRGECGGGGSSDTGRPFVEM
YSEIPEIIHMTEGRELVIPCRVTSPNITVTLKKFPLDTLIPDGK
RIIWDSRKGFIIISNATYKEIGLLTCEATVNGHLYKTNYLTHR
QTNTIIDVVLSPSHGIELSVGEKLVLNCTARTELNVGIDFNW
EYPSSKHQHKLVNRDLKTQSGSEMKKFLSTLTIDGVTRSD
QGLYTCAASSGLMTKKNSTFVRVHEK

EMBODIMENTS

- (476) The following non-limiting embodiments provide illustrative examples of the disclosure, but do not limit the scope of the disclosure.
- (477) Embodiment 1. A compound comprising: (a) a first domain, wherein the first domain modulates a phosphatase, wherein the phosphatase modulates Tie2; and (b) a second domain that specifically binds a receptor tyrosine kinase agonist.
- (478) Embodiment 2. The compound of embodiment 1, wherein the compound is an antibody.
- (479) Embodiment 3. The compound of any one of embodiments 1-2, wherein the compound is a multispecific antibody.
- (480) Embodiment 4. The compound of any one of embodiments 1-3, wherein the compound is a tetravalent antibody.
- (481) Embodiment 5. The compound of any one of embodiments 1-3, wherein the compound is a hexavalent antibody.
- (482) Embodiment 6. The compound of any one of embodiments 1-5, wherein the compound is a bispecific antibody.
- (483) Embodiment 7. The compound of any one of embodiments 1-4 and 6, wherein the compound is a tetravalent bispecific antibody.
- (484) Embodiment 8. The compound of any one of embodiments 1-3 and 5-6, wherein the compound is a hexavalent bispecific antibody.
- (485) Embodiment 9. The compound of any one of embodiments 1-8, wherein the compound inhibits the phosphatase that modulates Tie2.
- (486) Embodiment 10. The compound of any one of embodiments 1-9, wherein the compound inhibits HPTP- β .
- (487) Embodiment 11. The compound of any one of embodiments 1-10, wherein the compound inhibits VE-PTP.
- (488) Embodiment 12. The compound of any one of embodiments 1-11, wherein the compound activates Tie2.
- (489) Embodiment 13. The compound of any one of embodiments 1-12, wherein the compound inhibits the receptor tyrosine kinase agonist.
- (490) Embodiment 14. The compound of any one of embodiments 1-13, wherein the compound inhibits VEGF receptor signaling.
- (491) Embodiment 15. The compound of any one of embodiments 1-14, wherein the compound inhibits a VEGF.
- (492) Embodiment 16. The compound of any one of embodiments 1-15, wherein the compound inhibits VEGF-A.
- (493) Embodiment 17. The compound of any one of embodiments 1-16, wherein the compound inhibits the phosphatase that modulates Tie2, and inhibits the receptor tyrosine kinase agonist.
- (494) Embodiment 18. The compound of any one of embodiments 1-17, wherein the phosphatase is

HPTP- β , and the receptor tyrosine kinase agonist is a VEGF.

(495) Embodiment 19. The compound of any one of embodiments 1-18, wherein the phosphatase is HPTP- β , and the receptor tyrosine kinase agonist is VEGF-A.

(496) Embodiment 20. The compound of any one of embodiments 1-19, wherein the compound activates Tie2, and the receptor tyrosine kinase agonist is a VEGF.

(497) Embodiment 21. The compound of any one of embodiments 1-20, wherein the compound activates Tie2, and the receptor tyrosine kinase agonist is VEGF-A.

(498) Embodiment 22. The compound of any one of embodiments 1-21, wherein the phosphatase that modulates Tie2 signaling is a protein tyrosine phosphatase.

(499) Embodiment 23. The compound of any one of embodiments 1-22, wherein the phosphatase that modulates Tie2 signaling is a receptor-like protein tyrosine phosphatase.

(500) Embodiment 24. The compound of any one of embodiments 1-23, wherein the phosphatase that modulates Tie2 signaling is HPTP- β .

(501) Embodiment 25. The compound of any one of embodiments 1-23, wherein the phosphatase that modulates Tie2 signaling is VE-PTP.

(502) Embodiment 26. The compound of any one of embodiments 1-25, wherein the receptor tyrosine kinase agonist is a growth factor.

(503) Embodiment 27. The compound of any one of embodiments 1-26, wherein the receptor tyrosine kinase agonist is a cysteine-knot growth factor superfamily member.

(504) Embodiment 28. The compound of any one of embodiments 1-27, wherein the receptor tyrosine kinase agonist is a PDGF family member.

(505) Embodiment 29. The compound of any one of embodiments 1-28, wherein the receptor tyrosine kinase agonist is a pro-angiogenic factor.

(506) Embodiment 30. The compound of any one of embodiments 1-29, wherein the receptor tyrosine kinase agonist is a VEGF receptor agonist.

(507) Embodiment 31. The compound of any one of embodiments 1-30, wherein the receptor tyrosine kinase agonist is a VEGF.

(508) Embodiment 32. The compound of any one of embodiments 1-31, wherein the receptor tyrosine kinase agonist is VEGF-A.

(509) Embodiment 33. The compound of any one of embodiments 1-32, wherein the first domain binds to HPTP- β .

(510) Embodiment 34. The compound of any one of embodiments 1-33, wherein the first domain binds to VE-PTP.

(511) Embodiment 35. The compound of any one of embodiments 1-34, wherein the first domain binds to an extracellular domain of HPTP- β .

(512) Embodiment 36. The compound of any one of embodiments 1-34, wherein the first domain binds to a first FN3 repeat of an extracellular domain of HPTP- β .

(513) Embodiment 37. The compound of any one of embodiments 1-36, wherein the second domain binds to a VEGF.

(514) Embodiment 38. The compound of any one of embodiments 1-37, wherein the second domain binds to VEGF-A.

(515) Embodiment 39. The compound of any one of embodiments 1-38, wherein the first domain binds to HPTP- β , and the receptor tyrosine kinase agonist is a VEGF.

(516) Embodiment 40. The compound of any one of embodiments 1-39, wherein the first domain binds to HPTP- β , and the receptor tyrosine kinase agonist is VEGF-A.

(517) Embodiment 41. The compound of any one of embodiments 1-40, wherein the first domain comprises a sequence that is at least 80% identical to any one of SEQ ID NOS: 76-98.

(518) Embodiment 42. The compound of any one of embodiments 1-41, wherein the first domain comprises a sequence that is at least 80% identical to SEQ ID NO: 76, SEQ ID NO: 77, SEQ ID NO: 78, SEQ ID NO: 79, or SEQ ID NO: 80.

(519) Embodiment 43. The compound of any one of embodiments 1-42, wherein the first domain comprises a sequence that is at least 80% identical to SEQ ID NO: 81, SEQ ID NO: 82, SEQ ID NO: 83, SEQ ID NO: 84, SEQ ID NO: 85, SEQ ID NO: 86, or SEQ ID NO: 87.

(520) Embodiment 44. The compound of any one of embodiments 1-43, wherein the first domain comprises a sequence that is at least 80% identical to SEQ ID NO: 88, SEQ ID NO: 89, or SEQ ID NO: 90.

(521) Embodiment 45. The compound of any one of embodiments 1-44, wherein the first domain comprises a sequence that is at least 80% identical to SEQ ID NO: 91, SEQ ID NO: 92, or SEQ ID NO: 93.

(522) Embodiment 46. The compound of any one of embodiments 1-45, wherein the first domain comprises a sequence that is at least 80% identical to SEQ ID NO: 94, SEQ ID NO: 95, or SEQ ID NO: 96.

(523) Embodiment 47. The compound of any one of embodiments 1-46, wherein the first domain comprises a sequence that is at least 80% identical to SEQ ID NO: 97 or SEQ ID NO: 98.

(524) Embodiment 48. The compound of any one of embodiments 1-47, wherein the first domain comprises: (a) a sequence that is at least 80% identical to SEQ ID NO: 76, SEQ ID NO: 77, SEQ ID NO: 78, SEQ ID NO: 79, or SEQ ID NO: 80; (b) a sequence that is at least 80% identical to SEQ ID NO: 81, SEQ ID NO: 82, SEQ ID NO: 83, SEQ ID NO: 84, SEQ ID NO: 85, SEQ ID NO: 86, or SEQ ID NO: 87; (c) a sequence that is at least 80% identical to SEQ ID NO: 88, SEQ ID NO: 89, or SEQ ID NO: 90; (d) a sequence that is at least 80% identical to SEQ ID NO: 91, SEQ ID NO: 92, or SEQ ID NO: 93; (e) a sequence that is at least 80% identical to SEQ ID NO: 94, SEQ ID NO: 95, or SEQ ID NO: 96; and (f) a sequence that is at least 80% identical to SEQ ID NO: 97 or SEQ ID NO: 98.

(525) Embodiment 49. The compound of any one of embodiments 1-48, wherein the second domain comprises a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 28, or SEQ ID NO: 244.

(526) Embodiment 50. The compound of any one of embodiments 1-49, wherein the second domain comprises a sequence that is at least 80% identical to any one of SEQ ID NOS: 99-146.

(527) Embodiment 51. The compound of any one of embodiments 1-50, wherein the second domain comprises a sequence that is at least 80% identical to SEQ ID NO: 99, SEQ ID NO: 100, SEQ ID NO: 101, SEQ ID NO: 102, SEQ ID NO: 103, SEQ ID NO: 104, SEQ ID NO: 105, SEQ ID NO: 106, SEQ ID NO: 107, SEQ ID NO: 108, SEQ ID NO: 109, SEQ ID NO: 110, SEQ ID NO: 111, SEQ ID NO: 112, or SEQ ID NO: 113.

(528) Embodiment 52. The compound of any one of embodiments 1-51, wherein the second domain comprises a sequence that is at least 80% identical to SEQ ID NO: 114, SEQ ID NO: 115, SEQ ID NO: 116, SEQ ID NO: 117, SEQ ID NO: 118, SEQ ID NO: 119, SEQ ID NO: 120, SEQ ID NO: 121, SEQ ID NO: 122, or SEQ ID NO: 123.

(529) Embodiment 53. The compound of any one of embodiments 1-52, wherein the second domain comprises a sequence that is at least 80% identical to SEQ ID NO: 124, SEQ ID NO: 125, SEQ ID NO: 126, SEQ ID NO: 127, SEQ ID NO: 128, SEQ ID NO: 129, SEQ ID NO: 130, or SEQ ID NO: 131.

(530) Embodiment 54. The compound of any one of embodiments 1-53, wherein the second domain comprises a sequence that is at least 80% identical to SEQ ID NO: 132, SEQ ID NO: 133, SEQ ID NO: 134, SEQ ID NO: 135, SEQ ID NO: 136, or SEQ ID NO: 137.

(531) Embodiment 55. The compound of any one of embodiments 1-54, wherein the second domain comprises a sequence that is at least 80% identical to SEQ ID NO: 138, SEQ ID NO: 139, SEQ ID NO: 140, SEQ ID NO: 141, SEQ ID NO: 142, or SEQ ID NO: 143.

(532) Embodiment 56. The compound of any one of embodiments 1-55, wherein the second domain comprises a sequence that is at least 80% identical to SEQ ID NO: 144, SEQ ID NO: 145, or SEQ ID NO: 146.

(533) Embodiment 57. The compound of any one of embodiments 1-56, wherein the second domain comprises: (a) a sequence that is at least 80% identical to SEQ ID NO: 99, SEQ ID NO: 100, SEQ ID NO: 101, SEQ ID NO: 102, SEQ ID NO: 103, SEQ ID NO: 104, SEQ ID NO: 105, SEQ ID NO: 106, SEQ ID NO: 107, SEQ ID NO: 108, SEQ ID NO: 109, SEQ ID NO: 110, SEQ ID NO: 111, SEQ ID NO: 112, or SEQ ID NO: 113; (b) a sequence that is at least 80% identical to SEQ ID NO: 114, SEQ ID NO: 115, SEQ ID NO: 116, SEQ ID NO: 117, SEQ ID NO: 118, SEQ ID NO: 119, SEQ ID NO: 120, SEQ ID NO: 121, SEQ ID NO: 122, or SEQ ID NO: 123; (c) a sequence that is at least 80% identical to SEQ ID NO: 124, SEQ ID NO: 125, SEQ ID NO: 126, SEQ ID NO: 127, SEQ ID NO: 128, SEQ ID NO: 129, SEQ ID NO: 130, or SEQ ID NO: 131; (d) a sequence that is at least 80% identical to SEQ ID NO: 132, SEQ ID NO: 133, SEQ ID NO: 134, SEQ ID NO: 135, SEQ ID NO: 136, or SEQ ID NO: 137; (e) a sequence that is at least 80% identical to SEQ ID NO: 138, SEQ ID NO: 139, SEQ ID NO: 140, SEQ ID NO: 141, SEQ ID NO: 142, or SEQ ID NO: 143; and (f) a sequence that is at least 80% identical to SEQ ID NO: 144, SEQ ID NO: 145, or SEQ ID NO: 146.

(534) Embodiment 58. The compound of any one of embodiments 1-57, wherein the second domain comprises a sequence that is at least 80% identical to SEQ ID NO: 147, SEQ ID NO: 148, or any one of SEQ ID NOS: 233-242.

(535) Embodiment 59. The compound of any one of embodiments 1-58, wherein: (a) the first domain comprises a sequence that is at least 80% identical to any one of SEQ ID NOS: 76-98; and (b) the second domain comprises a sequence that is at least 80% identical to any one of SEQ ID NOS: 99-148 or any one of SEQ ID NOS: 233-242.

(536) Embodiment 60. The compound of any one of embodiments 1-59, wherein: (a) the first domain comprises: (i) a sequence that is at least 80% identical to SEQ ID NO: 76, SEQ ID NO: 77, SEQ ID NO: 78, SEQ ID NO: 79, or SEQ ID NO: 80; (ii) a sequence that is at least 80% identical to SEQ ID NO: 81, SEQ ID NO: 82, SEQ ID NO: 83, SEQ ID NO: 84, SEQ ID NO: 85, SEQ ID NO: 86, or SEQ ID NO: 87; (iii) a sequence that is at least 80% identical to SEQ ID NO: 88, SEQ ID NO: 89, or SEQ ID NO: 90; (iv) a sequence that is at least 80% identical to SEQ ID NO: 91, SEQ ID NO: 92, or SEQ ID NO: 93; (v) a sequence that is at least 80% identical to SEQ ID NO: 94, SEQ ID NO: 95, or SEQ ID NO: 96; and (vi) a sequence that is at least 80% identical to SEQ ID NO: 97 or SEQ ID NO: 98; and (b) the second domain comprises: (i) a sequence that is at least 80% identical to SEQ ID NO: 99, SEQ ID NO: 100, SEQ ID NO: 101, SEQ ID NO: 102, SEQ ID NO: 103, SEQ ID NO: 104, SEQ ID NO: 105, SEQ ID NO: 106, SEQ ID NO: 107, SEQ ID NO: 108, SEQ ID NO: 109, SEQ ID NO: 110, SEQ ID NO: 111, SEQ ID NO: 112, or SEQ ID NO: 113; (ii) a sequence that is at least 80% identical to SEQ ID NO: 114, SEQ ID NO: 115, SEQ ID NO: 116, SEQ ID NO: 117, SEQ ID NO: 118, SEQ ID NO: 119, SEQ ID NO: 120, SEQ ID NO: 121, SEQ ID NO: 122, or SEQ ID NO: 123; (iii) a sequence that is at least 80% identical to SEQ ID NO: 124, SEQ ID NO: 125, SEQ ID NO: 126, SEQ ID NO: 127, SEQ ID NO: 128, SEQ ID NO: 129, SEQ ID NO: 130, or SEQ ID NO: 131; (iv) a sequence that is at least 80% identical to SEQ ID NO: 132, SEQ ID NO: 133, SEQ ID NO: 134, SEQ ID NO: 135, SEQ ID NO: 136, or SEQ ID NO: 137; (v) a sequence that is at least 80% identical to SEQ ID NO: 138, SEQ ID NO: 139, SEQ ID NO: 140, SEQ ID NO: 141, SEQ ID NO: 142, or SEQ ID NO: 143; and (vi) a sequence that is at least 80% identical to SEQ ID NO: 144, SEQ ID NO: 145, or SEQ ID NO: 146.

(537) Embodiment 61. The compound of any one of embodiments 1-60, wherein: (a) the first domain comprises a sequence that is at least 80% identical to SEQ ID NO: 76, SEQ ID NO: 77, SEQ ID NO: 78, SEQ ID NO: 79, or SEQ ID NO: 80; and (b) the second domain comprises a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 147, SEQ ID NO: 148, SEQ ID NO: 244, or any one of SEQ ID NOS: 233-242.

(538) Embodiment 62. The compound of any one of embodiments 1-61, wherein: (a) the first domain comprises a sequence that is at least 80% identical to SEQ ID NO: 81, SEQ ID NO: 82, SEQ ID NO: 83, SEQ ID NO: 84, SEQ ID NO: 85, SEQ ID NO: 86, or SEQ ID NO: 87; and (b)

the second domain comprises a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 147, SEQ ID NO: 148, SEQ ID NO: 244, or any one of SEQ ID NOS: 233-242.

(539) Embodiment 63. The compound of any one of embodiments 1-62, wherein: (a) the first domain comprises a sequence that is at least 80% identical to SEQ ID NO: 88, SEQ ID NO: 89, or SEQ ID NO: 90; and (b) the second domain comprises a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 147, SEQ ID NO: 148, SEQ ID NO: 244, or any one of SEQ ID NOS: 233-242.

(540) Embodiment 64. The compound of any one of embodiments 1-63, wherein: (a) the first domain comprises a sequence that is at least 80% identical to SEQ ID NO: 91, SEQ ID NO: 92, or SEQ ID NO: 93; and (b) the second domain comprises a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 147, SEQ ID NO: 148, SEQ ID NO: 244, or any one of SEQ ID NOS: 233-242.

(541) Embodiment 65. The compound of any one of embodiments 1-64, wherein: (a) the first domain comprises a sequence that is at least 80% identical to SEQ ID NO: 94, SEQ ID NO: 95, or SEQ ID NO: 96; and (b) the second domain comprises a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 147, SEQ ID NO: 148, SEQ ID NO: 244, or any one of SEQ ID NOS: 233-242.

(542) Embodiment 66. The compound of any one of embodiments 1-65, wherein: (a) the first domain comprises a sequence that is at least 80% identical to SEQ ID NO: 97 or SEQ ID NO: 98; and (b) the second domain comprises a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 147, SEQ ID NO: 148, SEQ ID NO: 244, or any one of SEQ ID NOS: 233-242.

(543) Embodiment 67. The compound of any one of embodiments 1-66, wherein: (a) the first domain comprises: (i) a sequence that is at least 80% identical to SEQ ID NO: 76, SEQ ID NO: 77, SEQ ID NO: 78, SEQ ID NO: 79, or SEQ ID NO: 80; (ii) a sequence that is at least 80% identical to SEQ ID NO: 81, SEQ ID NO: 82, SEQ ID NO: 83, SEQ ID NO: 84, SEQ ID NO: 85, SEQ ID NO: 86, or SEQ ID NO: 87; (iii) a sequence that is at least 80% identical to SEQ ID NO: 88, SEQ ID NO: 89, or SEQ ID NO: 90; (iv) a sequence that is at least 80% identical to SEQ ID NO: 91, SEQ ID NO: 92, or SEQ ID NO: 93; (v) a sequence that is at least 80% identical to SEQ ID NO: 94, SEQ ID NO: 95, or SEQ ID NO: 96; and (vi) a sequence that is at least 80% identical to SEQ ID NO: 97 or SEQ ID NO: 98; and (b) the second domain comprises a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 147, SEQ ID NO: 148, SEQ ID NO: 244, or any one of SEQ ID NOS: 233-242.

(544) Embodiment 68. The compound of any one of embodiments 1-67, wherein the compound comprises: (a) a heavy chain sequence that is at least 80% identical to SEQ ID NO: 149, SEQ ID NO: 150, SEQ ID NO: 151, or SEQ ID NO: 231; and (b) a light chain sequence that is at least 80% identical to SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, or SEQ ID NO: 20.

(545) Embodiment 69. The compound of any one of embodiments 1-4, 6-7, and 9-68, wherein the compound is a tetravalent bispecific antibody comprising: (a) a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 28, or SEQ ID NO: 244; (b) a sequence that is at least 80% identical to SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, or SEQ ID NO: 16; (c) a sequence that is at least 80% identical to any one of SEQ ID NOS: 31-75; and (d) a sequence that is at least 80% identical to SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, or SEQ ID NO: 20.

(546) Embodiment 70. The compound of any one of embodiments 1-4, 6-7 and 9-69, wherein the compound is a tetravalent bispecific antibody comprising: (a) a first chain, wherein the first chain comprises a linker, wherein the linker comprises a sequence that is at least 80% identical to any one of SEQ ID NOS: 31-75, wherein the linker comprises a N-terminus and a C-terminus, wherein the N-terminus of the linker is attached to a C terminus of a sequence that is at least 80% identical to SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, or SEQ ID NO: 16, and the C-terminus of the linker is attached to a N-terminus of a sequence that is at least 80% identical to SEQ ID NO: 22,

SEQ ID NO: 23, SEQ ID NO: 28, or SEQ ID NO: 244; and (b) a second chain, wherein the second chain comprises a sequence that is at least 80% identical to SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, or SEQ ID NO: 20.

(547) Embodiment 71. The compound of any one of embodiments 1-4, 6-7, 9-67, and 69, wherein the compound is a tetravalent bispecific antibody comprising: (a) a first chain, wherein the first chain comprises a sequence that is at least 80% identical to SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, or SEQ ID NO: 16; and (b) a second chain, wherein the second chain comprises a linker, wherein the linker comprises a sequence that is at least 80% identical to any one of SEQ ID NOS: 31-75, wherein the linker comprises a N-terminus and a C-terminus, wherein the N-terminus of the linker is attached to a C terminus of a sequence that is at least 80% identical to SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, or SEQ ID NO: 20, and the C-terminus of the linker is attached to a N-terminus of a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 28, or SEQ ID NO: 244.

(548) Embodiment 72. The compound of any one of embodiments 1-3, 5-6, and 8-67, wherein the compound is a hexavalent bispecific antibody comprising: (a) a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 28, or SEQ ID NO: 244; (b) a sequence that is at least 80% identical to SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, or SEQ ID NO: 16; (c) a sequence that is at least 80% identical to any one of SEQ ID NOS: 31-75; and (d) a sequence that is at least 80% identical to SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, or SEQ ID NO: 20.

(549) Embodiment 73. The compound of any one of embodiments 1-3, 5-6, 8-67, and 72, wherein the compound is a hexavalent bispecific antibody comprising: (a) a first chain, wherein the first chain comprises a linker, wherein the linker comprises a sequence that is at least 80% identical to any one of SEQ ID NOS: 31-75, wherein the linker comprises a N-terminus and a C-terminus, wherein the N-terminus of the linker is attached to a C terminus of a sequence that is at least 80% identical to SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, or SEQ ID NO: 16, and the C-terminus of the linker is attached to a N-terminus of a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 28, or SEQ ID NO: 244; and (b) a second chain, wherein the second chain comprises a linker, wherein the linker comprises a sequence that is at least 80% identical to any one of SEQ ID NOS: 31-75, wherein the linker comprises a N-terminus and a C-terminus, wherein the N-terminus of the linker is attached to a C terminus of a sequence that is at least 80% identical to SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, or SEQ ID NO: 20, and the C-terminus of the linker is attached to a N-terminus of a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 28, or SEQ ID NO: 244.

(550) Embodiment 74. The compound of any one of embodiments 1-67, wherein the compound comprises: (a) a heavy chain sequence that is at least 80% identical to SEQ ID NO: 245 or any one of SEQ ID NOS: 13-16; and (b) a light chain sequence that is at least 80% identical to SEQ ID NO: 218, SEQ ID NO: 219, SEQ ID NO: 232, or SEQ ID NO: 243.

(551) Embodiment 75. The compound of any one of embodiments 1-67, wherein the compound comprises: (a) a heavy chain sequence that is at least 80% identical to SEQ ID NO: 149, SEQ ID NO: 150, SEQ ID NO: 151, or SEQ ID NO: 231; and (b) a light chain sequence that is at least 80% identical to SEQ ID NO: 218, SEQ ID NO: 219, SEQ ID NO: 232, or SEQ ID NO: 243.

(552) Embodiment 76. The compound of any one of embodiments 1-75, wherein a binding affinity (K.sub.D) of the compound to HPTP- β is about 30 fM to about 70 nM.

(553) Embodiment 77. The compound of any one of embodiments 1-76, wherein a binding affinity (K.sub.D) of the compound to the VEGF is about 30 fM to about 70 nM.

(554) Embodiment 78. The compound of any one of embodiments 1-75, wherein a binding affinity (K.sub.D) of the compound to HPTP- β is about 30 fM to about 70 nM, and a binding affinity (K.sub.D) of the compound to VEGF is about 30 fM to about 70 nM.

(555) Embodiment 79. A method of treating a condition in a subject in need thereof, the method

comprising administering to the subject a therapeutically-effective amount of the compound of any one of embodiments 1-78.

(556) Embodiment 80. The method of embodiment 79, wherein the condition is an ocular condition.

(557) Embodiment 81. The method of any one of embodiments 79-80, wherein the condition is diabetic retinopathy.

(558) Embodiment 82. The method of any one of embodiments 79-80, wherein the condition is neovascularization.

(559) Embodiment 83. The method of any one of embodiments 79-80, wherein the condition is vascular leak.

(560) Embodiment 84. The method of any one of embodiments 79-80, wherein the condition is increased intraocular pressure.

(561) Embodiment 85. The method of any one of embodiments 79-80, wherein the condition is ocular edema.

(562) Embodiment 86. The method of any one of embodiments 79-80 and 85, wherein the condition is diabetic macular edema.

(563) Embodiment 87. The method of any one of embodiments 79-80, wherein the condition is ocular hypertension.

(564) Embodiment 88. The method of any one of embodiments 79-80, wherein the condition is ocular inflammation.

(565) Embodiment 89. The method of any one of embodiments 79-80, wherein the condition is glaucoma.

(566) Embodiment 90. The method of any one of embodiments 79-89, wherein the administration is to an eye of the subject.

(567) Embodiment 91. The method of any one of embodiments 79-90, wherein the administration is intravitreal.

(568) Embodiment 92. The method of any one of embodiments 79-89, wherein the administration is subcutaneous.

(569) Embodiment 93. The method of any one of embodiments 79-90, wherein the administration is topical.

(570) Embodiment 94. The method of any one of embodiments 79-93, wherein the subject is human.

(571) Embodiment 95. The method of any one of embodiments 79-94, wherein the therapeutically-effective amount is from about 0.25 mg to about 200 mg.

(572) Embodiment 96. The method of any one of embodiments 79-95, wherein the therapeutically-effective amount is from about 1 mg/kg to about 10 mg/kg.

(573) Embodiment 97. The method of any one of embodiments 79-95, wherein the therapeutically-effective amount is from about 1 mg to about 50 mg.

(574) Embodiment 98. The method of any one of embodiments 79-95, wherein the therapeutically-effective amount is from about 50 mg to about 200 mg.

(575) Embodiment 99. The compound of any one of embodiments 1-67, wherein the compound comprises: (a) a heavy chain sequence that is at least 80% identical to SEQ ID NO: 254, SEQ ID NO: 255, SEQ ID NO: 256, or SEQ ID NO: 257; and (b) a light chain sequence that is at least 80% identical to SEQ ID NO: 250, SEQ ID NO: 251, SEQ ID NO: 252, or SEQ ID NO: 253.

(576) Embodiment 100. The compound of any one of embodiments 1-4, 6-7, and 9-68, wherein the compound is a tetravalent bispecific antibody comprising: (a) a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 28, or SEQ ID NO: 244; (b) a sequence that is at least 80% identical to SEQ ID NO: 246, SEQ ID NO: 247, SEQ ID NO: 248, or SEQ ID NO: 249; (c) a sequence that is at least 80% identical to any one of SEQ ID NOS: 31-75; and (d) a sequence that is at least 80% identical to SEQ ID NO: 250, SEQ ID NO: 251, SEQ ID NO: 252, or

SEQ ID NO: 253.

(577) Embodiment 101. The compound of any one of embodiments 1-4, 6-7 and 9-69, wherein the compound is a tetravalent bispecific antibody comprising: (a) a first chain, wherein the first chain comprises a linker, wherein the linker comprises a sequence that is at least 80% identical to any one of SEQ ID NOS: 31-75, wherein the linker comprises a N-terminus and a C-terminus, wherein the N-terminus of the linker is attached to a C terminus of a sequence that is at least 80% identical to SEQ ID NO: 246, SEQ ID NO: 247, SEQ ID NO: 248, or SEQ ID NO: 249, and the C-terminus of the linker is attached to a N-terminus of a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 28, or SEQ ID NO: 244; and (b) a second chain, wherein the second chain comprises a sequence that is at least 80% identical to SEQ ID NO: 250, SEQ ID NO: 251, SEQ ID NO: 252, or SEQ ID NO: 253.

(578) Embodiment 102. The compound of any one of embodiments 1-4, 6-7, 9-67, and 69, wherein the compound is a tetravalent bispecific antibody comprising: (a) a first chain, wherein the first chain comprises a sequence that is at least 80% identical to SEQ ID NO: 246, SEQ ID NO: 247, SEQ ID NO: 248, or SEQ ID NO: 249; and (b) a second chain, wherein the second chain comprises a linker, wherein the linker comprises a sequence that is at least 80% identical to any one of SEQ ID NOS: 31-75, wherein the linker comprises a N-terminus and a C-terminus, wherein the N-terminus of the linker is attached to a C terminus of a sequence that is at least 80% identical to SEQ ID NO: 250, SEQ ID NO: 251, SEQ ID NO: 252, or SEQ ID NO: 253, and the C-terminus of the linker is attached to a N-terminus of a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 28, or SEQ ID NO: 244.

(579) Embodiment 103. The compound of any one of embodiments 1-3, 5-6, and 8-67, wherein the compound is a hexavalent bispecific antibody comprising: (a) a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 28, or SEQ ID NO: 244; (b) a sequence that is at least 80% identical to SEQ ID NO: 246, SEQ ID NO: 247, SEQ ID NO: 248, or SEQ ID NO: 249; (c) a sequence that is at least 80% identical to any one of SEQ ID NOS: 31-75; and (d) a sequence that is at least 80% identical to SEQ ID NO: 250, SEQ ID NO: 251, SEQ ID NO: 252, or SEQ ID NO: 253.

(580) Embodiment 104. The compound of any one of embodiments 1-3, 5-6, 8-67, and 72, wherein the compound is a hexavalent bispecific antibody comprising: (a) a first chain, wherein the first chain comprises a linker, wherein the linker comprises a sequence that is at least 80% identical to any one of SEQ ID NOS: 31-75, wherein the linker comprises a N-terminus and a C-terminus, wherein the N-terminus of the linker is attached to a C terminus of a sequence that is at least 80% identical to SEQ ID NO: 246, SEQ ID NO: 247, SEQ ID NO: 248, or SEQ ID NO: 249, and the C-terminus of the linker is attached to a N-terminus of a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 28, or SEQ ID NO: 244; and (b) a second chain, wherein the second chain comprises a linker, wherein the linker comprises a sequence that is at least 80% identical to any one of SEQ ID NOS: 31-75, wherein the linker comprises a N-terminus and a C-terminus, wherein the N-terminus of the linker is attached to a C terminus of a sequence that is at least 80% identical to SEQ ID NO: 250, SEQ ID NO: 251, SEQ ID NO: 252, or SEQ ID NO: 253, and the C-terminus of the linker is attached to a N-terminus of a sequence that is at least 80% identical to SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 28, or SEQ ID NO: 244.

(581) Embodiment 105. The compound of any one of embodiments 1-67, wherein the compound comprises: (a) a heavy chain sequence that is at least 80% identical to SEQ ID NO: 262 or any one of SEQ ID NOS: 246-249; and (b) a light chain sequence that is at least 80% identical to SEQ ID NO: 258, SEQ ID NO: 259, SEQ ID NO: 261, or SEQ ID NO: 260.

(582) Embodiment 106. The compound of any one of embodiments 1-3, 5-6, and 8-67, wherein the compound comprises: (a) a heavy chain sequence that is at least 80% identical to SEQ ID NO: 254, SEQ ID NO: 255, SEQ ID NO: 256, or SEQ ID NO: 257; and (b) a light chain sequence that is at least 80% identical to SEQ ID NO: 258, SEQ ID NO: 259, SEQ ID NO: 261, or SEQ ID NO: 260.

Claims

1. A method of treating an ocular condition in a subject in need thereof, comprising administering to the subject a therapeutically effective amount of an antibody comprising: (a) a first domain that activates Tie2, wherein the first domain comprises an amino acid sequence that is any one of SEQ ID NOs: 76-80, an amino acid sequence that is any one of SEQ ID NOs: 81-87, an amino acid sequence that is any one of SEQ ID NOs: 88-90, an amino acid sequence that is any one of SEQ ID NOs: 91-93, an amino acid sequence that is any one of SEQ ID NOs: 94-96, and an amino acid sequence that is any one of SEQ ID NOs: 97-98; and (b) a second domain that specifically binds a receptor tyrosine kinase agonist, wherein the second domain comprises an amino acid sequence that is any one of SEQ ID NOs: 99-113, an amino acid sequence that is any one of SEQ ID NOs: 114-123, an amino acid sequence that is any one of SEQ ID NOs: 124-131, an amino acid sequence that is any one of SEQ ID NOs: 132-137, an amino acid sequence that is any one of SEQ ID NOs: 138-143, and an amino acid sequence that is any one of SEQ ID NOs: 144-146.
 2. The method of claim 1, wherein the ocular condition is diabetic retinopathy.
 3. The method of claim 1, wherein the ocular condition is neovascularization.
 4. The method of claim 1, wherein the ocular condition is vascular leak.
 5. The method of claim 1, wherein the ocular condition is increased intraocular pressure.
 6. The method of claim 1, wherein the ocular condition is ocular edema.
 7. The method of claim 1, wherein the ocular condition is diabetic macular edema.
 8. The method of claim 1, wherein the ocular condition is wet age-related macular degeneration.
 9. The method of claim 1 wherein the ocular condition is ocular inflammation.
 10. The method of claim 1, wherein the ocular condition is retinal vein occlusion.
 11. The method of claim 1, wherein the administering is to an eye of the subject.
 12. The method of claim 1, wherein the administering is intravitreal.
 13. The method of claim 1, wherein the administering is subcutaneous.
 14. The method of claim 1, wherein the subject is a human.
 15. The method of claim 1, wherein the therapeutically effective amount is from about 0.25 mg to about 200 mg.
 16. The method of claim 1, wherein the therapeutically effective amount is from about 1 mg/kg to about 10 mg/kg.
 17. The method of claim 1, wherein the antibody inhibits HPTP- β .
 18. The method of claim 1, wherein the antibody inhibits VE-PTP.
 19. The method of claim 1, wherein the antibody inhibits VEGF.
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